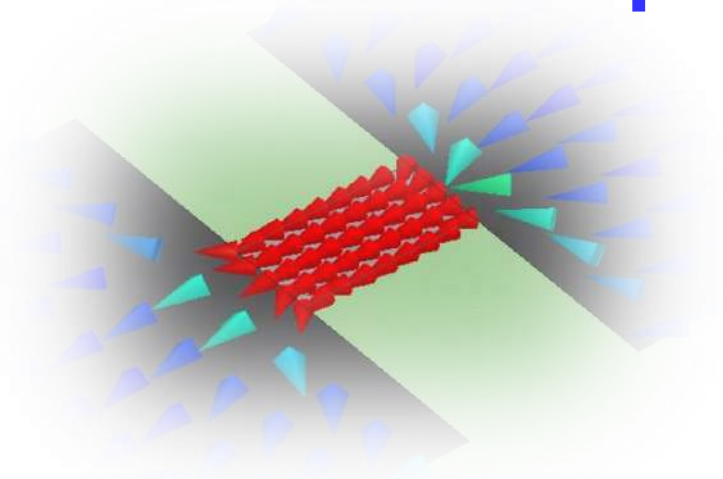




Energy-participation approach to Josephson circuit quantization

arXiv:1902.10355 (Ch. 4)
Manuscript in preparation
Github



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Yale University * Present: IBM Research ⁺ ENS, Paris ⁺ KIT, Germany

Acknowledgements: S.M. Girvin, R.J. Schoelkopf, A. Blais, S. Nigg, H. Paik, F. Solgun, RSL, Qulab, ...

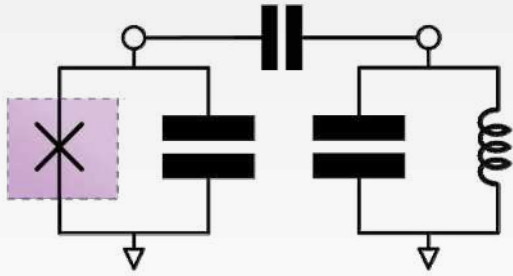


Yale Institute for Nanoscience
and Quantum Engineering

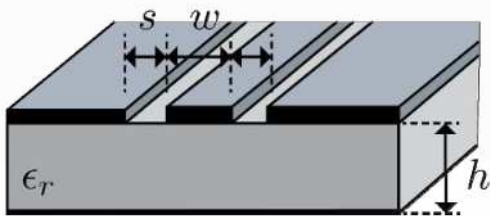
How to design a superconducting
quantum circuit / computer?

Circuit quantization

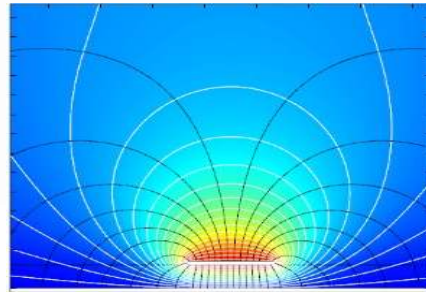
Lumped-element approximation



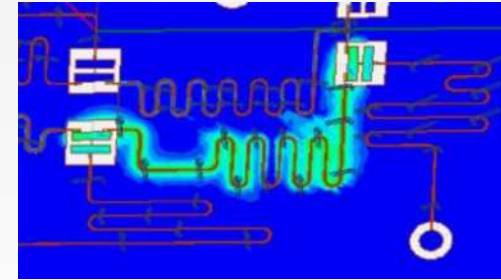
analytical



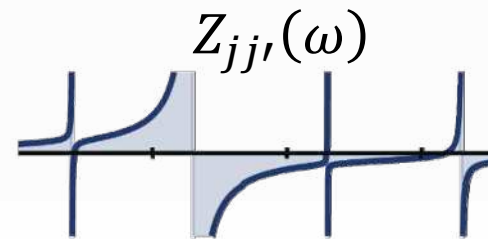
FE/boundary



Full Maxwell equations



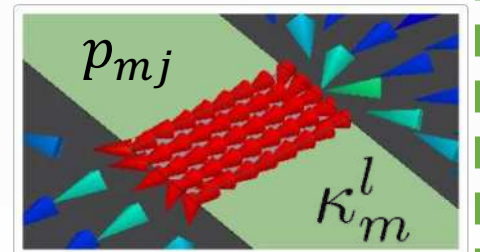
impedance^{*}



eigen and driven

energy[†]

Image: Brumby *et al.*, QST (2018)



eigen

**Captures more information,
complexity, accuracy**

* Nigg, Paik, *et al.* (2012); Bourassa *et al.* (2012); Solgun *et al.* (2014, 2015, 2017) ...

† Mineev *et al.*, arXiv:1902.10355, in prep. (2019)

A unified framework to handle all these questions.

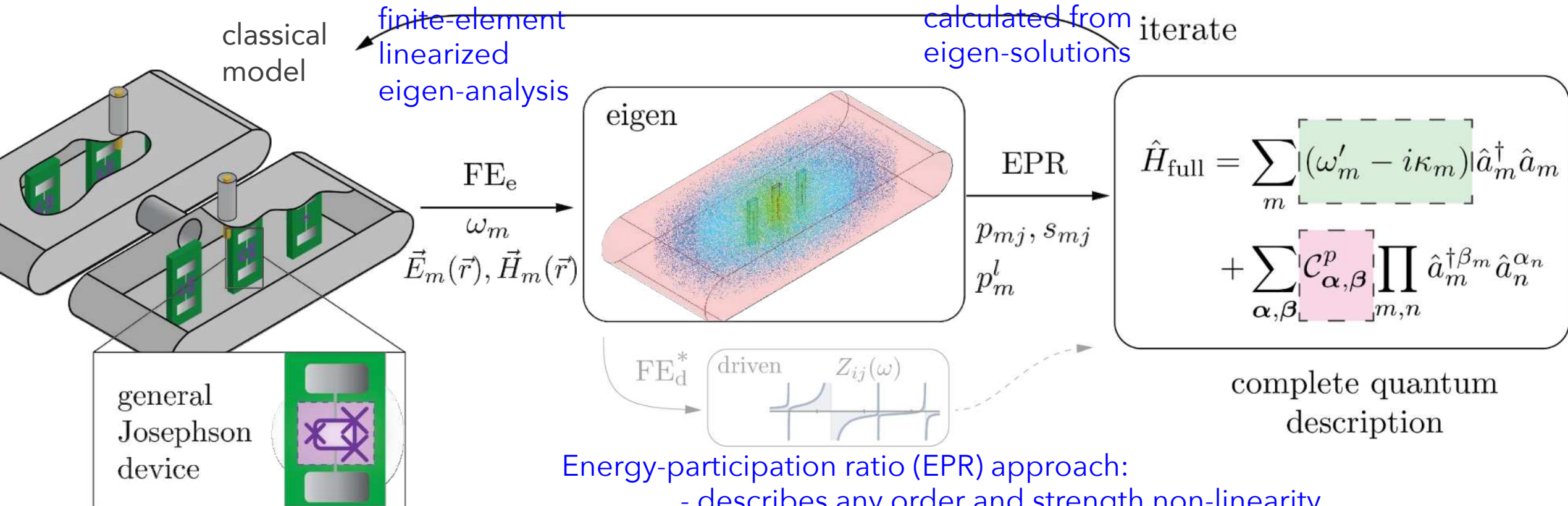
The solution reduces to asking:

Where is the energy?

What *fraction of the energy* of the mode is stored in the non-linear/dissipative element?

$$0 \leq p, p^l \leq 1$$

Overview of energy approach



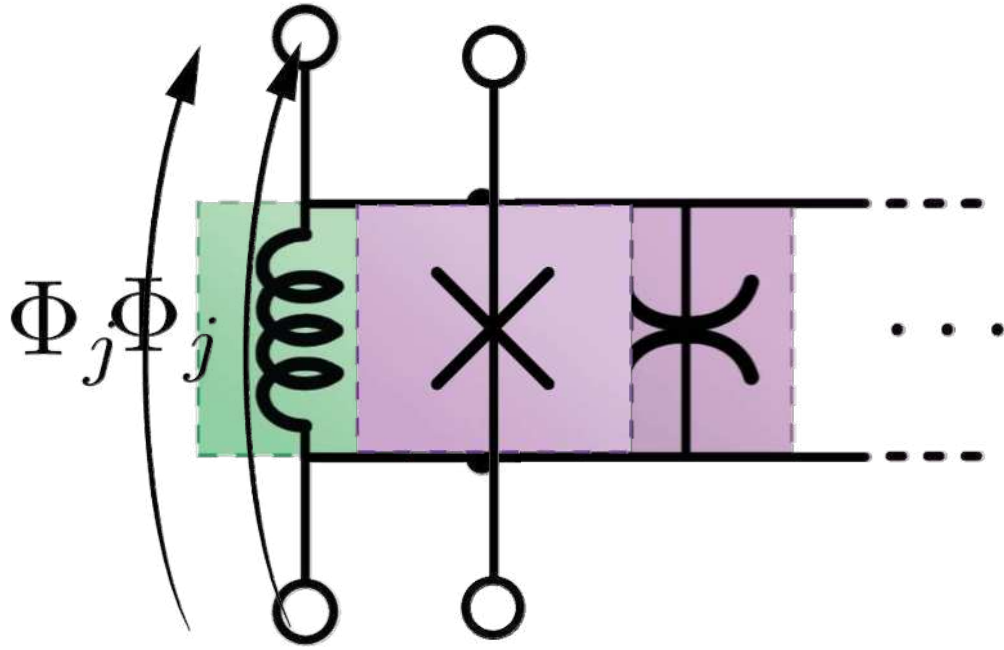
Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in python (github.com/zlatko-minev)

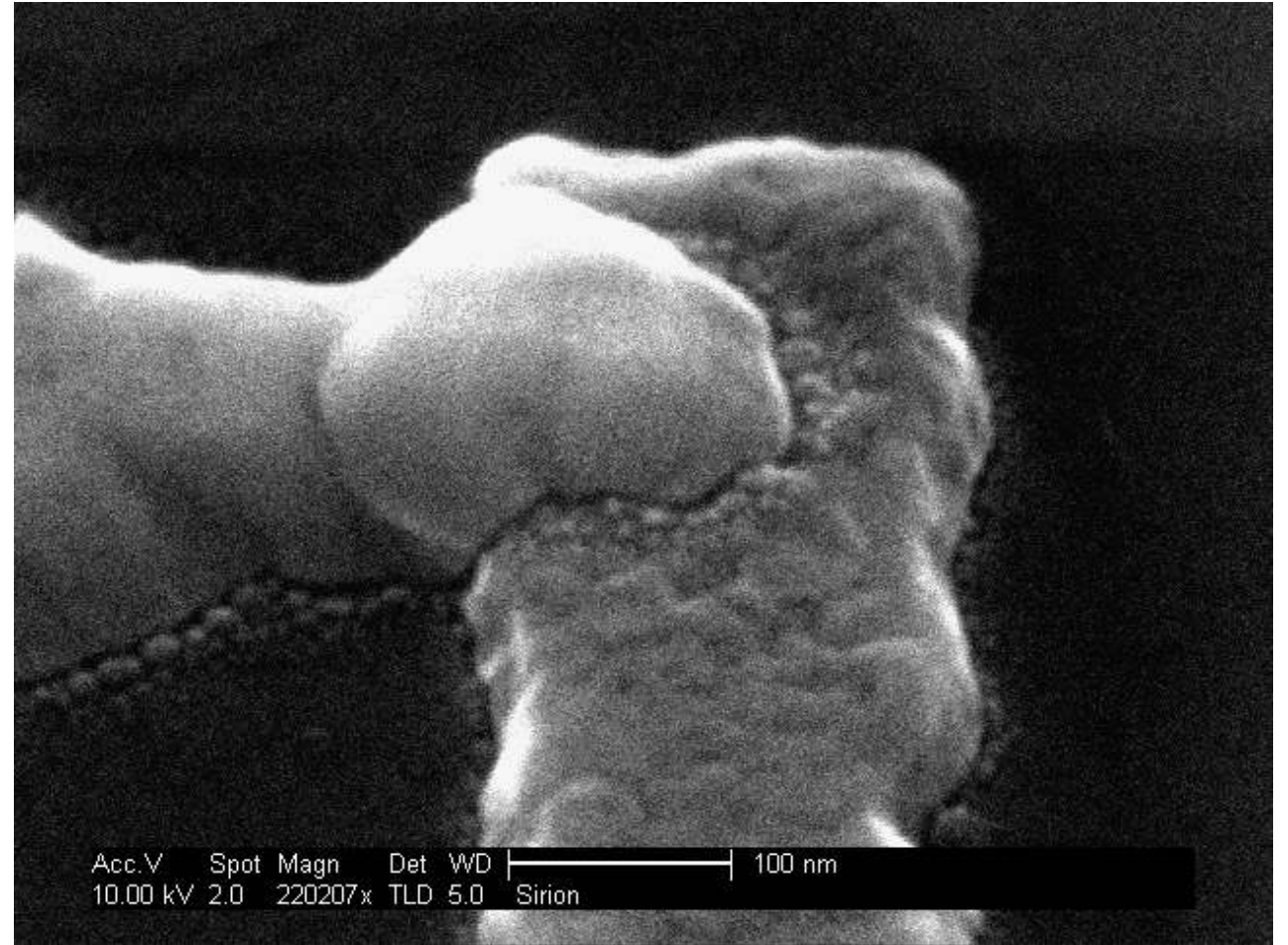
Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017), ...

Non-linear element: Josephson tunnel junction

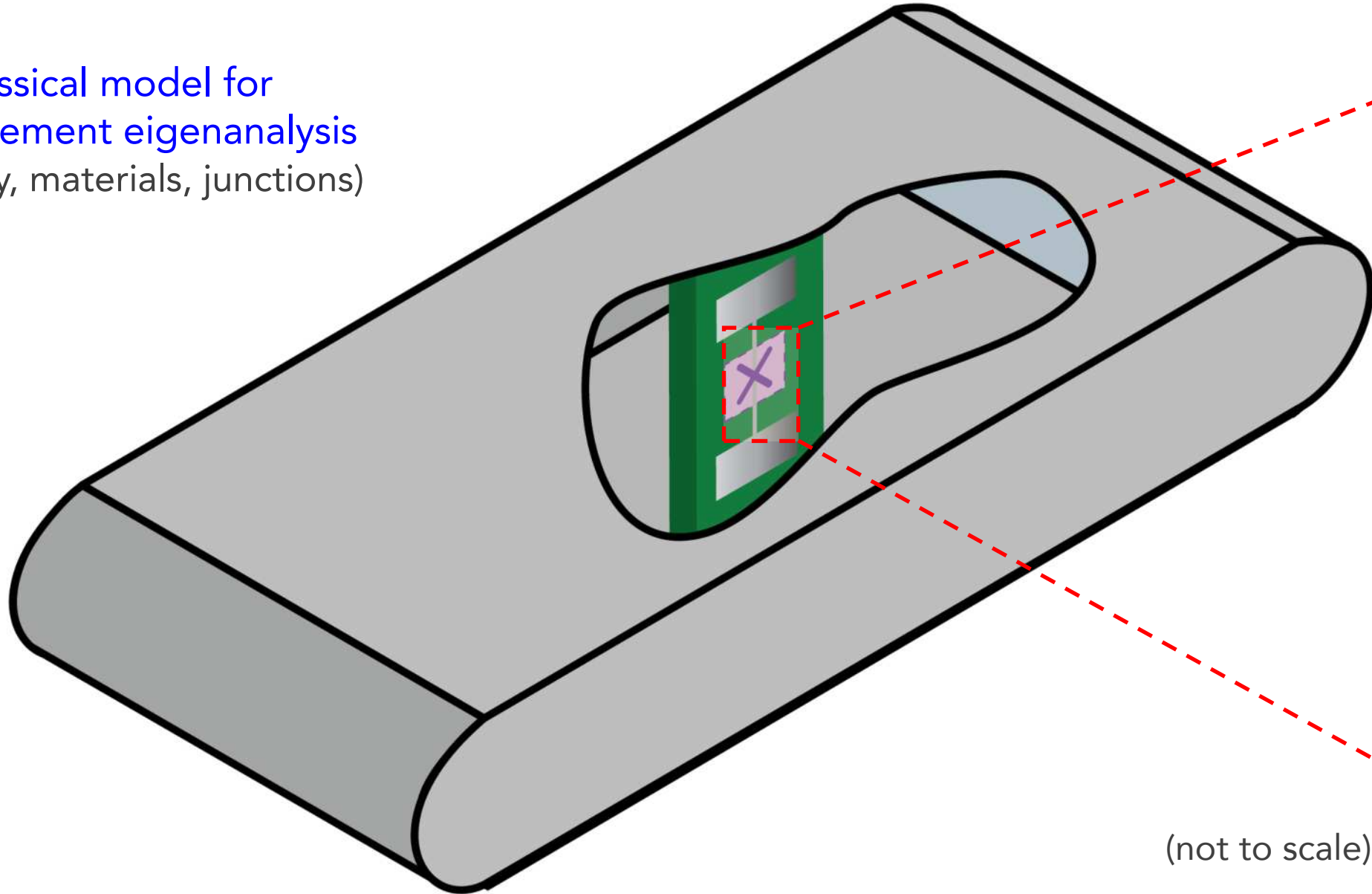


$$\mathcal{E}_j(\Phi_j) = \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$



Transmon qubit coupled to cavity

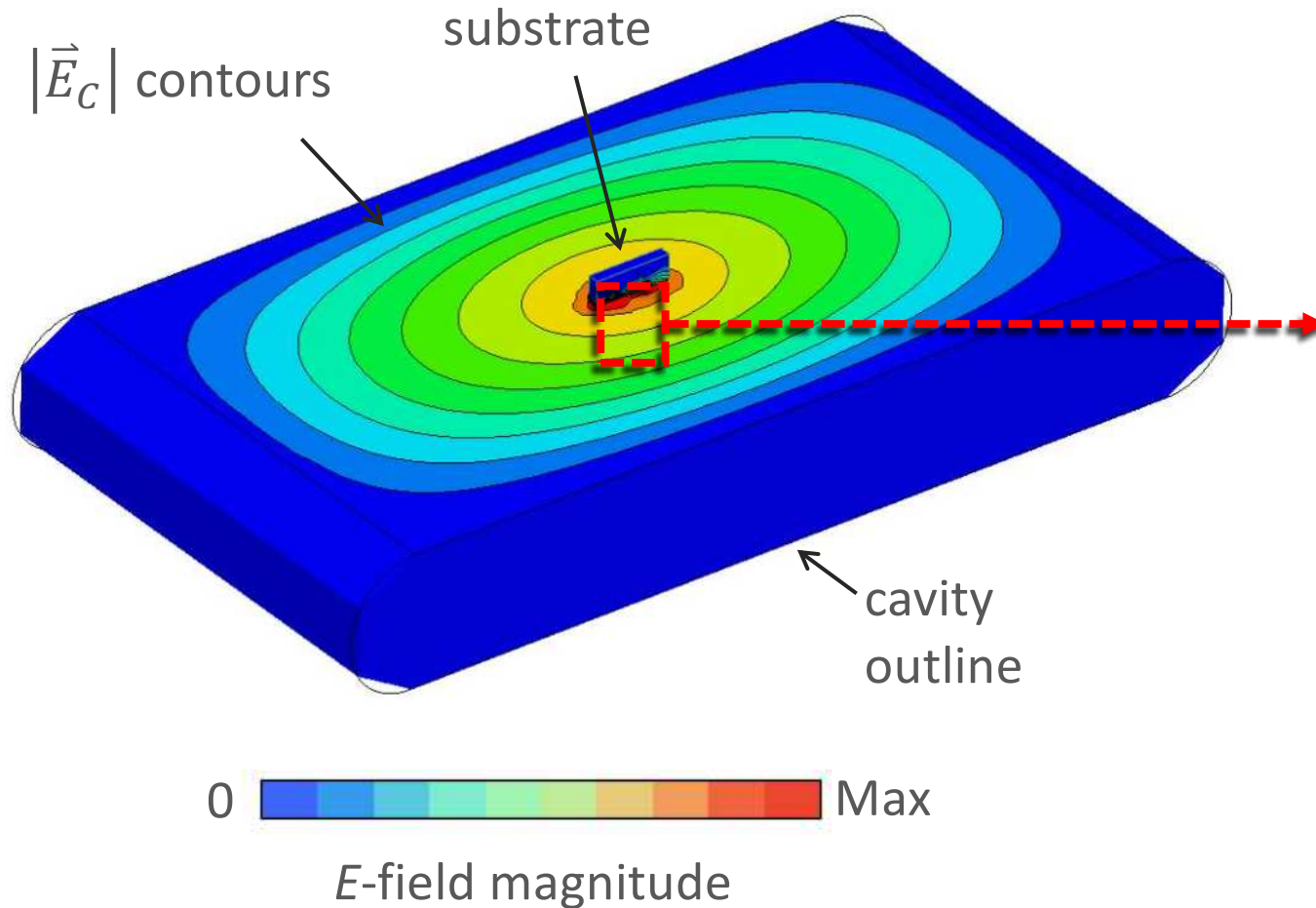
Classical model for
finite-element eigenanalysis
(geometry, materials, junctions)



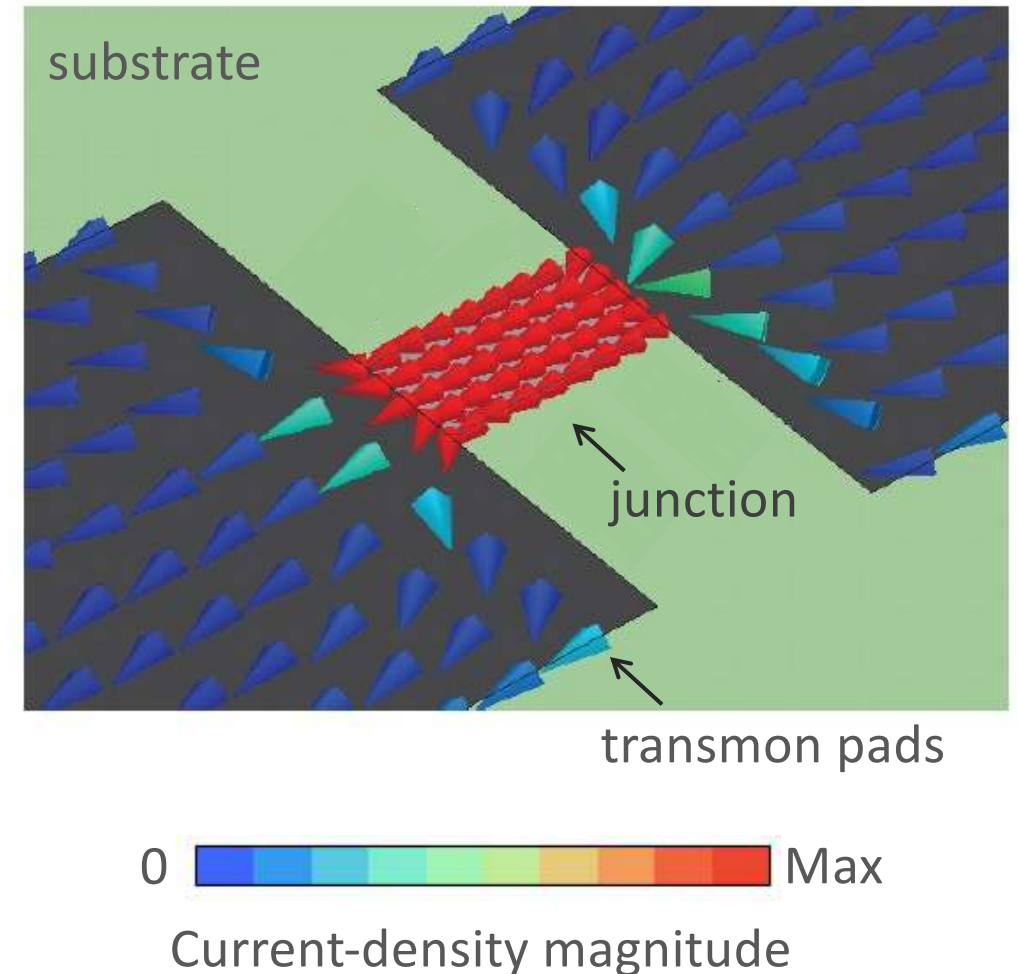
(not to scale)

$\mathcal{H}_{\text{lin}}$ eigen modes

Cavity mode (7.0 GHz)



Qubit mode (linearized, 5 GHz)

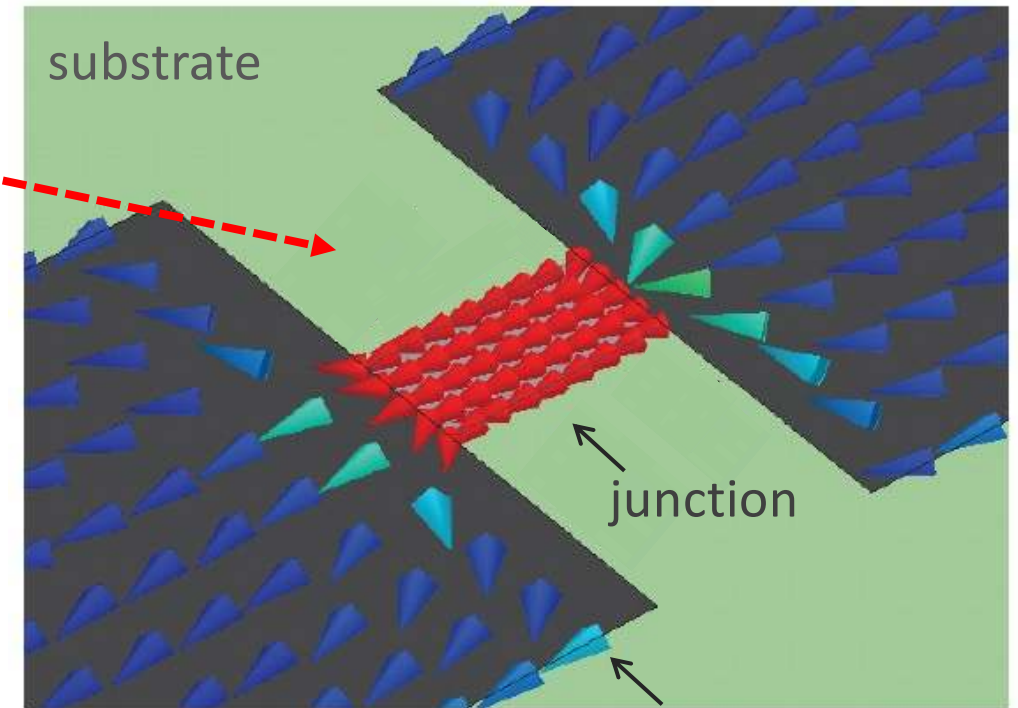


Energy participation of the junction

$$p_m = \frac{\text{Energy stored in junction}}{\text{Inductive energy stored in mode } m}$$

Note: The fraction above is enclosed in a red dashed box in the original image, and a red dashed arrow points from this box to the 'junction' label in the adjacent image.

Qubit mode (linearized, 5 GHz)

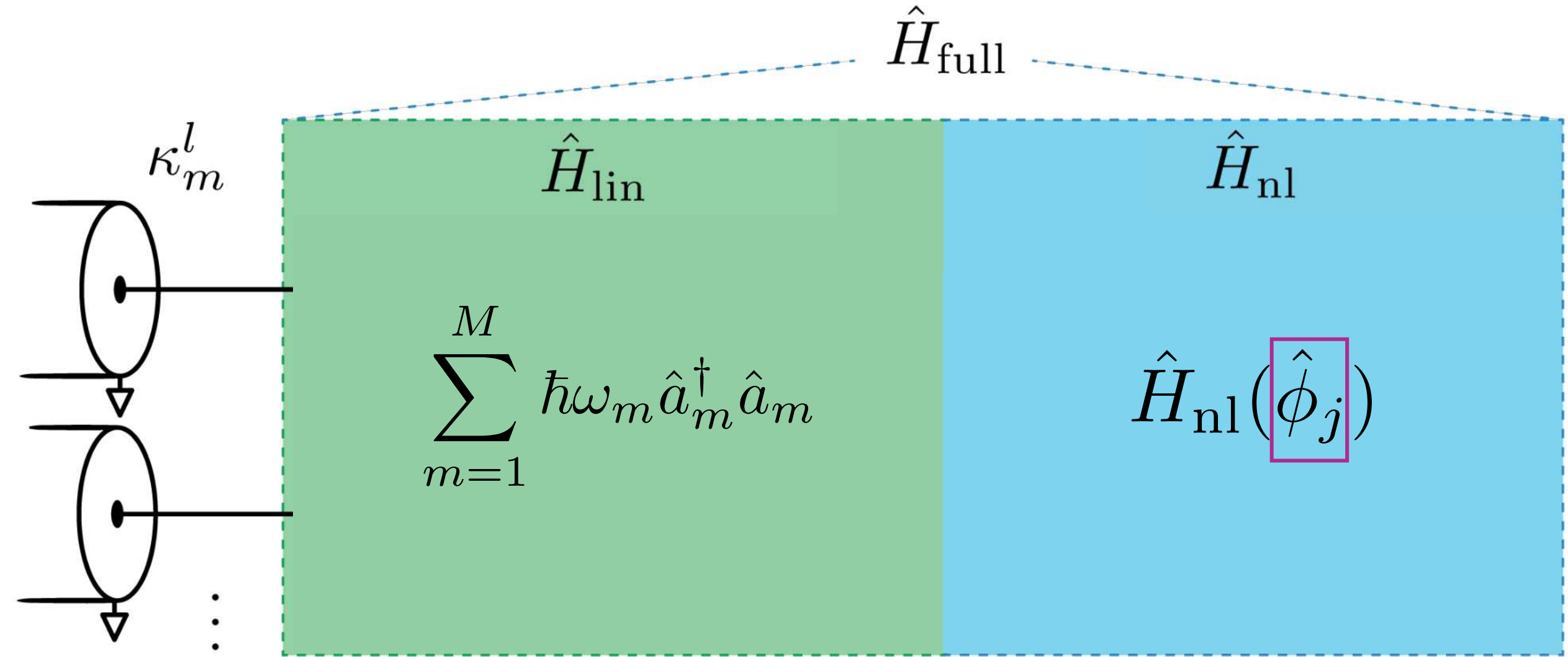


transmon pads



Current-density magnitude

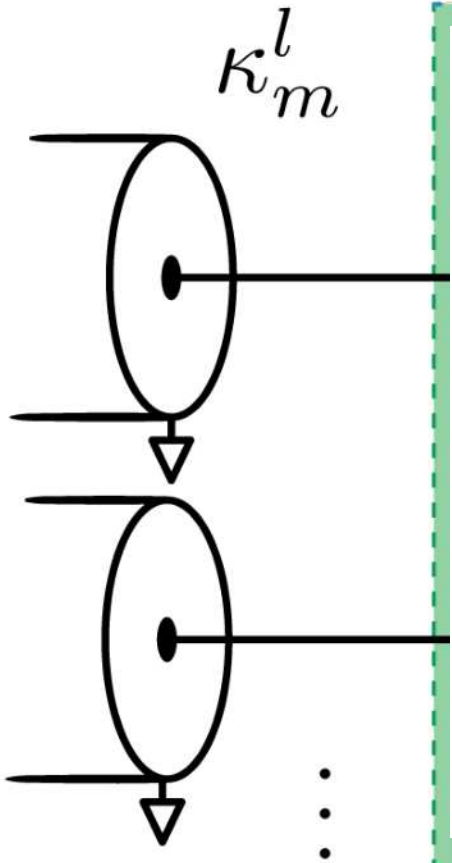
Decomposition of a general circuit



second quantization in eigen basis of linearized circuit

Decomposition of a general circuit

$$\hat{H}_{\text{full}}$$



What is zero-point fluctuation of the phase of junction j due to mode m ?

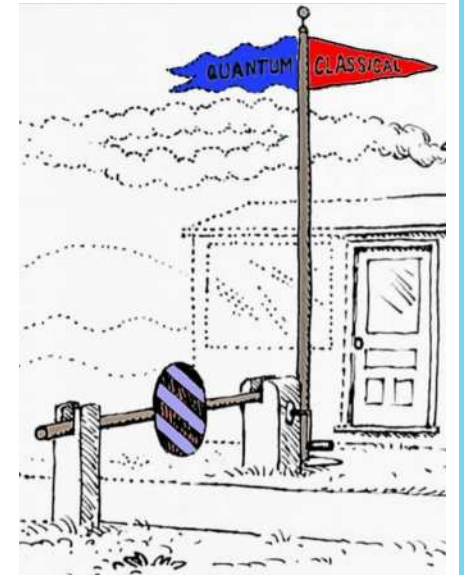
$$\hat{\phi}_j = \sum_{m=1}^M \phi_{mj} (\hat{a}_m^\dagger + \hat{a}_m)$$

Decomposition of a general circuit

$$\hat{H}_{\text{full}}$$

What fraction of the energy of mode m is stored in junction j ?

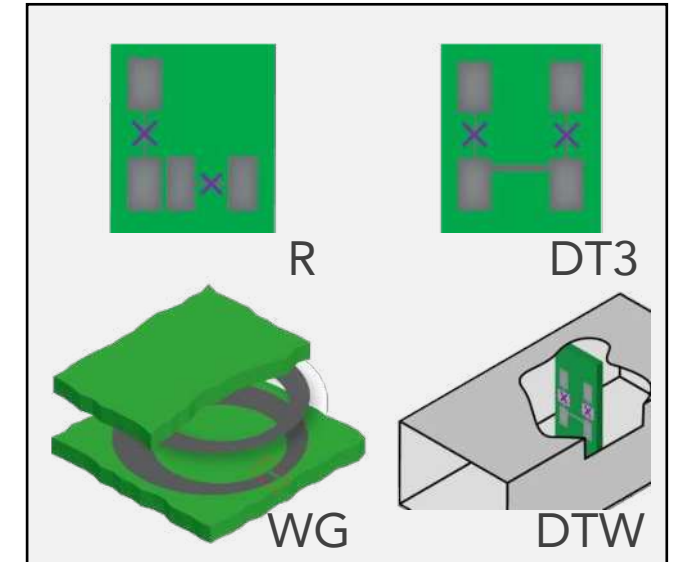
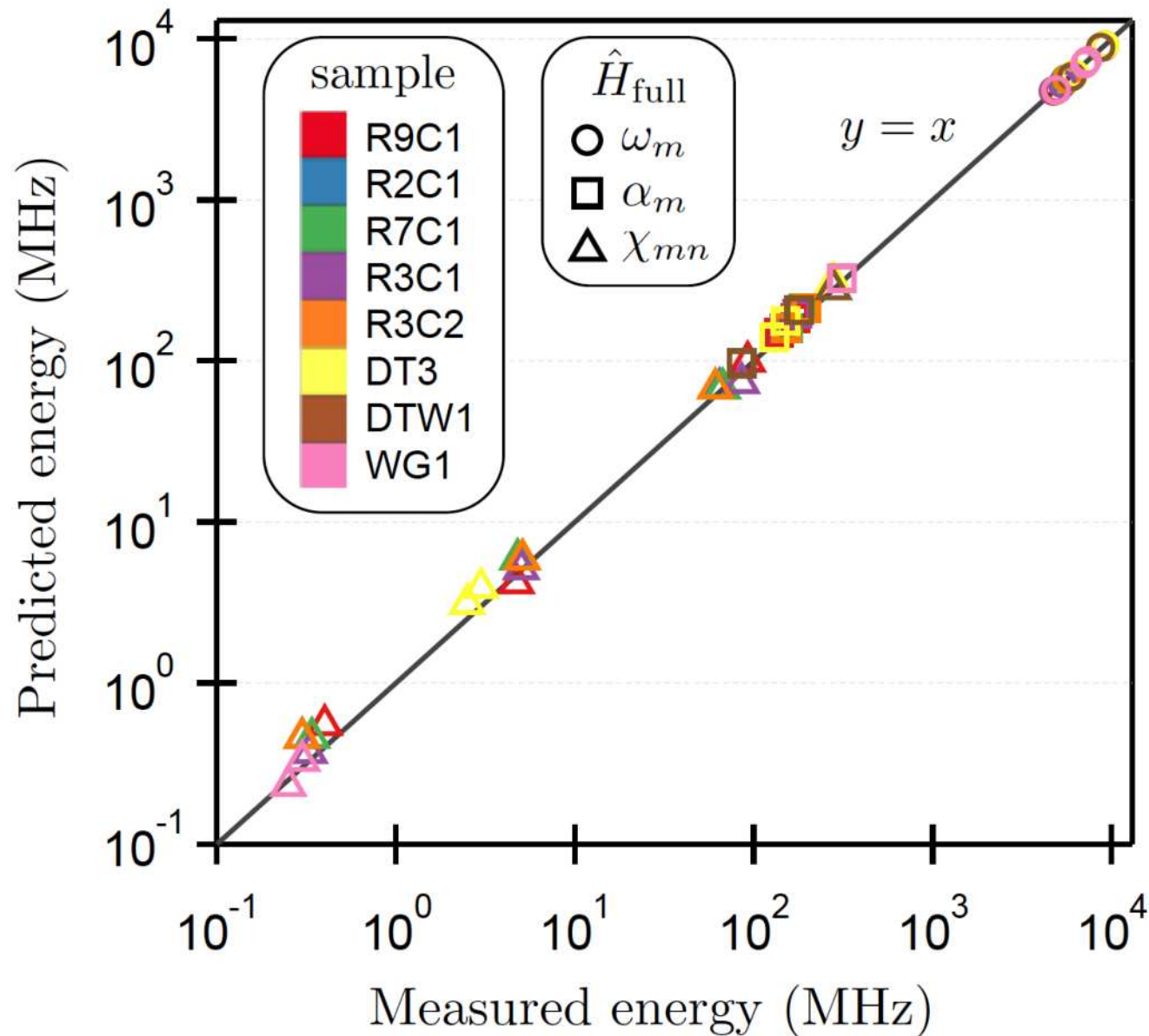
$$\frac{1}{\hbar} \phi_{mj}^2 = p_{mj} \frac{\omega_m}{2E_j}$$



for $j > 1$, root requires sign bit $s_{mj} = \pm 1$

Drawing: Zurek, Physics Today (1991)

Theory vs. experiment: agreement over 5 orders of magnitude



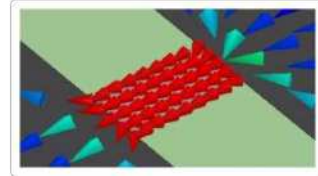
R: Minev *et al.* (2018)

WG: Minev *et al.* (2013, 2016)

DT3, DTW: Minev *et al.* (2019)

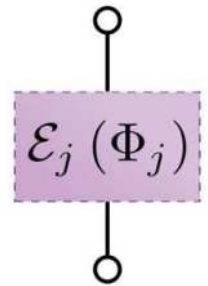
Energy-Participation-Ratio (EPR) Approach

Simple and unified analysis of dissipation and Hamiltonian
single-simulation efficient
obviate driven simulations and notion of $Z_{ij}(\omega)$



General

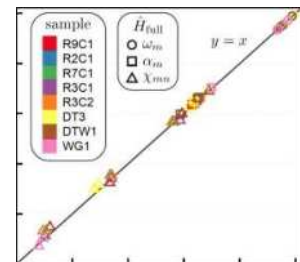
handle arbitrary architectures and non-linear devices
e.g., nanowires, ...



Accuracy

experimental ten to percent level over 5 orders in $\hat{H}_{\text{full}}$

Suited for automated, robust analysis of large systems
fully automated and tested (see GitHub)



arXiv:1902.10355 (Ch. 4); to appear soon; see also E28.7

Application to J and CR gate, ...

EXTRA SLIDES

Fundamental constraints with EPR

$$\hat{H}_{\text{full}}$$

$$\kappa_m^l$$

What circuits/parameters can and cannot be realized?

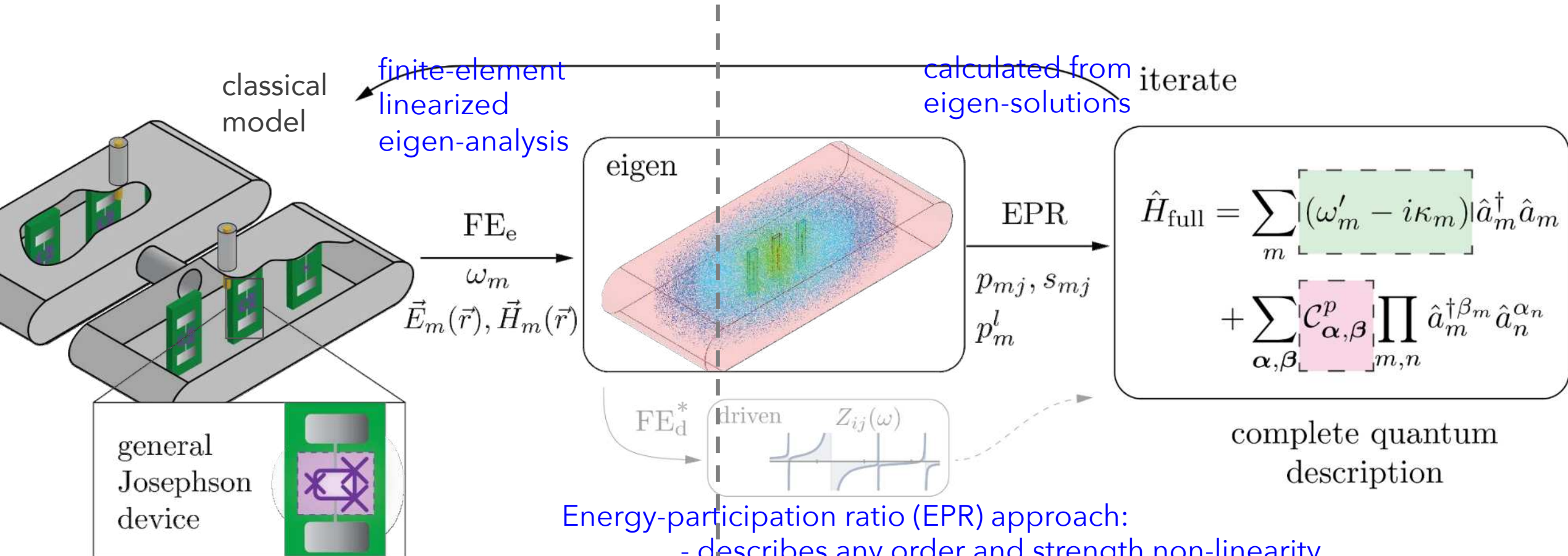
$$\sum_{m=1}^M p_{mj} = 1$$

$$0 \leq \sum_{j=1}^J p_{mj} \leq 1$$

...

Convergence checks

Overview of energy approach



Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in python (github.com/zlatko-minev)

Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017), ...

What fraction of the energy of the mode
is 'present' in the dissipative element?

$$0 \leq p^l \leq 1$$

Non-linear interactions with EPR

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar \omega_m \omega_n}{4E_j} p_{mj} p_{nj}$$

Energy
participation
overlap

(only variable subject
to significant variation)

$$\chi_{mn} \hat{a}_m^\dagger \hat{a}_m \hat{a}_n^\dagger \hat{a}_n$$

Outline

The problem

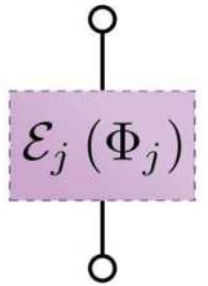
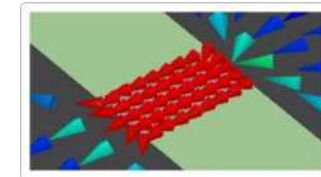
Energy-participation approach

overview

simple example

CR gate

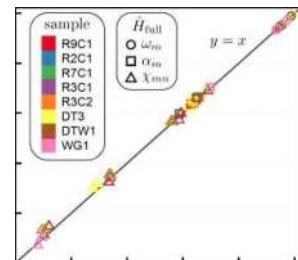
advantages & general formulation



Experimental results

samples: multi-qubit, 2.5D flip-chip, 3D, waveguides, ...

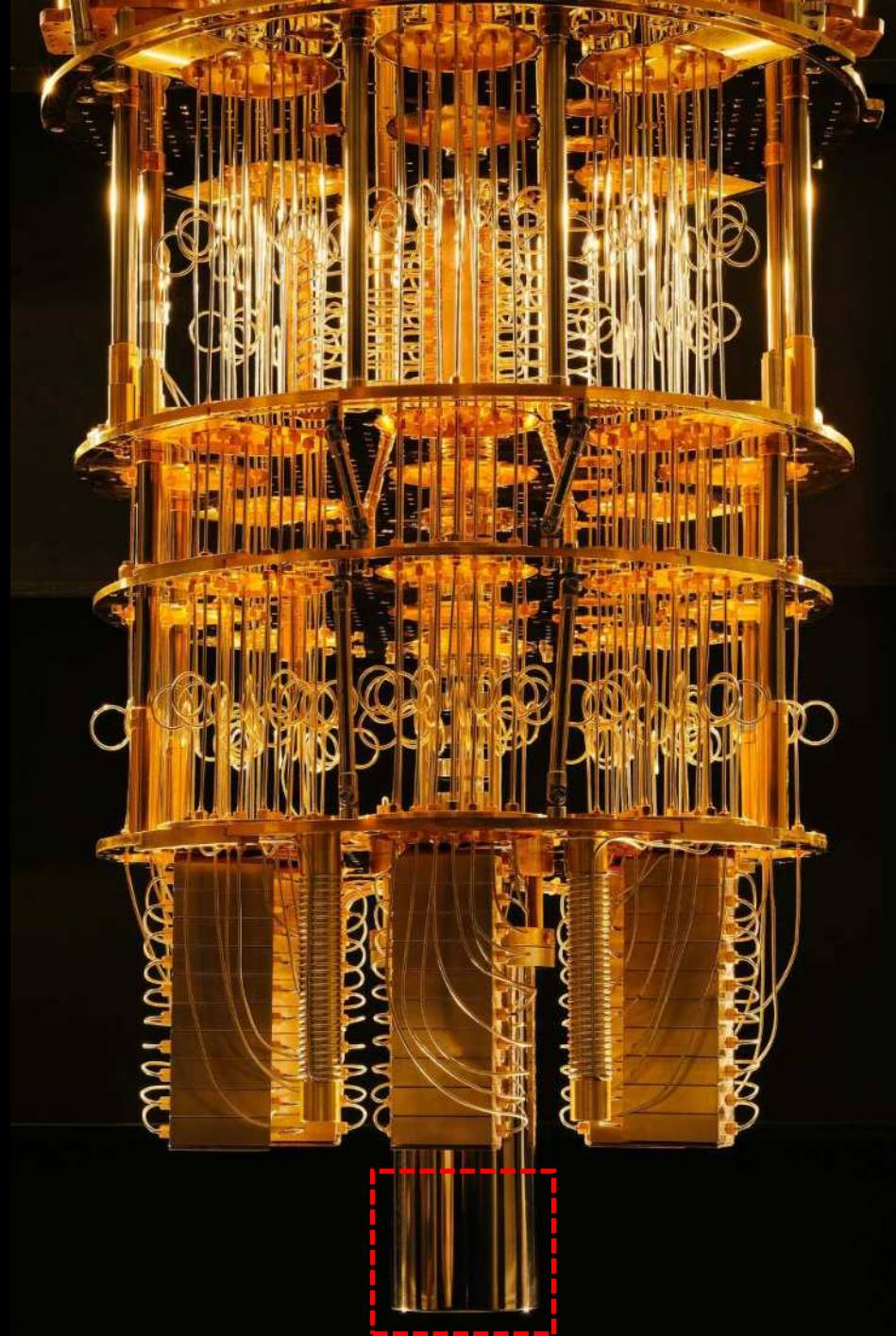
precision: theory vs. experiment

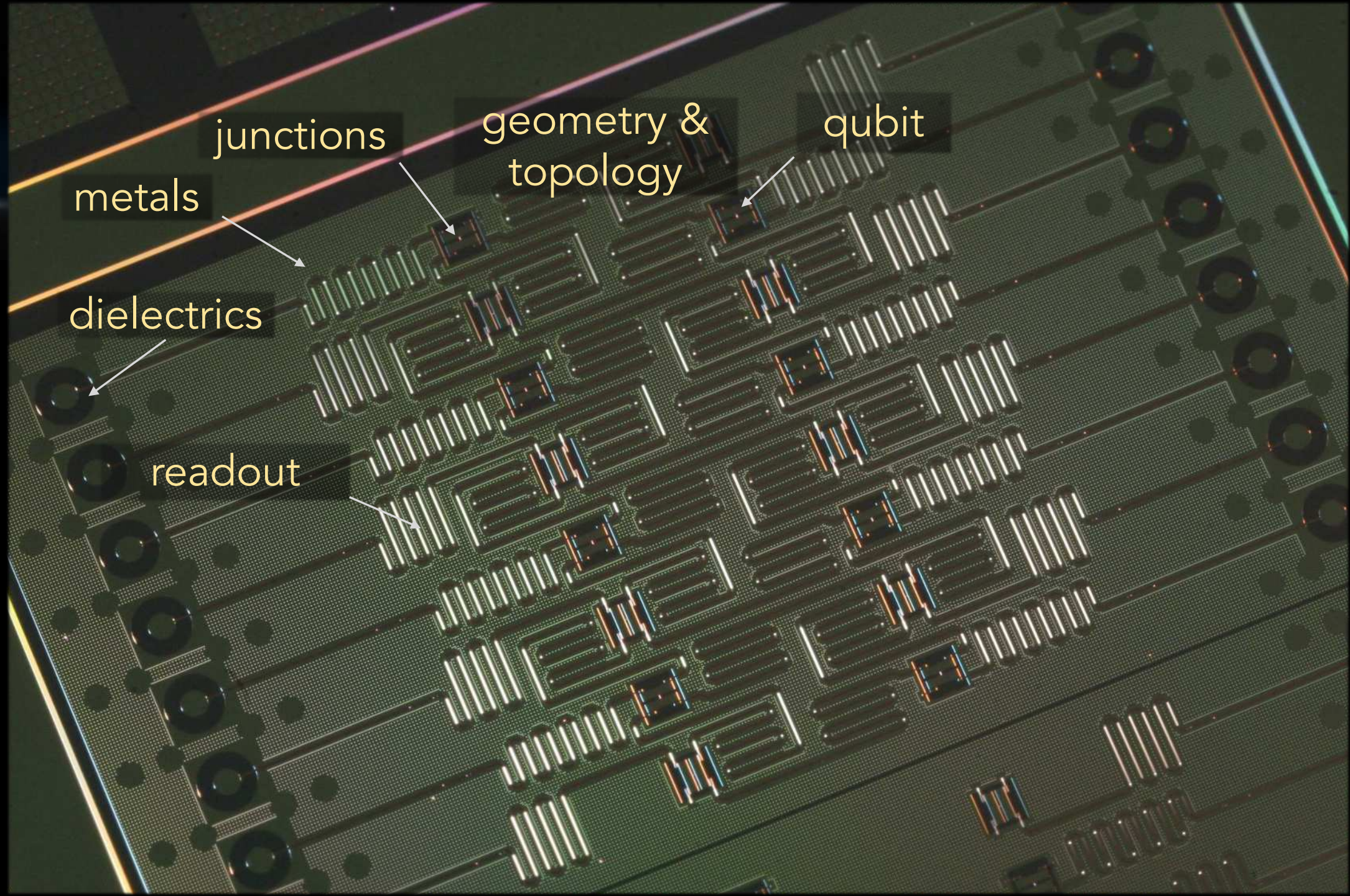
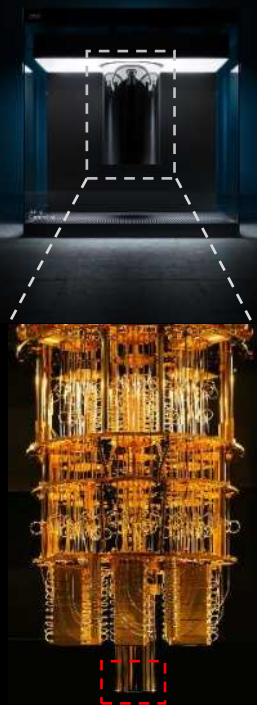


IBM

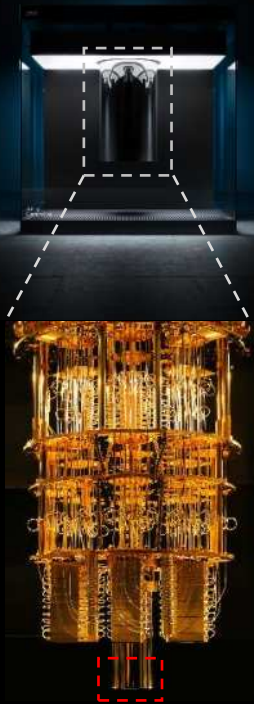
IBM Q
System One





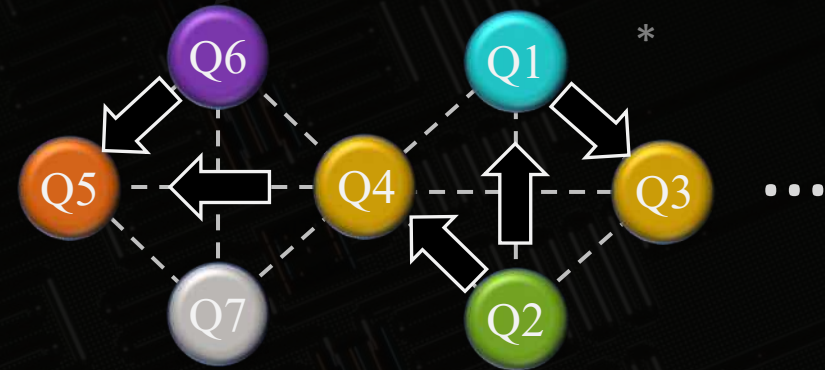


From chip to algorithm

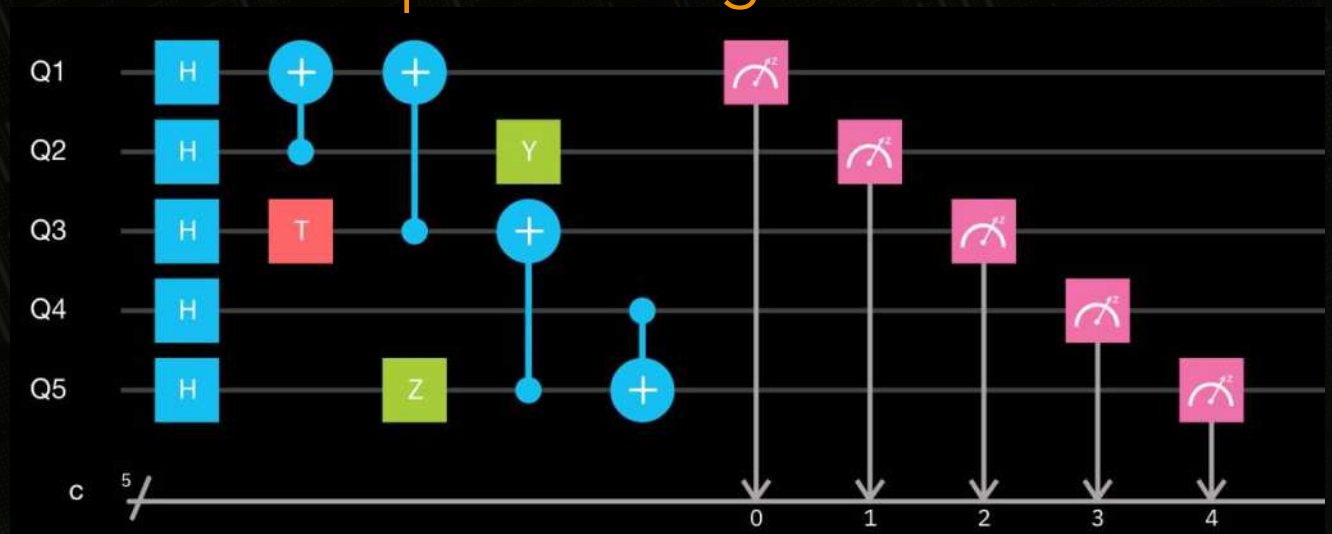


junctions
metals
dielectrics
readout

geometry & topology
qubit connectivity graph

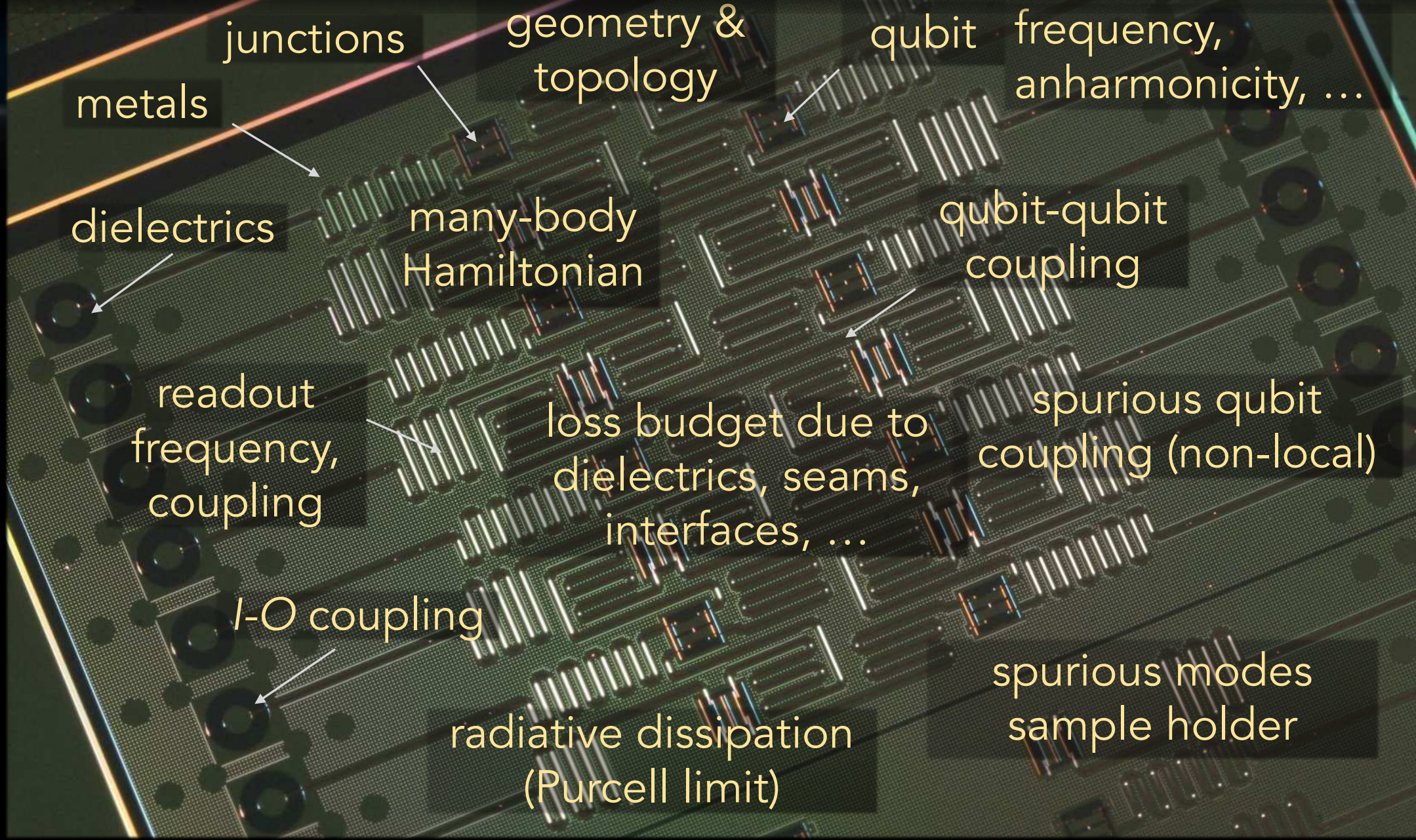
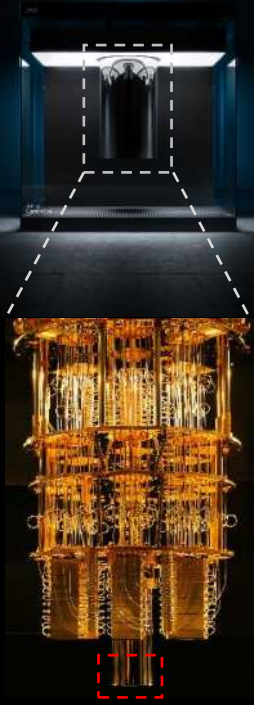


quantum algorithm⁺



Images adapted from: * Kandala *et al.*, Nature (2017) + IBM quantum experience

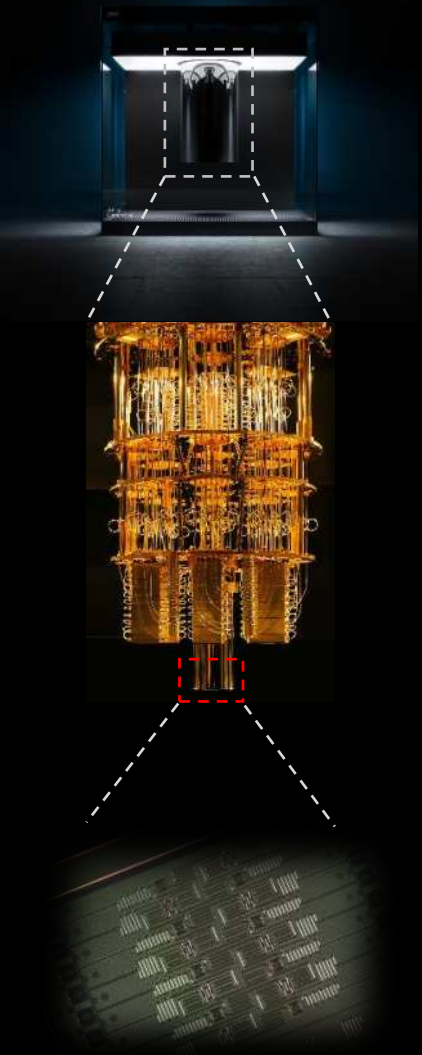
What needs to be designed?



A unified framework to handle all these questions.

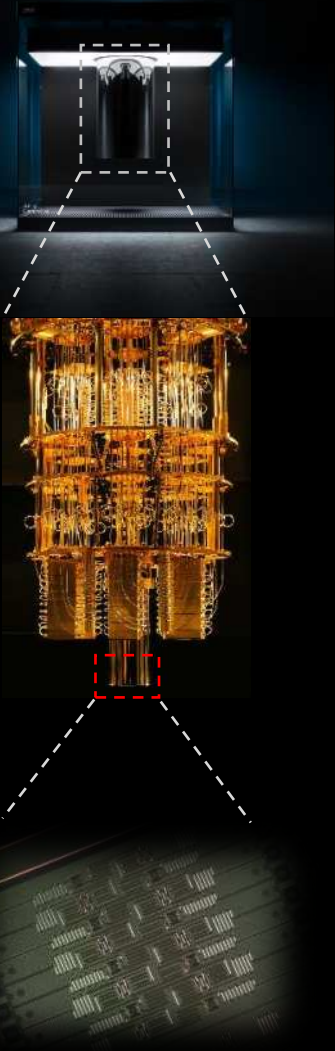
The solution reduced to:

Where is the energy?



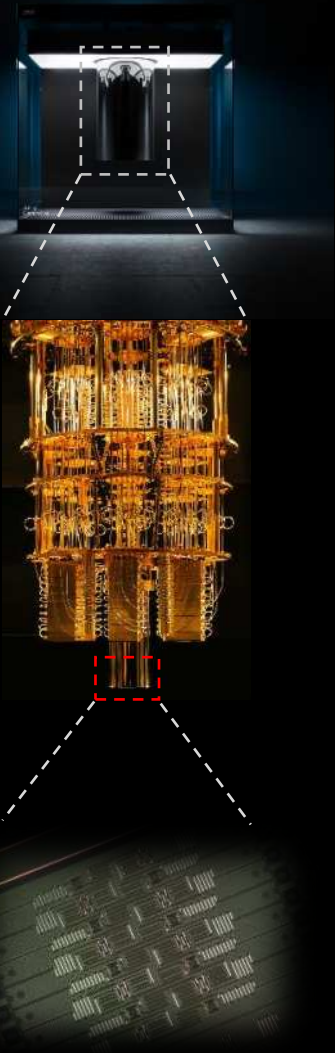
Solution reduced to:

What fraction of the energy of the mode is stored in the junction?

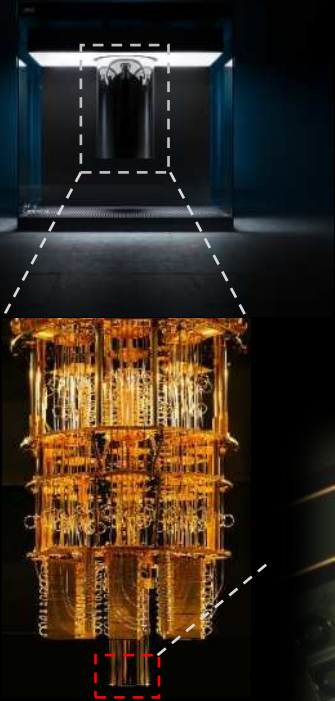


Solution reduced to:

What fraction of the energy of the mode
is stored in the dissipative element?



classical



2D

energy
participation

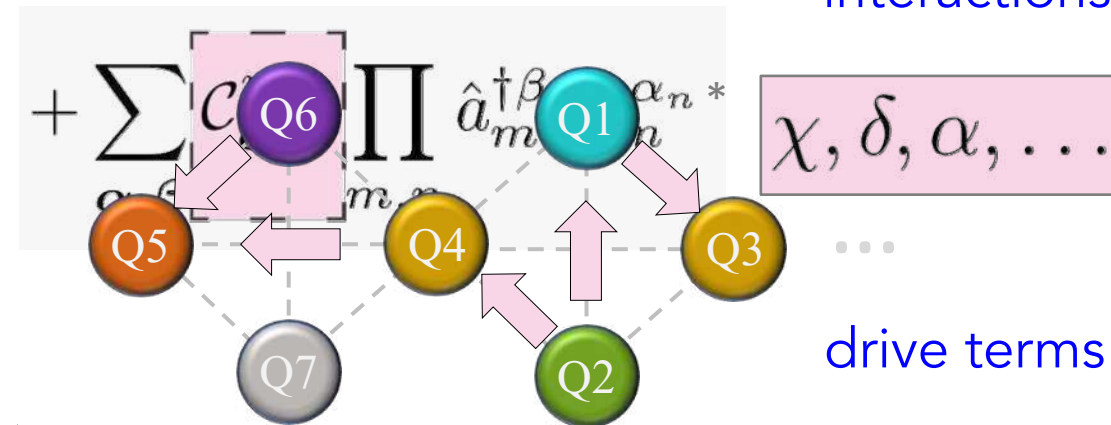
geometry, circuits, materials,
electromagnetic fields,
...

quantum

bosonic modes coupled
to environment

$$\hat{H}_{\text{full}} = \sum_m \boxed{(\omega'_m - i\kappa_m)} \hat{a}_m^\dagger \hat{a}_m$$

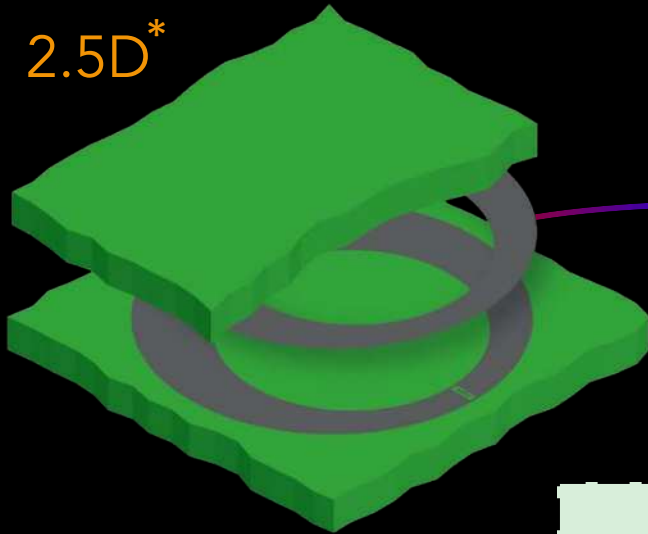
interactions



classical

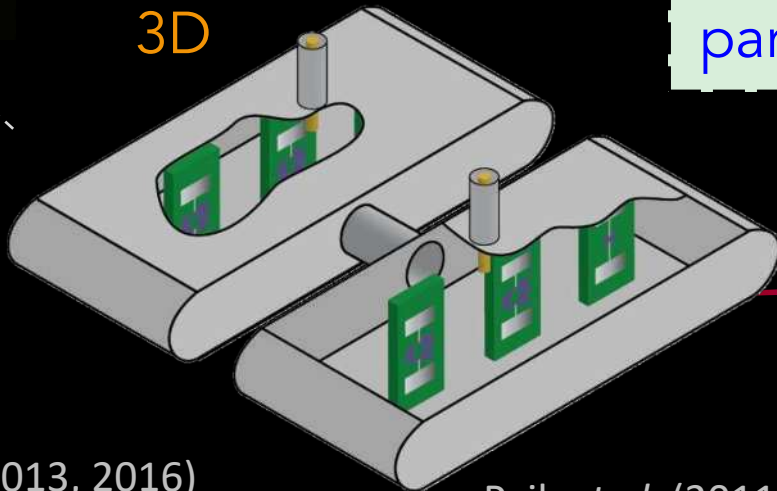
quantum

2.5D*



energy participation

3D



Paik et al. (2011), ...

$$\hat{H}_{\text{full}} = \sum_m \boxed{(\omega'_m - i\kappa_m)} \hat{a}_m^\dagger \hat{a}_m$$

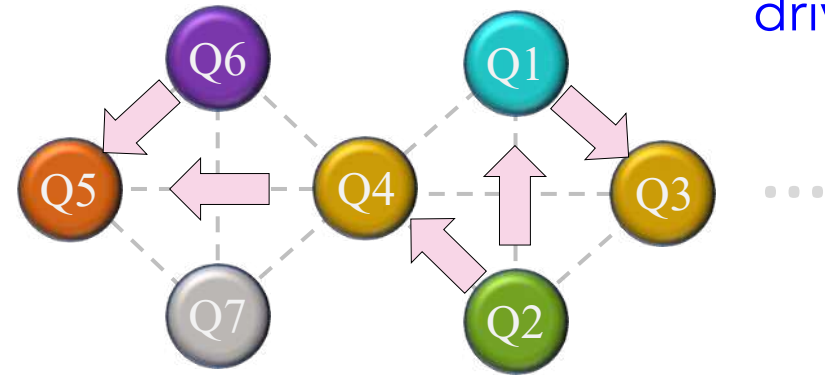
bosonic modes coupled to environment

$$+ \sum_{\alpha, \beta} \boxed{C_{\alpha, \beta}^p} \prod_{m, n} \hat{a}_m^{\dagger \beta_m} \hat{a}_n^{\alpha_n}$$

interactions

$\chi, \delta, \alpha, \dots$

drive terms



* Minev et al. (2013, 2016)
For flip chip, also see:
Li et al. (2010),
Gambetta et al. (2017), Rosenberg et al. (2017), ...

A simple example:

transmon qubit coupled to resonator

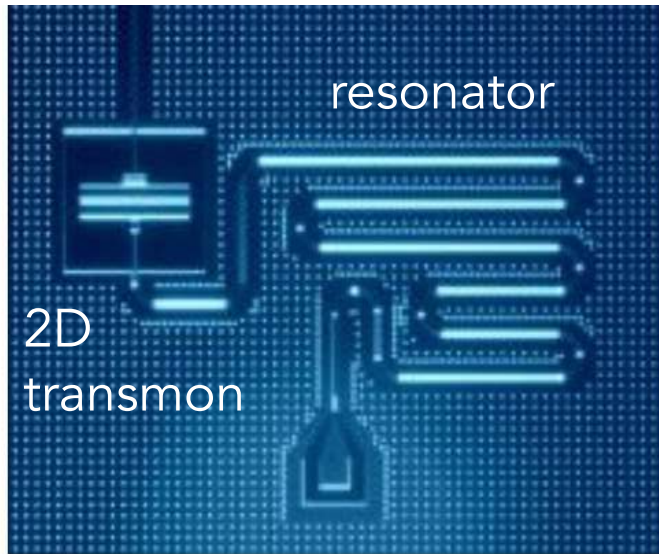
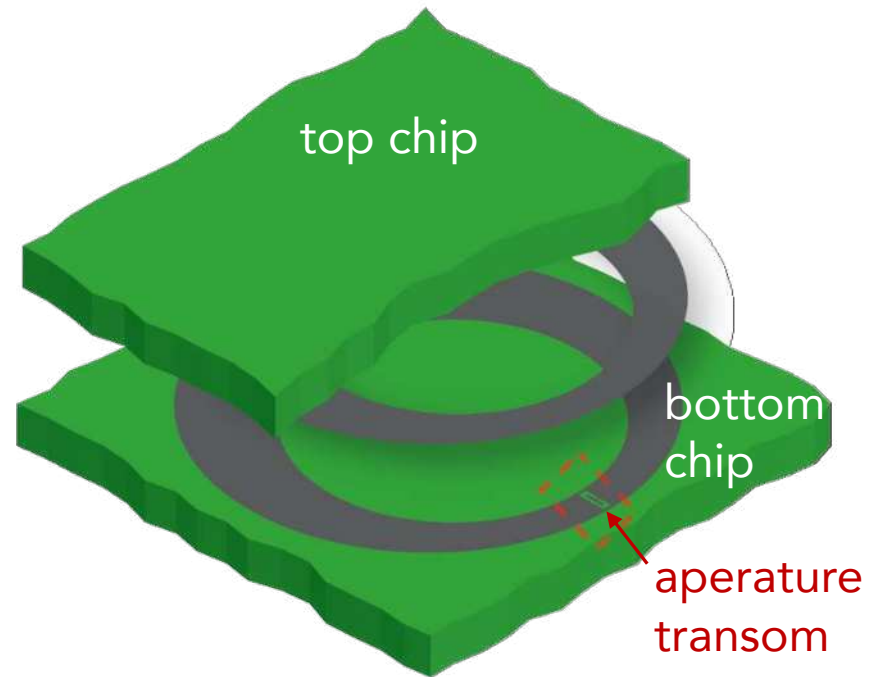
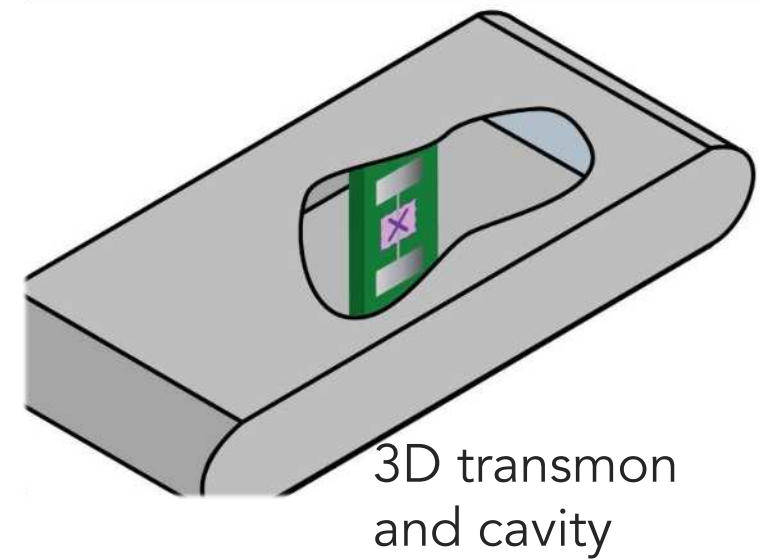


Image: Gambetta, Chow, and Steffen (2017)

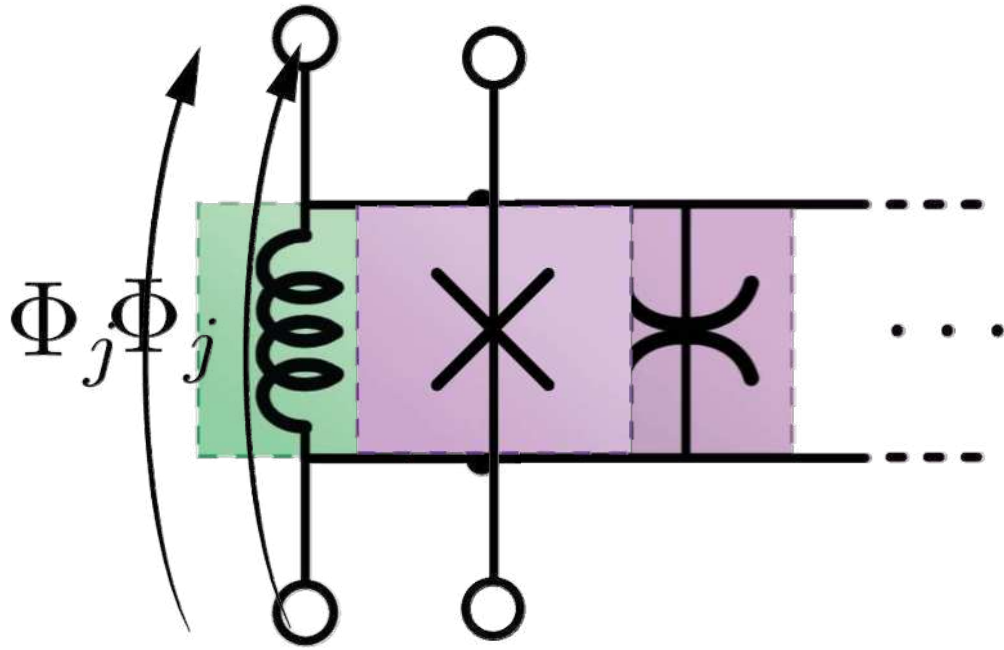


Minev *et al.* (2013, 2016)



Paik *et al.* (2011), ...

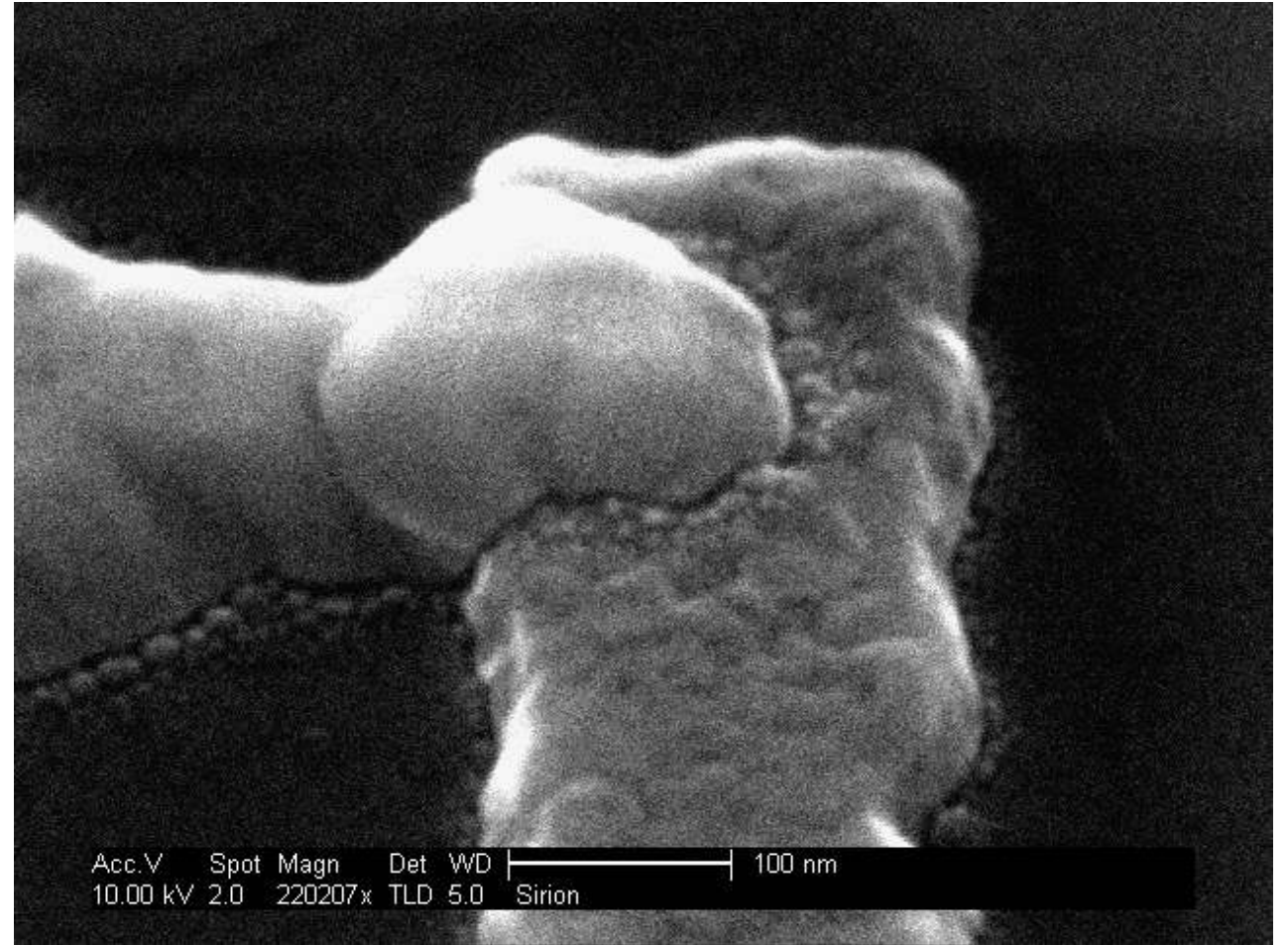
Non-linear element: Josephson tunnel junction



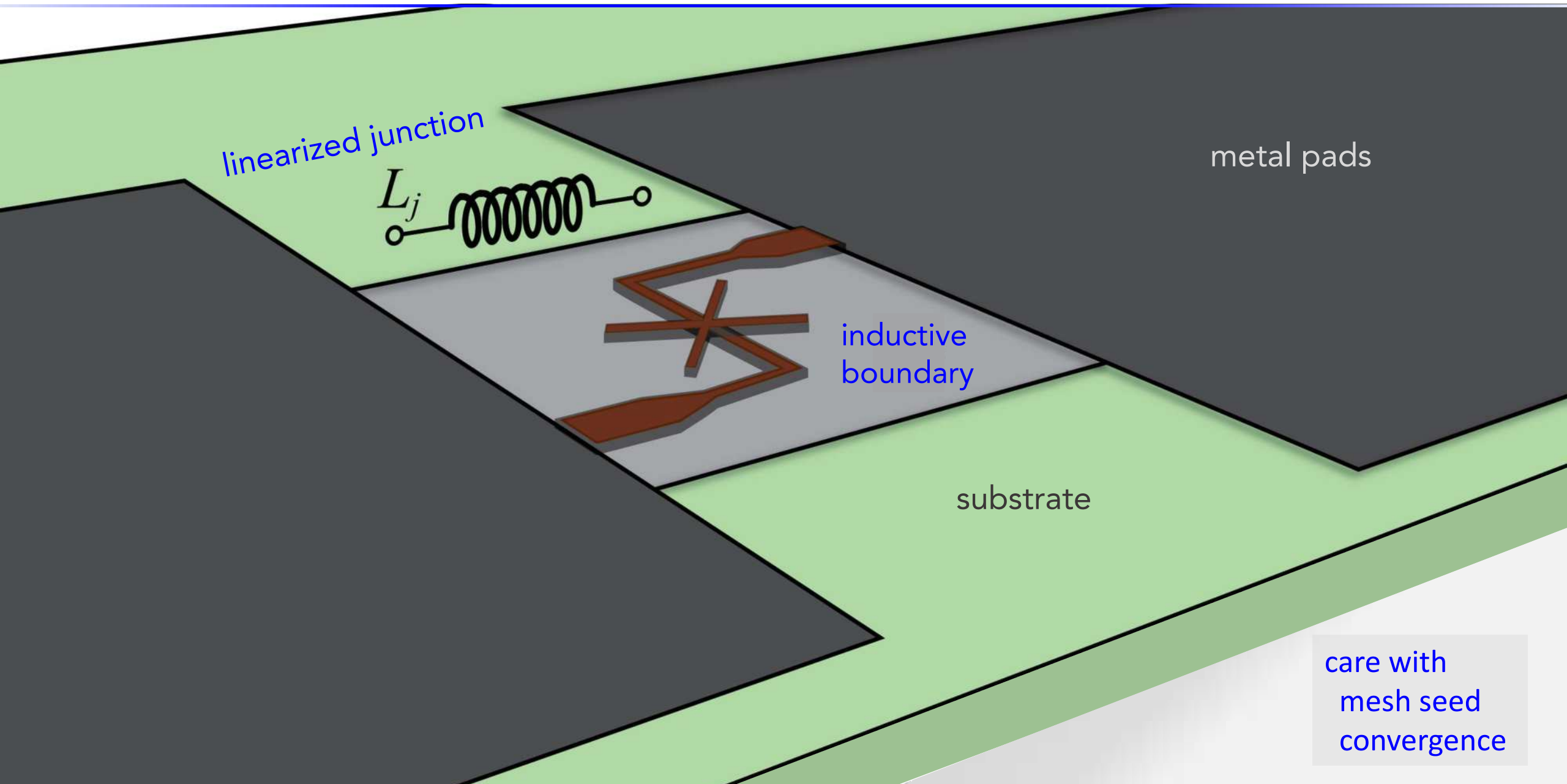
$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$\phi_0 \equiv \hbar/2e \Phi_j + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$

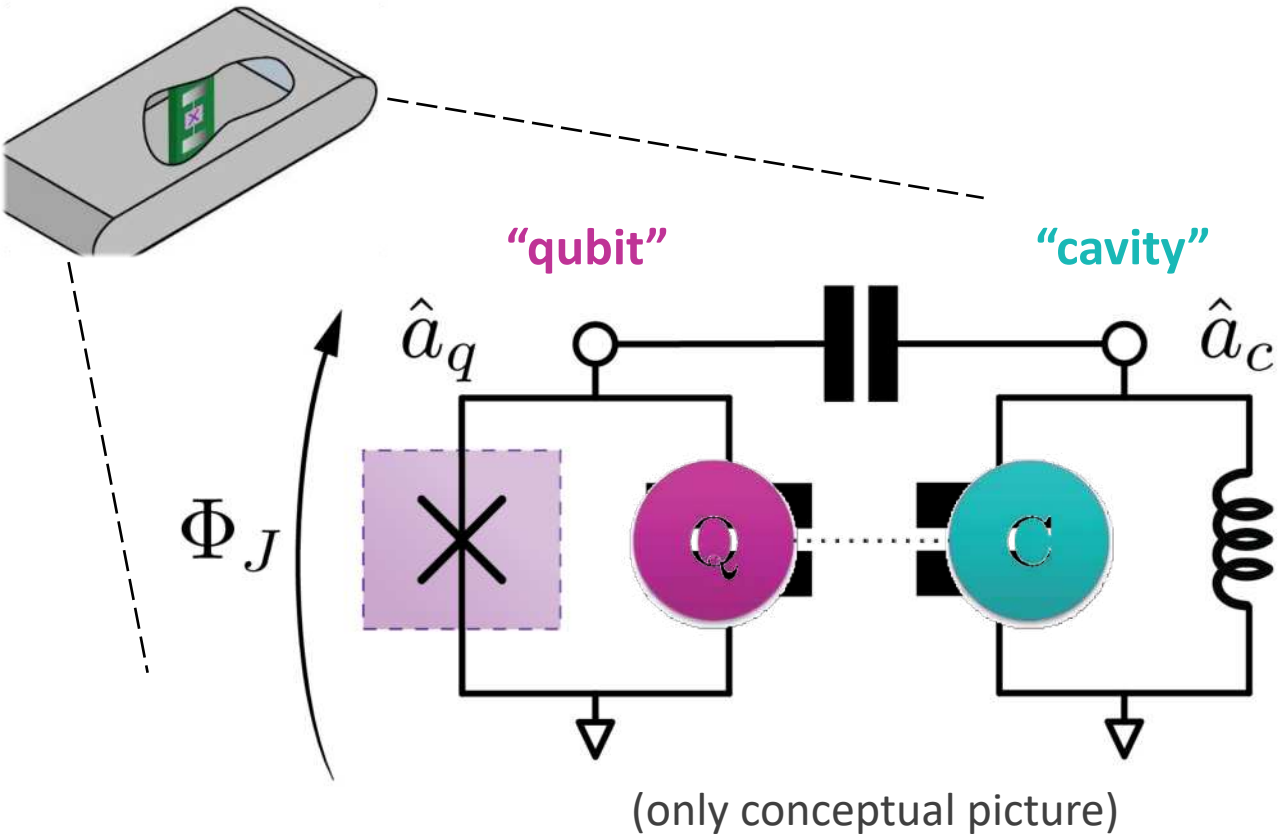


Finite-element model of linearized Junction



care with
mesh seed
convergence

Explanation of quantization



$$\hbar\omega_c \hat{a}_c^\dagger \hat{a}_c + \hbar\omega_q \hat{a}_q^\dagger \hat{a}_q$$

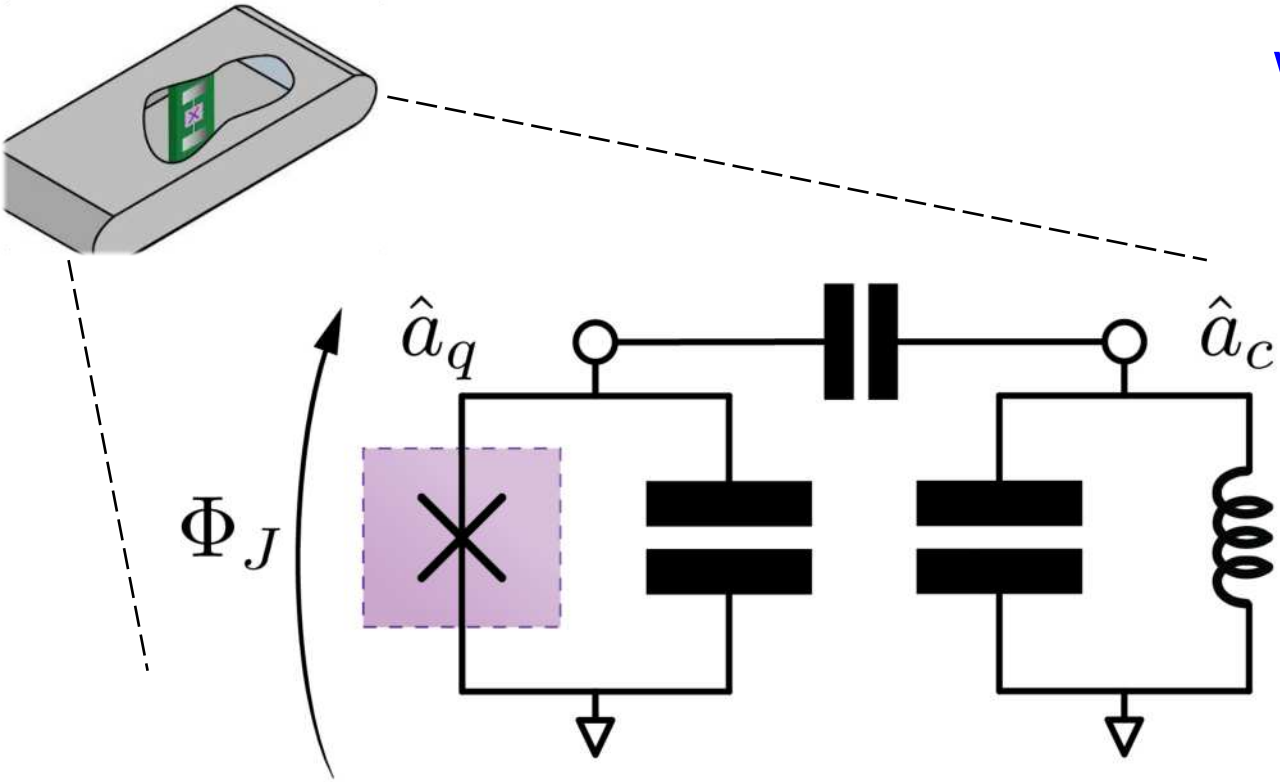
$$\hat{H}_{\text{full}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$

$$-E_J \left(\cos(\hat{\phi}_J) + \hat{\phi}_J^2/2 \right)$$

$$\hat{\phi}_J = \phi_c^{\text{ZPF}} (\hat{a}_c^\dagger + \hat{a}_c) + \phi_q^{\text{ZPF}} (\hat{a}_q^\dagger + \hat{a}_q)$$

second quantization in eigen basis of linearized circuit

Energy-participation ratio (EPR) and the quantum bridge



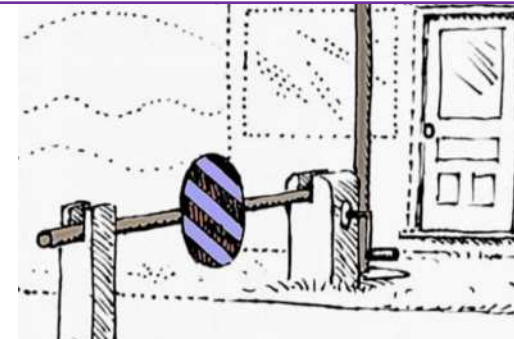
What fraction of the energy of a mode m is stored in the junction?

$$p_m = \frac{\text{Energy stored in junction}}{\text{Inductive energy stored in mode } m}$$

$$\frac{\langle n_m | : \frac{1}{2} E_J \phi_J^2 :_{\text{ZPF}} | n_m \rangle}{\langle n_m | \frac{1}{2} \hat{H}_{\text{lin}} | n_m \rangle} = \frac{E_J \phi_m^2}{2 \hbar \omega_m}$$

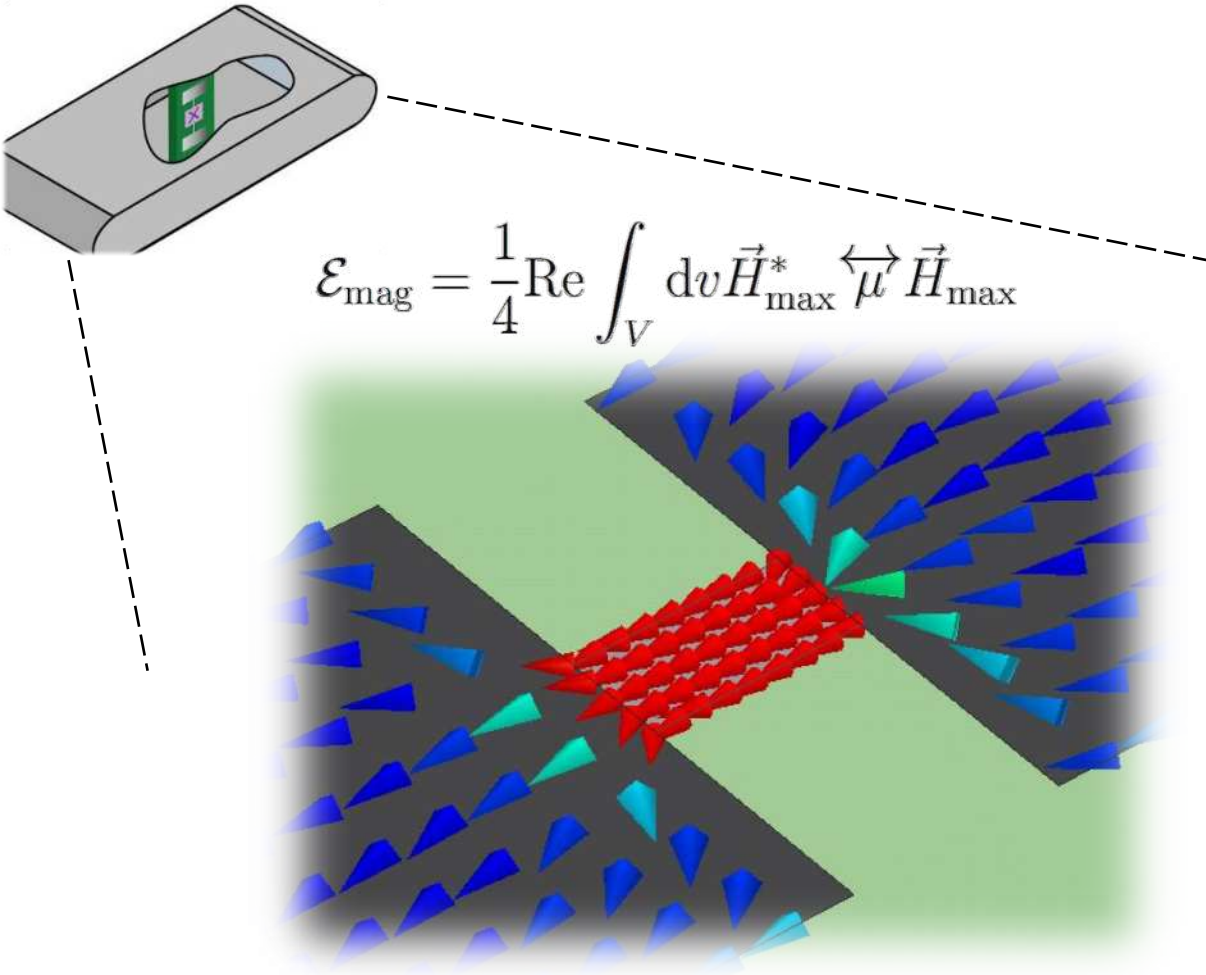


$$\hat{\phi}_J = \phi_c^{\text{ZPF}} (\hat{a}_c^\dagger + \hat{a}_c) + \phi_q^{\text{ZPF}} (\hat{a}_q^\dagger + \hat{a}_q)$$



Drawing:
Zurek, Physics Today (1991)

Calculation of the EPR from the classical solution of $\mathcal{H}_{\text{lin}}$



What fraction of the energy of a mode m is stored in the junction?

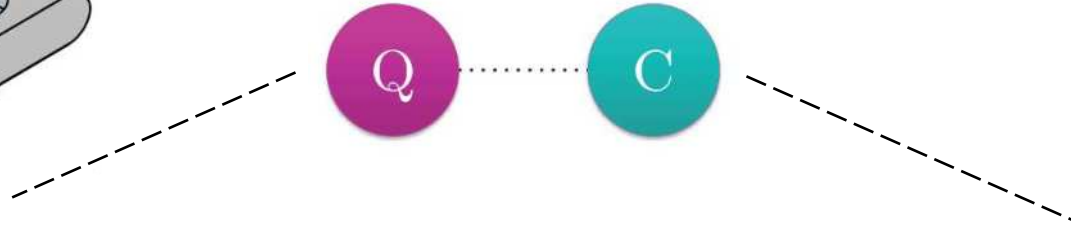
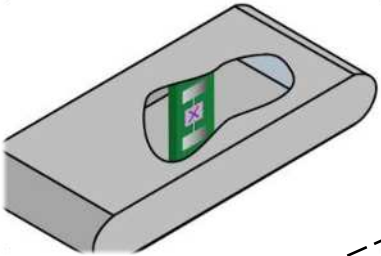
$$p_m = \frac{\text{Energy stored in junction}}{\text{Inductive energy stored in mode } m}$$
$$= \frac{\frac{1}{2} L_j I_{mj}^2}{\mathcal{E}_{\text{ind},m}}$$

Resolution of the energy

$$\mathcal{E}_{\text{mag}} + \mathcal{E}_{\text{kin}} = \mathcal{E}_{\text{ind}} = \mathcal{E}_{\text{cap}} = \mathcal{E}_{\text{elec}}$$

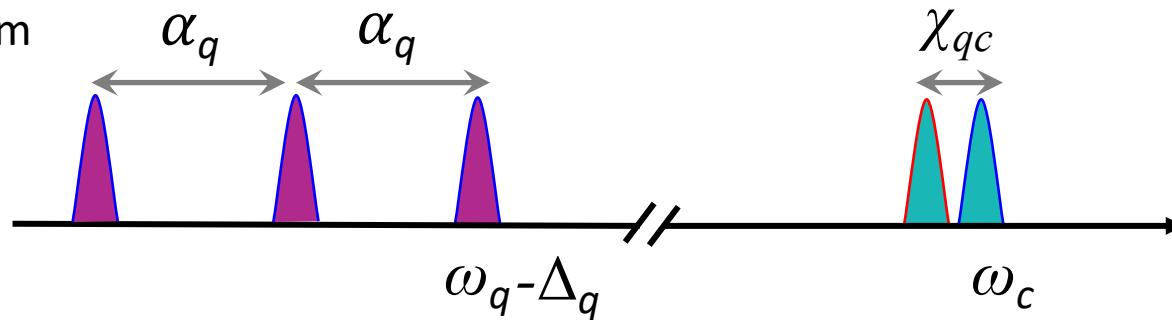
$$p_m = \frac{\mathcal{E}_{\text{elec}} - \mathcal{E}_{\text{mag}}}{\mathcal{E}_{\text{elec}}}$$

Quantum Hamiltonian from the EPR



$$\hat{H}_{\text{eff}} = (\omega_q - \Delta_q) \hat{n}_q + (\omega_q \hat{n}_q + \omega_c \hat{n}_c - \chi_{qc} \hat{n}_q \hat{n}_c - \frac{1}{2} \alpha_q \hat{n}_q (\hat{n}_q - 1) - \frac{1}{2} \alpha_c \hat{n}_c (\hat{n}_c - 1) ,$$

Transition spectrum



for simplicity, showing up to $\mathcal{O}(\phi^6)$ in RWA

Qubit/cavity anharmonicity

$$\alpha_{q/c} = p_{q/c}^2 \frac{\hbar \omega_{q/c}^2}{8 E_J}$$

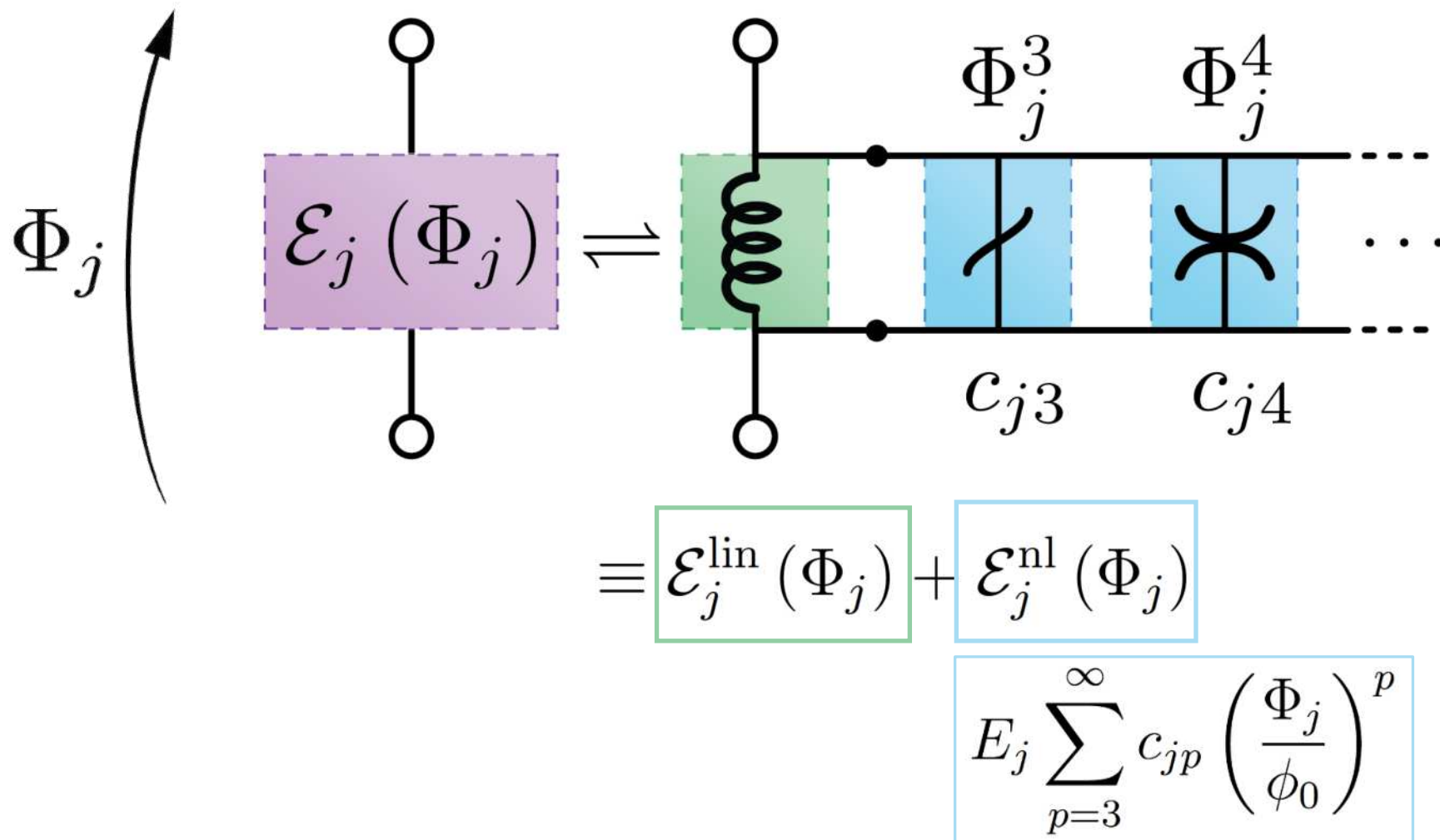
Qubit-cavity dispersive shift

$$\chi_{qc} = p_q p_c \frac{\hbar \omega_q \omega_c}{4 E_J}$$

Qubit Lamb shift

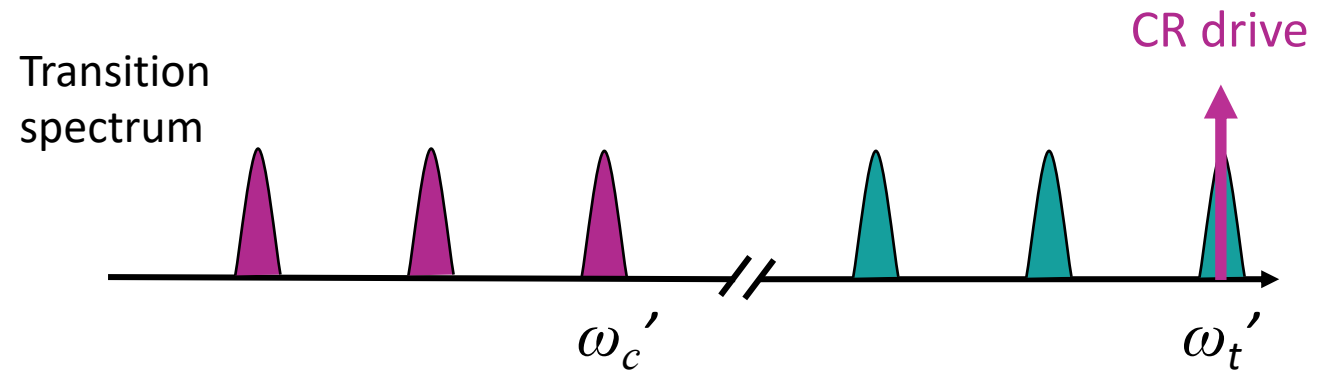
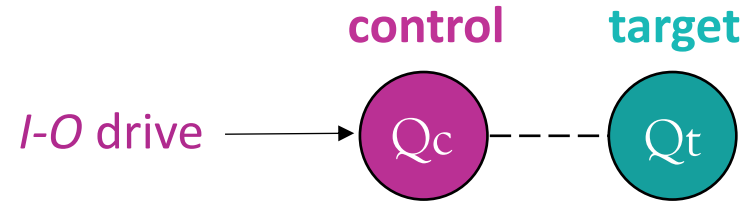
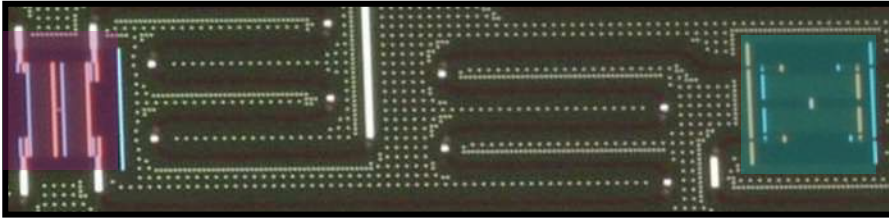
$$\Delta_q = \alpha_q - \frac{1}{2} \chi_{qc}$$

General Josephson device



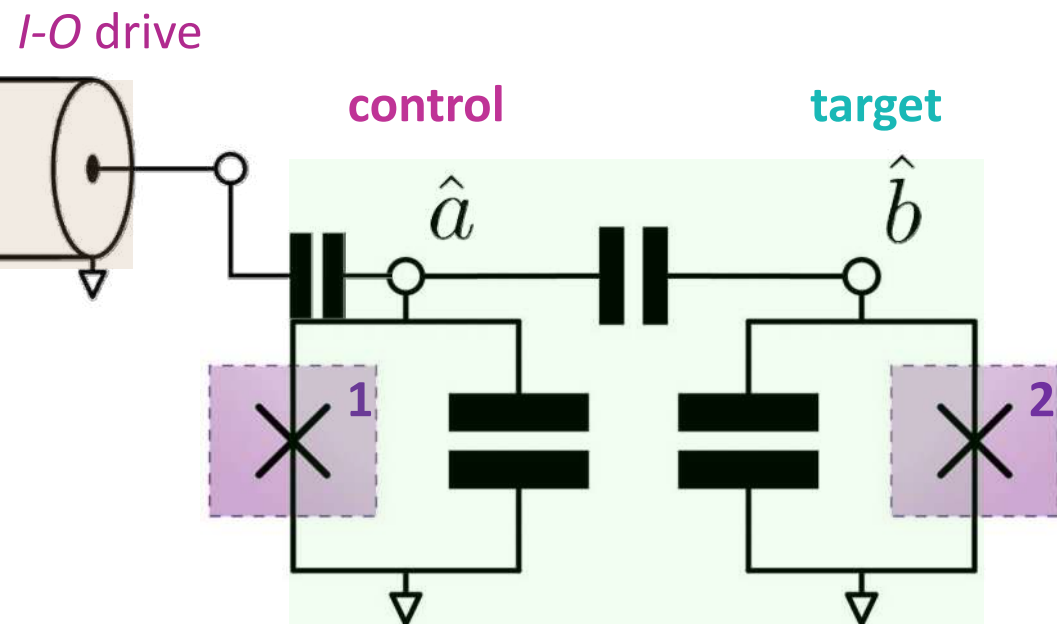
series with respect to operational equilibrium point

Cross-resonance (CR) entangling gate



dispersive regime; χ_{qc} small

Cross-resonance (CR) entangling gate



Participation matrix

$$\begin{pmatrix} p_{a1} & p_{a2} \\ p_{b1} & p_{b2} \end{pmatrix} \approx \begin{pmatrix} 0.92 & 2 \times 10^{-4} \\ 2 \times 10^{-4} & 0.92 \end{pmatrix}$$

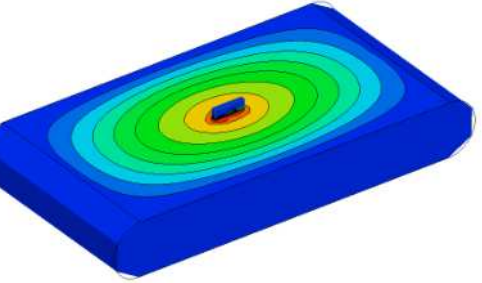
dispersive regime; χ_{qc} small

$$\hat{H} = \underbrace{\omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b}}_{\text{Eigen modes}} - \underbrace{\sum_{j=1}^2 \frac{E_j}{4!} \left(\phi_{aj} (\hat{a} + \hat{a}^\dagger) + \phi_{bj} (\hat{b} + \hat{b}^\dagger) \right)^4 + \dots}_{\text{Non-linear}} - \underbrace{i\epsilon_a(t) (\hat{a}^\dagger - \hat{a}) - i\epsilon_b(t) (\hat{b}^\dagger - \hat{b})}_{\text{Drives}}.$$

$$-\frac{E_j}{4!} \hat{H}_{\text{CR}}^3 \phi_b = \left[\frac{1}{2} \hat{a}^\dagger \hat{a} \hat{\sigma}_z \frac{1}{2} \hat{\sigma}_x \right] (\xi \hat{b} + \xi^* \hat{b}^\dagger)$$

$$\omega_{zx} \approx \sqrt{p_{b1}} \times \frac{1}{2} \text{ GHz}$$

Dissipation budget and the energy-participation ratio (EPR)



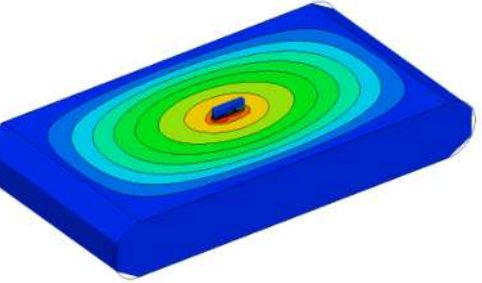
Quality of
mode m

EPR of lossy element l
in mode m

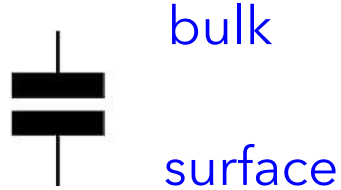
Quality of
lossy element l

$$Q_m^{-1} = \sum_l p_m^l Q_l^{-1}$$

Lossy energy-participation ratios (EPRs)



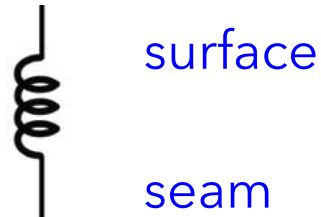
Capacitive



$$p_{\text{cap}}^l = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{1}{4} \text{Re} \int_{V_l} dv \vec{E}_{\text{max}}^* \overleftrightarrow{\epsilon} \vec{E}_{\text{max}}$$

$$p_{\text{cap,surf}}^l = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{t_l \epsilon_l}{4} \text{Re} \int_{\text{surf}_l} ds \left| \vec{E}_{\text{max}} \right|^2$$

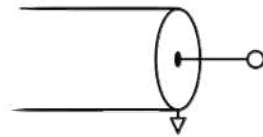
Inductive



$$p_{\text{ind,surf}}^l = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 \mu_l}{4} \text{Re} \int_{\text{surf}_l} ds \left| \vec{H}_{\text{max},\parallel} \right|^2$$

$$p_{\text{ind,seam}}^l = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 t_l \mu_l}{4} \text{Re} \int_{\text{seam}_l} dl \left| \vec{H}_{\text{max},\perp} \right|^2$$

Radiative



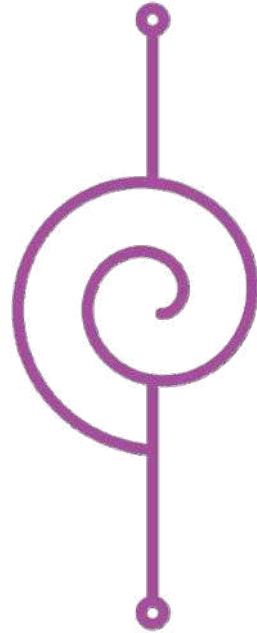
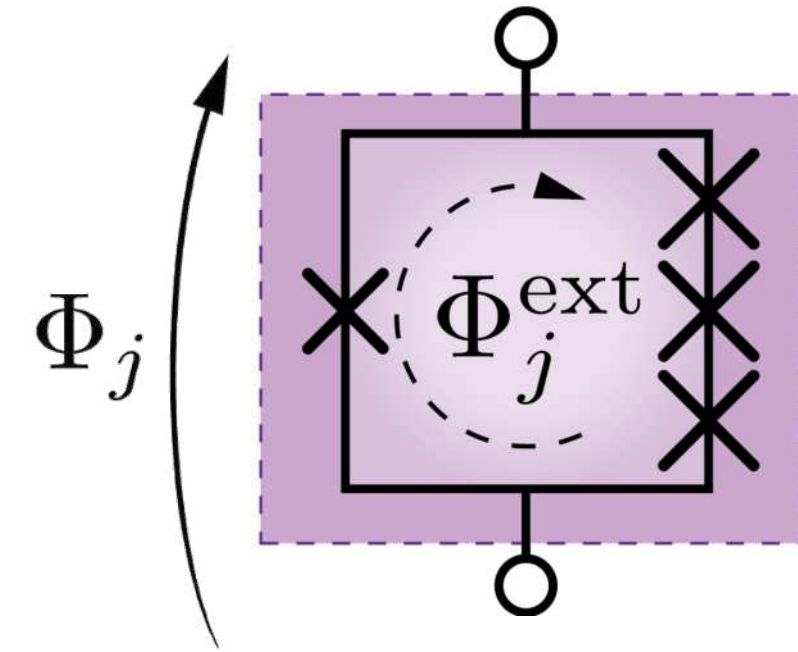
$$Q_{mp} = \frac{\omega_m \mathcal{E}_m(0)}{\frac{1}{2} R I_{mp}^2}$$

Pozar, Microwave engineering (1990); Gao, Thesis (2008), Zmuidzinas, ARCMP (2012); ...

Geerlings, Thesis (2013); Minev *et al.* (2013); Brecht *et al.*, APL (2015); Wang *et al.*, APL (2015); Dial SST *et al.* (2016); ...

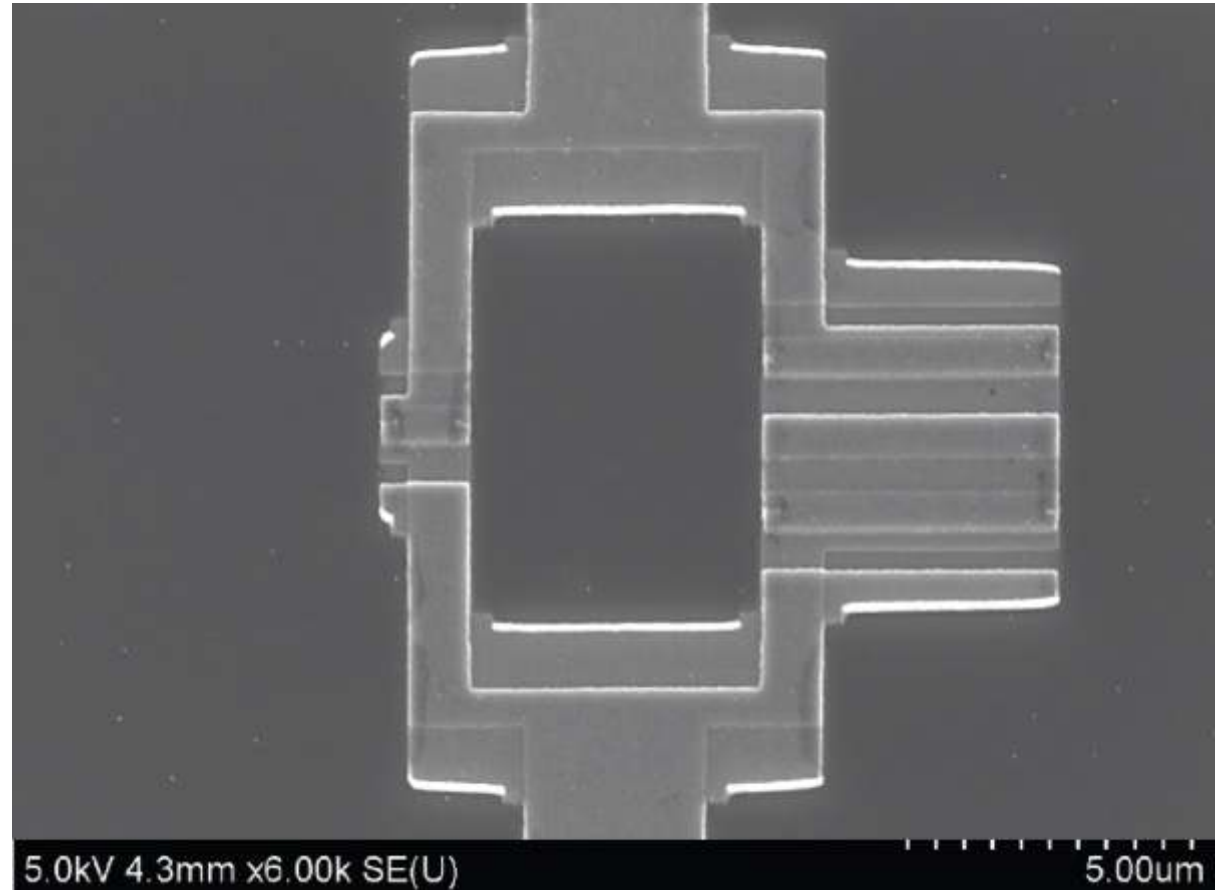
General circuits and non-linear devices

Composite Josephson device: example



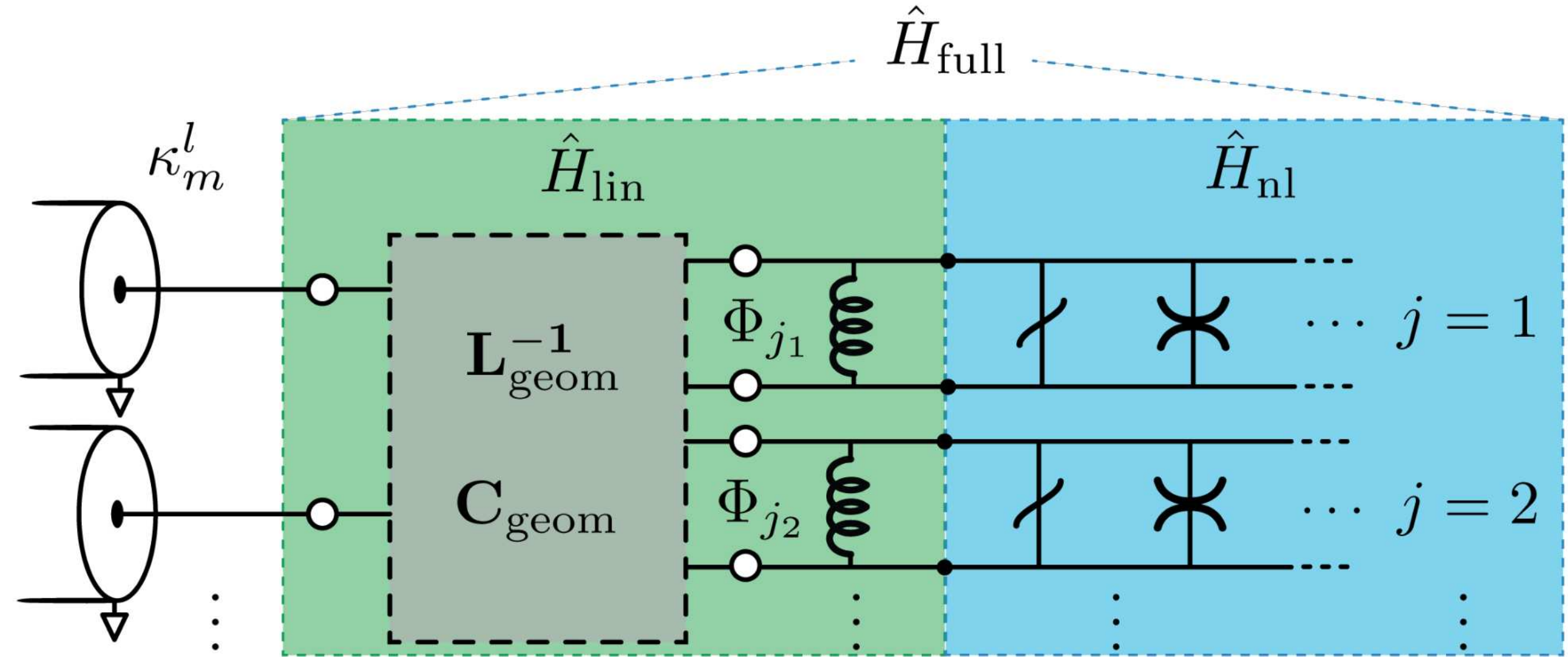
Energy function

$$\mathcal{E}_j (\Phi_j; \Phi_j^{\text{ext}})$$



SEM image: Frattini & Sivak; see APL (2017)

Decomposition of the general circuit



Non-linear interaction with EPR

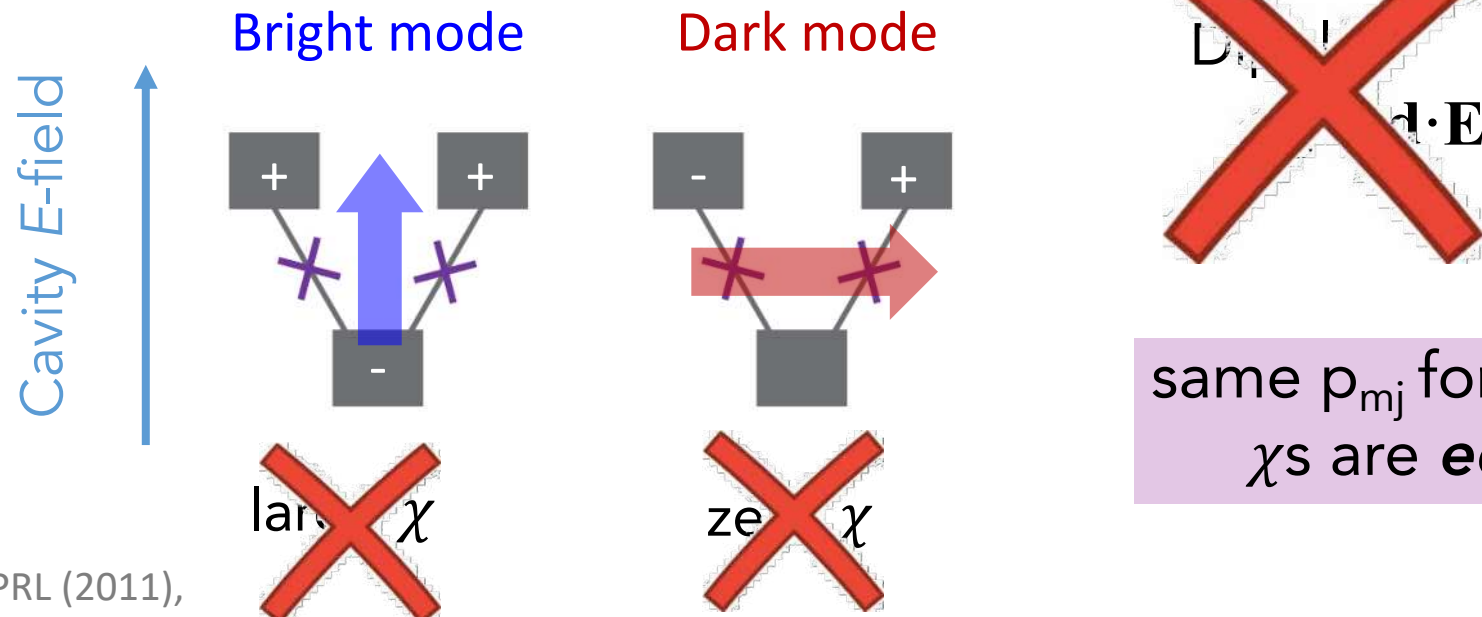
Non-linear interactions in terms of energy overlap

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar \omega_m \omega_n}{4E_j} p_{mj} p_{nj}$$

Energy participation overlap

(only variable subject to significant variation)

Intuition with dipole and EPR



Physical insights with EPR

Non-linear interactions in terms of energy overlap

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar \omega_m \omega_n}{4E_j} p_{mj} p_{nj}$$

Energy participation overlap
(only variable subject to significant variation)

Fundamental constraints

What circuits/parameters can and cannot be realized?

$$\sum_{m=1}^M p_{mj} = 1 \quad 0 \leq \sum_{j=1}^J p_{mj} \leq 1 \quad \sum_{m=1}^M s_{mj} s_{mj'} \sqrt{p_{mj} p_{mj'}} = 0$$

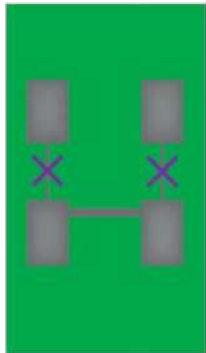
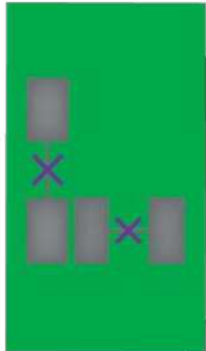
Arbitrary strength non-linear interaction can be calculated

Experimental results

2Q, 1C theory vs. experiment

Legend

measured
predicted
error (%)

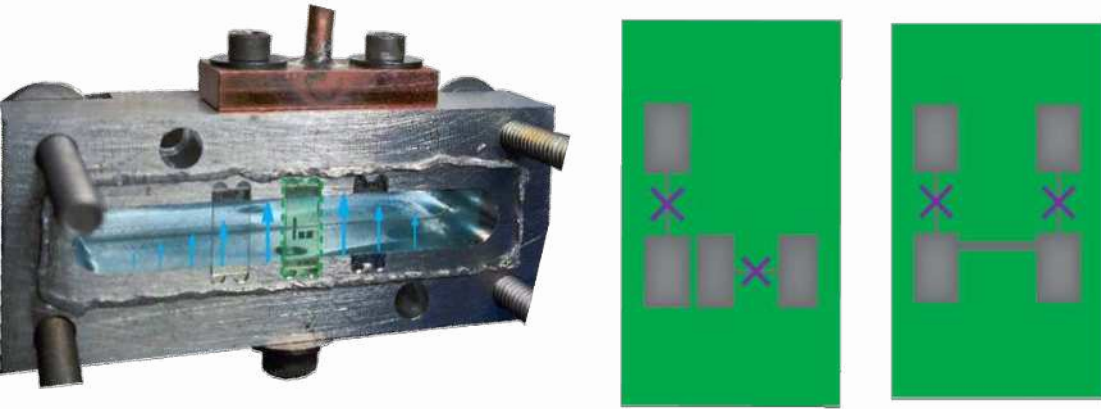


Device	Frequency (MHz)			Anharmonicity (MHz)		Cross-Kerr (MHz)			<i>I-O</i> coupling
	$\omega_D/2\pi$	$\omega_B/2\pi$	$\omega_C/2\pi$	$\alpha_D/2\pi$	$\alpha_B/2\pi$	$\chi_{DB}/2\pi$	$\chi_{BC}/2\pi$	$\chi_{DC}/2\pi$	Q_C
R9C1	4951	5664	9158	138	170	92.	4.7	0.4	5.20×10^3
	4866	5691	9154	150	185	99.	4.2	0.55	7.40×10^3
	-1.7%	0.5%	-0.04%	8%	8%	7%	-12%	27%	29%
R2C1	4823	5567	8947	150	192	64.5	4.8	0.3	4.97×10^3
	4770	5640	8950	161	211	67.7	5.88	0.46	5.44×10^3
	-1.1%	1.3%	0.03%	6.8%	9%	4.7%	18%	35%	9 %
R7C1	4726	5475	8999	156	189	67.	4.8	0.34	2.68×10^3
	4770	5640	8950	161	211	67.7	5.88	0.46	3.07×10^3
	0.9%	2.9%	-0.55%	3.1%	10%	1%	18%	26%	13%
R3C2	4845	5620	8979	152	195	61	5.1	0.3	2.11×10^3
	4770	5640	8950	161	211	67.7	5.88	0.46	1.78×10^3
	-1.5%	0.4%	-0.3%	5.6%	7.6%	9.9%	13%	35%	-19%
R3C1	4688	5300	9003	148	174	85	5.	0.33	2.43×10^3
	4745	5265	8922	159	198	73	5.1	0.37	5.65×10^3
	1.2%	-0.7%	-0.9%	6.9%	12.1%	-16%	2%	9%	57%
DT3	6160	7110	9170	130	150	278	3.	2.5	9.17×10^3
	6100	7141	9155	140	177	312	3.9	3.1	7.33×10^3
	-1.0%	0.4%	-0.15%	7%	15%	11%	23%	19%	-25%

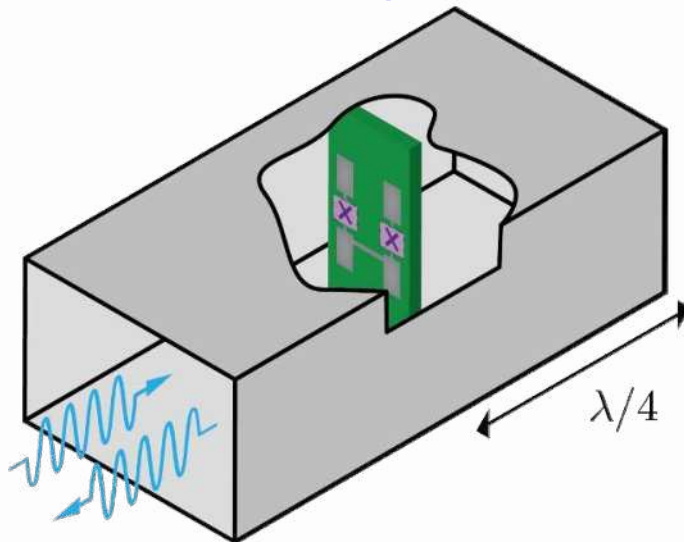
Aside from coarse adjustment on L_J , no adjustable parameters

Measured architectures and devices

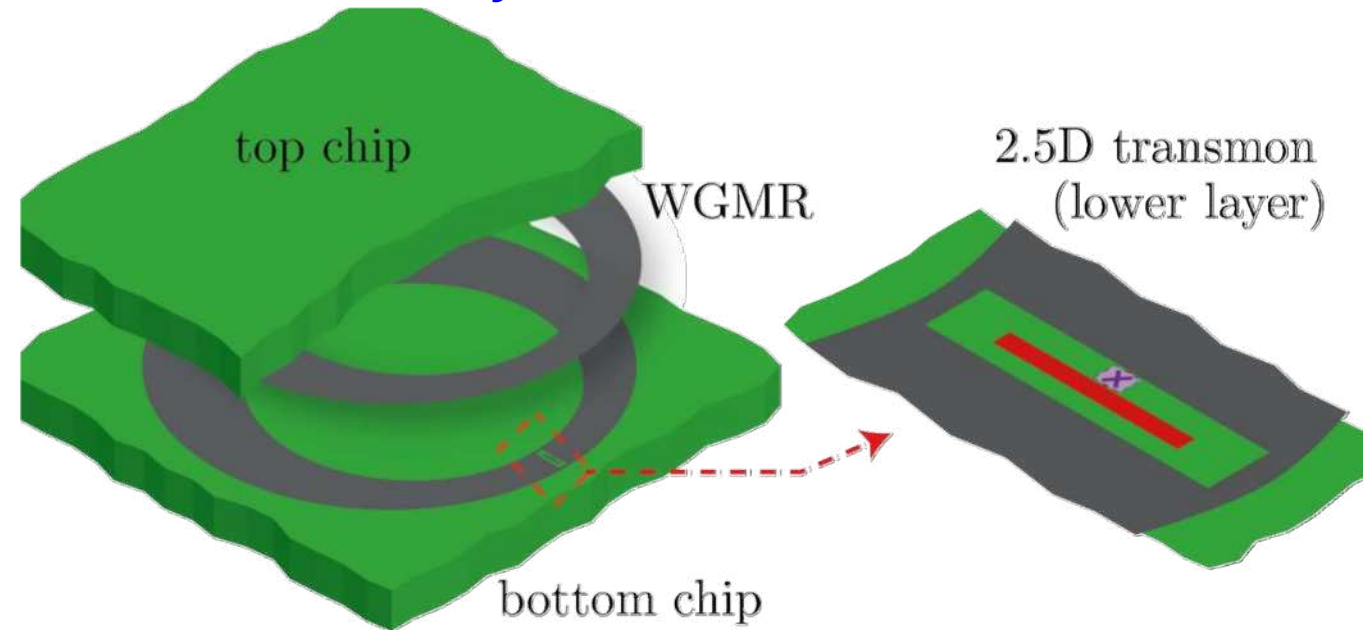
Three-dimensional (3D) *



Waveguide



Multilayer (2.5D) cQED



Minev *et al.*, APL (2013)

Minev *et al.*, WO/2016/138395 (2015)

Minev *et al.*, Phys. Rev. App. (2016)

* Minev *et al.*, arXiv:1803.00545 (2018); Related: Gambetta *et al.*, PRL (2011), Srinivasan *et al.*, PRL (2011), Dumur *et al.*, PRB (2015), Zhang *et al.*, Nature JQI (2017) ...

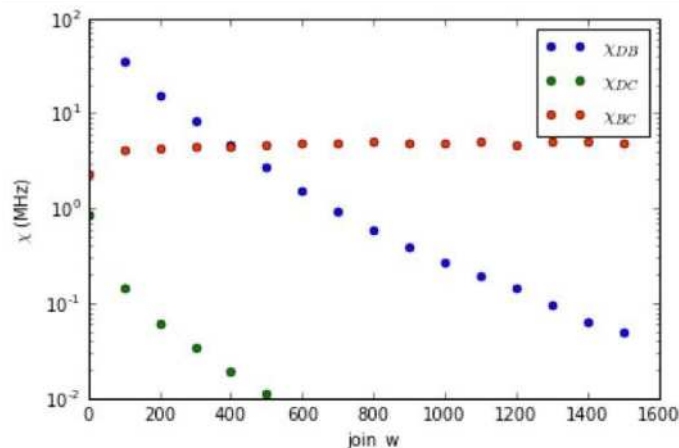
How to use the EPR approach

- Fully automated
 - control of HFSS
 - calculations
- Full numerical treatment
- Convergence checks

See my open source
project on github



Used at Yale, ENS Paris, IBM,
Berkeley, Saclay, and others



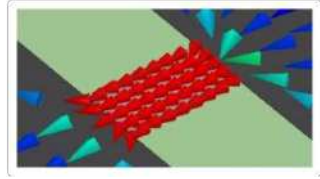
```
1 from pyEPR import *
2
3 # 1. Project and design. Open link to HFSS controls.
4 project_info = Project_Info('c:/sims',
5                             project_name = 'two_qubit_one_cavity',
6                             design_name = 'Alice_Bob'
7                             )
8
9 # 2a. Junctions. Specify junctions in HFSS model
10 project_info.junctions['jAlice'] = {'Lj_variable': 'LJAlice', 'rect': 'qubitA'}
11 project_info.junctions['jBob'] = {'Lj_variable': 'LJBob', 'rect': 'qubitB'}
12
13 # 2b. Dissipative elements.
14 project_info.dissipative.dielectrics_bulk = ['si_substrate']
15 project_info.dissipative.dielectric_surfaces = ['interface']
16
17 # 3. Run analysis
18 epr_hfss = pyEPR_HFSS(project_info)
19 epr_hfss.do_EPR_analysis()
20
21 # 4. Hamiltonian analysis
22 epr = pyEPR_Analysis(epr_hfss.data_filename)
23 epr.analyze_all_variations(cos_trunc = 8, fock_trunc = 7)
24 epr.plot_Hresults()
```

Conclusions / future directions

Simple and unified dissipation and Hamiltonian analysis

single-simulation efficient

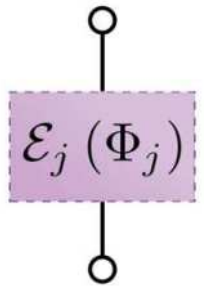
obviate driven simulations and notion of $Z_{ij}(\omega)$



General

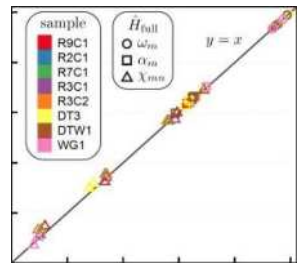
handle arbitrary architectures and non-linear devices

e.g., nanowires, ...



Accuracy

experimental ten to percent level over 5 orders in $\hat{H}_{\text{full}}$



Suited for automated, robust analysis of large systems

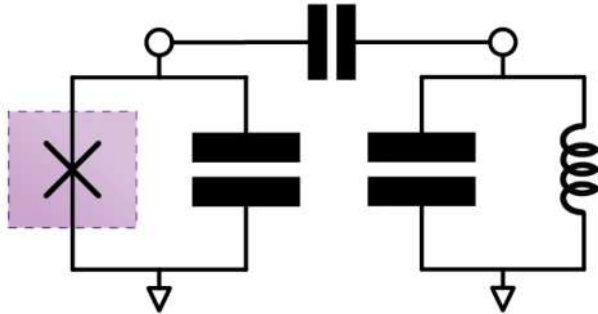
fully automated and tested

arXiv:very-soon; Minev, Ph.D. Thesis, Ch. 4 (2018)

Application to J and CR gate, ...

Where did g and $J(\text{ay})$ go?

Full

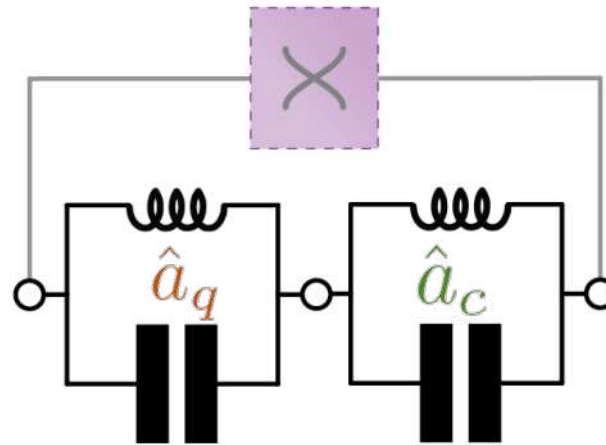
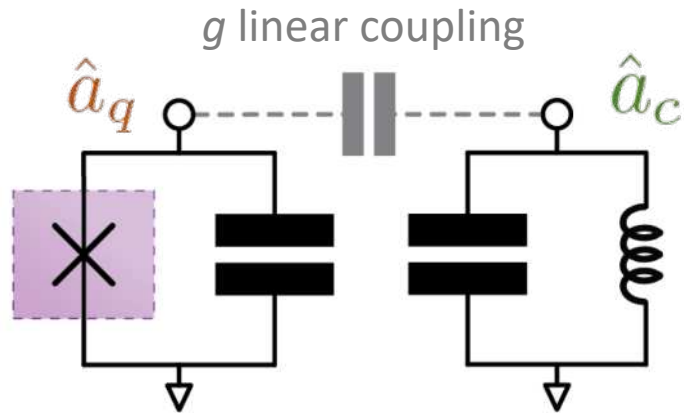


Classical simulation contains g coupling and diagonalizes it away; i.e., the linearized modes returned by the simulation are the linear-hybridized (g -dressed) modes.

Essentially work in a more efficient basis for second quantization; i.e., what basis are the ladder operators $\hat{a}_q$, $\hat{a}_c$ in?

g basis

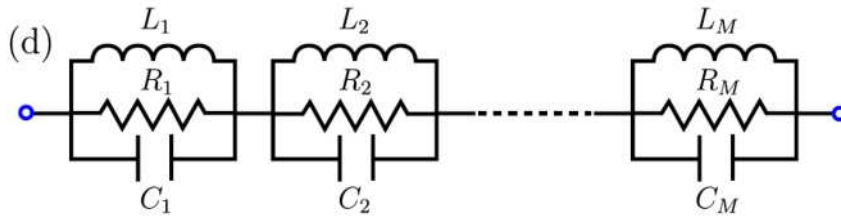
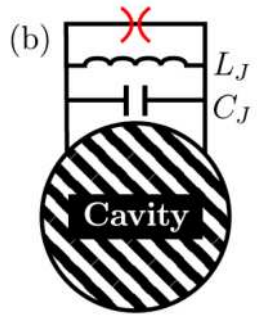
eigen basis



purely non-linear and
quantum coupling

Relation to impedance

Single junction case



$$\phi_m^{\text{ZPF}} = \sqrt{\frac{\hbar Z_a^{\text{eff}}}{2}} = \sqrt{\frac{\hbar}{\text{Im}\omega_a Y'(\omega_a)}}$$

$$L_m = p_m L_J / 2$$

$$C_m^{-1} = \omega_m^2 L_m$$

Multi-junction case

$$\hat{\phi}_l^{(k)} = \sum_{p=1}^M \frac{Z_{lk}(\omega_p)}{Z_{kk}(\omega_p)} \sqrt{\frac{\hbar}{2}} Z_{kp}^{\text{eff}} (a_p + a_p^\dagger),$$

where $Z_{kp}^{\text{eff}} = 2 / [\omega_p \text{Im} Y_k'(\omega_p)]$

Arbitrary non-linear terms

$$\hat{H}_{\text{full}} = \sum_m [(\omega'_m - i\kappa_m)] \hat{a}_m^\dagger \hat{a}_m \\ + \sum_{\alpha, \beta} [C_{\alpha, \beta}^p] \prod_{m, n} \hat{a}_m^{\dagger \beta_m} \hat{a}_n^{\alpha_n}$$

$$\frac{p!}{\alpha! \beta! k! 2^k} \sum_{j=1}^J E_j C_{jp} \phi_{mj}^{\alpha} \phi_{mj}^{\beta} \phi_{j, \text{tot}}^{2k}$$

where $\alpha \equiv (\alpha_1, \dots, \alpha_M)$ and $\beta \equiv (\beta_1, \dots, \beta_M)$

$$k \equiv \frac{1}{2} (p - (|\alpha| + |\beta|))$$

EPR matrix

Definition

$$\mathbf{P} \equiv \begin{pmatrix} p_{11} & \cdots & p_{1J} \\ \vdots & \ddots & \vdots \\ p_{M1} & \cdots & p_{MJ} \end{pmatrix},$$

Non-linear interaction amplitude matrices

$$\begin{aligned} \mathbf{X} &= \frac{\hbar}{4} \mathbf{\Omega P E_J^{-1} P^T \Omega} && \text{Kerr matrix,} \\ \boldsymbol{\alpha} &= \frac{1}{2} \text{diag}(\mathbf{X}) && \text{Anharmonicity,} \\ \boldsymbol{\Delta} &= \frac{1}{2} \mathbf{X} \mathbf{1}_M && \text{Lamb shift,} \end{aligned}$$

where

$$\mathbf{E}_j^{-1} \equiv \begin{pmatrix} E_1^{-1} & & \\ & \ddots & \\ & & E_J^{-1} \end{pmatrix} \quad \mathbf{\Omega} = \begin{pmatrix} \omega_1 & & \\ & \ddots & \\ & & \omega_M \end{pmatrix}$$

Generalization to arbitrary order

$$\chi_p \equiv \begin{pmatrix} \chi_{11}^{(p)} & \cdots & \chi_{1M}^{(p)} \\ \vdots & \ddots & \vdots \\ \chi_{M1}^{(p)} & \cdots & \chi_{MM}^{(p)} \end{pmatrix}$$

$$\boxed{\chi_p = \hbar c_p (\mathbf{\Omega P}) \mathbf{E}_j^{-1} \phi_{\sigma, tot}^{p-4} (\mathbf{\Omega P})^T,}$$

$$\boldsymbol{\chi} \equiv \sum_{p=4,6,\dots} \chi_p \quad \boldsymbol{\alpha}_p = \frac{1}{2} \text{diag} \chi_p \quad \boldsymbol{\Delta}_p = c_p (\mathbf{\Omega P}) \phi_{\sigma, tot}^{p-2} \mathbf{1}_M$$

Field calculations

Hamiltonian

$$p_m = \frac{\mathcal{E}_{\text{elec}} - \mathcal{E}_{\text{mag}}}{\mathcal{E}_{\text{elec}}}.$$

energies

$$\mathcal{E}_{\text{elec}} = \frac{1}{4} \text{Re} \int_V dv \vec{E}_{\text{max}}^* \overleftrightarrow{\epsilon} \vec{E}_{\text{max}},$$

$$\mathcal{E}_{\text{mag}} = \frac{1}{4} \text{Re} \int_V dv \vec{H}_{\text{max}}^* \overleftrightarrow{\mu} \vec{H}_{\text{max}},$$

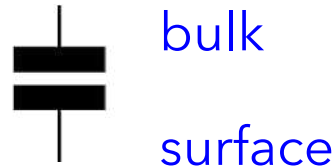
where

$$\vec{E}(x, y, z, t) = \text{Re} \vec{E}_{\text{max}}(x, y, z) e^{i\omega_m t}$$

Generalized to multi-junction

Dissipation budget

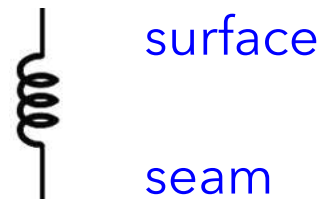
Capacitive



$$p_{\text{cap}}^l = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{1}{4} \text{Re} \int_{V_l} dv \vec{E}_{\text{max}}^* \overleftrightarrow{\epsilon} \vec{E}_{\text{max}}$$

$$p_{\text{cap,surf}}^l = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{t_l \epsilon_l}{4} \text{Re} \int_{\text{surf}_l} ds \left| \vec{E}_{\text{max}} \right|^2$$

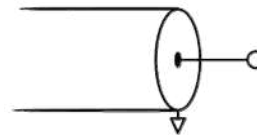
Inductive



$$p_{\text{ind,surf}}^l = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 \mu_l}{4} \text{Re} \int_{\text{surf}_l} ds \left| \vec{H}_{\text{max},\parallel} \right|^2$$

$$p_{\text{ind,seam}}^l = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 t_l \mu_l}{4} \text{Re} \int_{\text{seam}_l} dl \left| \vec{H}_{\text{max},\perp} \right|^2$$

Radiative



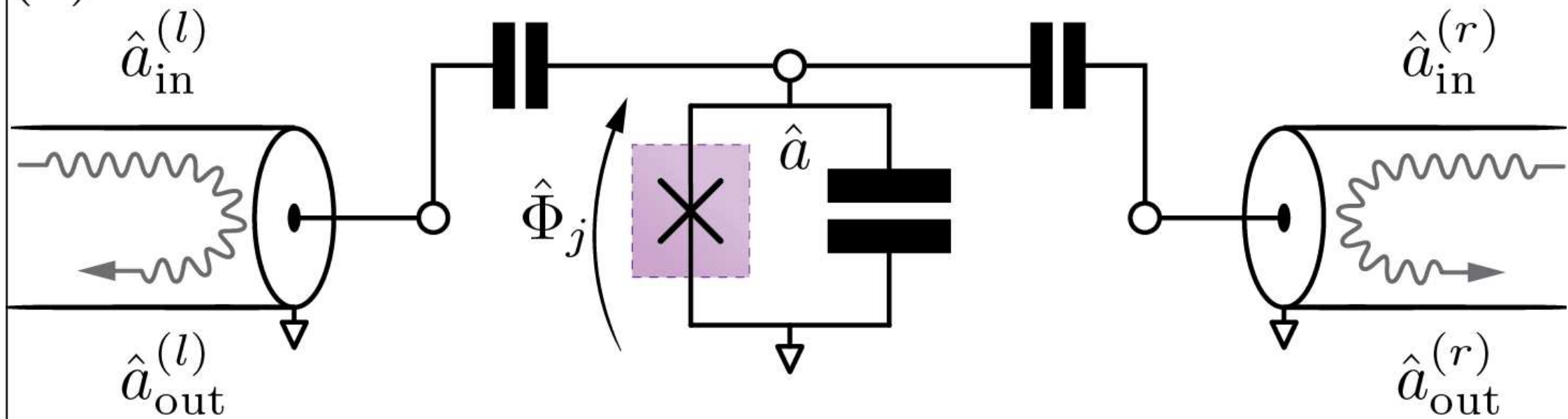
$$Q_{mp} = \frac{\omega_m \mathcal{E}_m(0)}{\frac{1}{2} R I_{mp}^2}$$

Pozar, Microwave engineering (1990); Gao, Thesis (2008)

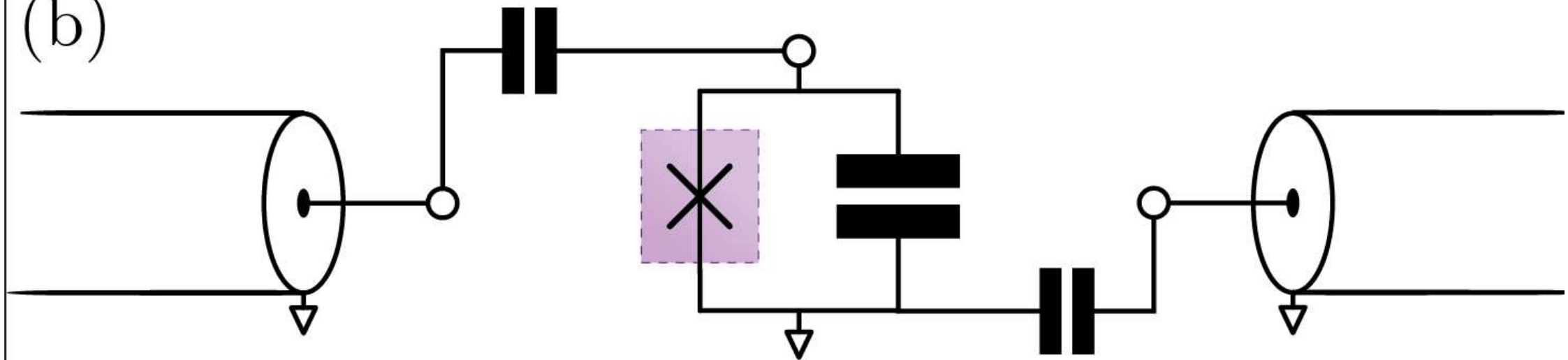
Zmuidzinas, ARCMP (2012); Geerlings, Thesis (2013);

Brecht *et al.*, APL (2015); Wang *et al.*, APL (2015); Dial SST *et al.* (2016); ...

(a)



(b)

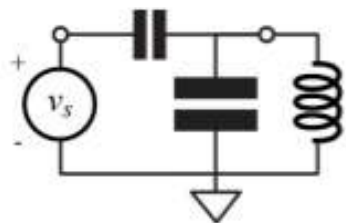


Device	Frequency (MHz)		Anharmonicity (MHz)		Cross-Kerr (MHz)
	$\omega_D/2\pi$	$\omega_B/2\pi$	$\alpha_D/2\pi$	$\alpha_B/2\pi$	$\chi_{DB}/2\pi$
DTW1	6010	8670	85	180	278
	5824	8878	97	206	281
	-3.2%	2.3%	12%	13%	1.1%

Table III. Two-qubit, one-waveguide devices. Summary of measured and calculated Hamiltonian parameters for the device described in Sec. #TODO. Indices D and B denote the dark and bright modes respectively. For each device, the first (second) row summarizes the measured, m , (calculated, c) values. The third row summarizes the agreement, $(c - m)/c$.

Device	Frequency (MHz)			Anharmonicity (MHz)	Cross-Kerr (MHz)		I - O coupling
	$\omega_Q/2\pi$	$\omega_S/2\pi$	$\omega_C/2\pi$	$\alpha_D/2\pi$	$\chi_{QS}/2\pi$	$\chi_{QC}/2\pi$	Q_C
WG1	4890	7070	7267	310	0.25	0.30	20×10^3
	4820	7020	7340	325	0.29	0.33	16×10^3
	-1.4%	-0.7%	1.0%	5.1%	13%	9%	-22%

Table II. Flip-chip (2.5D), one-qubit, one-storage-cavity, one-readout-cavity devices. Summary of measured and calculated Hamiltonian and input-output (I - O) coupling parameters for the device described in Sec. #TODO. Indices Q, S, C denote the qubit, storage, and readout cavity modes respectively. The input-output quality factor to the readout cavity is denoted Q_C . For each device, the first (second) row summarizes the measured, m , (calculated, c) values. The third row summarizes the agreement, $(c - m)/c$.



Charge dispersion with EPR

$$\omega_0 = 6.50 \text{ GHz}$$

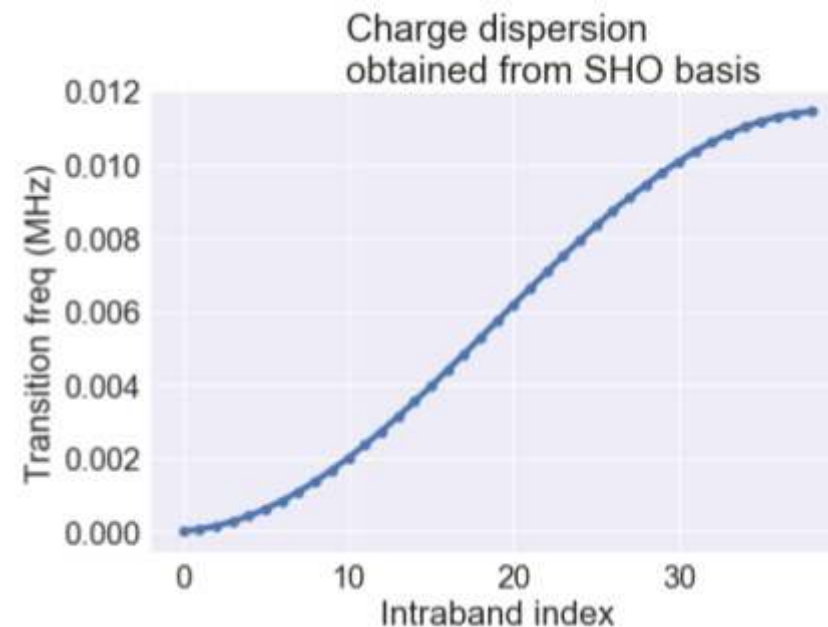
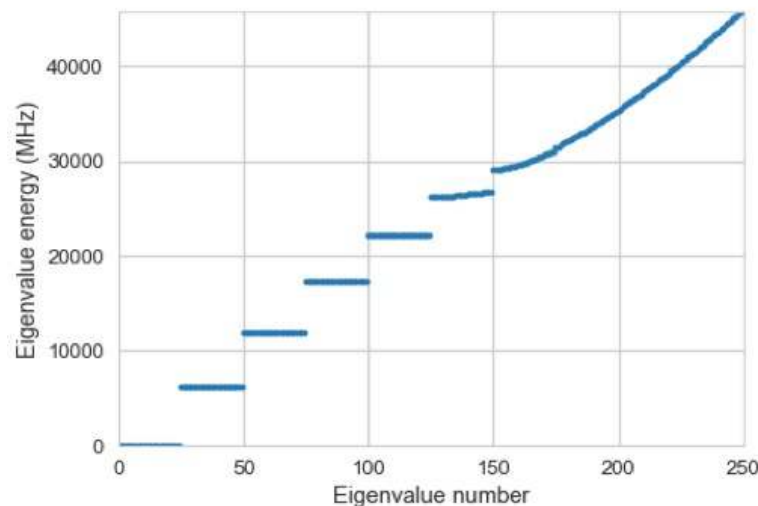
$$\alpha^{(4)} = 322.00 \text{ MHz}$$

$$\phi_{\text{ZPF}} = 0.45$$

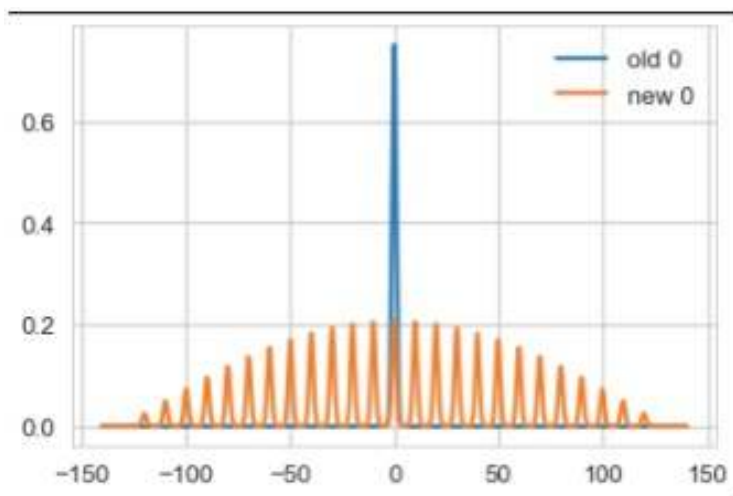
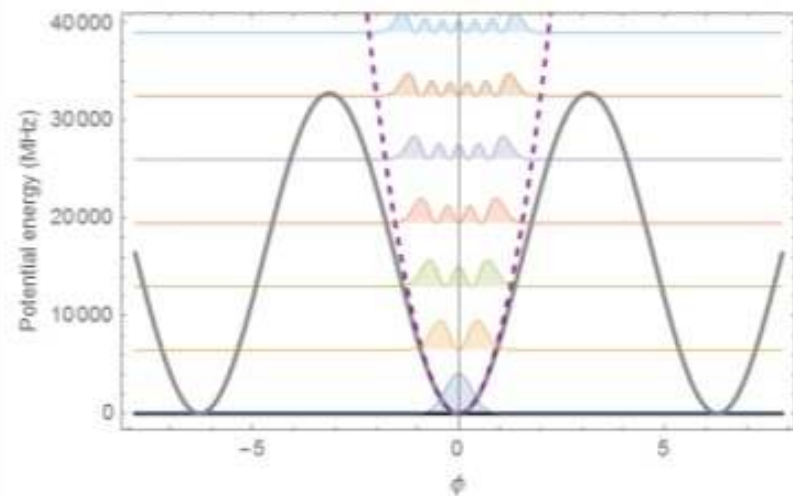
$$E_J = 16.40 \text{ GHz}$$

$$E_C = 0.32 \text{ GHz} \quad (E_J/E_C = 50.90)$$

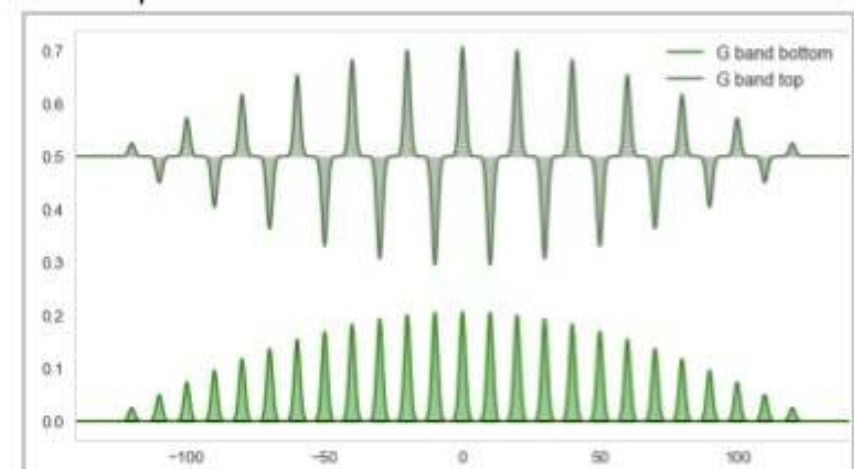
$$\text{Dispersion} = -12.90 \text{ KHz}$$



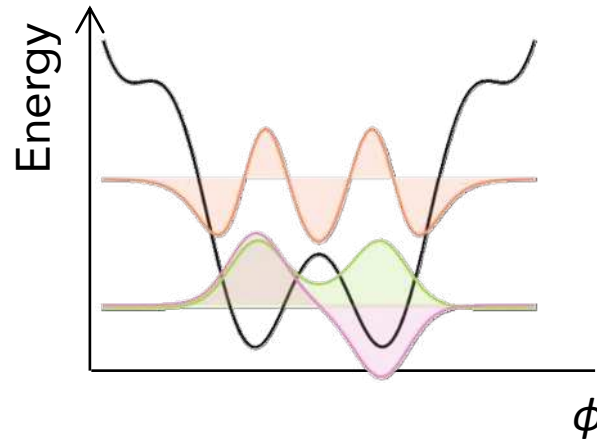
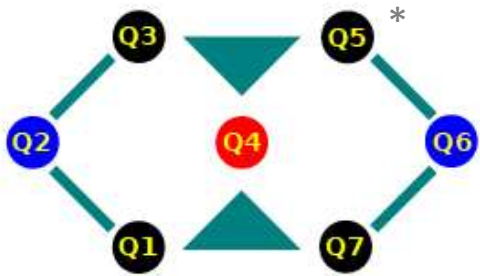
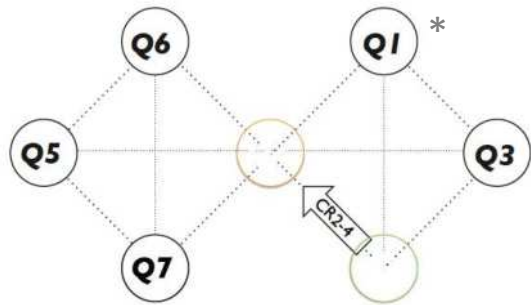
$|\psi|^2$



ϕ

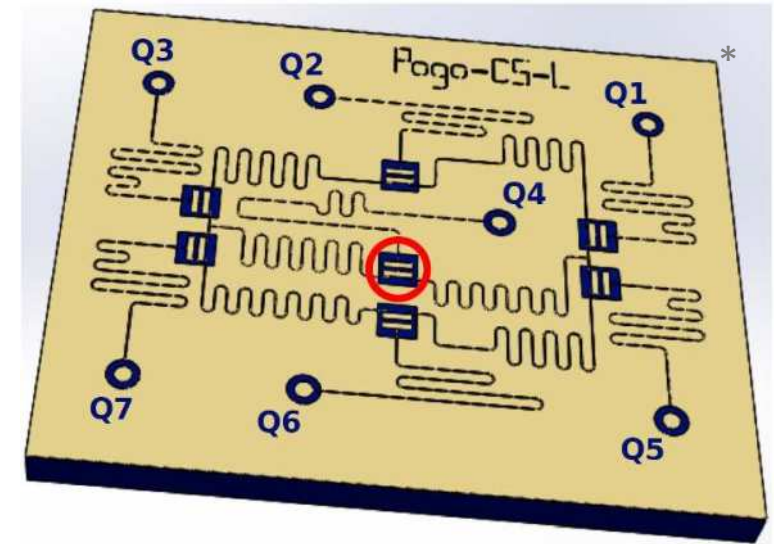
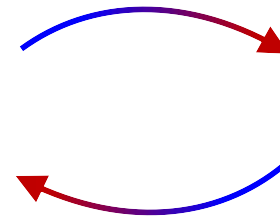


Is it possible to find optimal
distributed circuit given $\mathcal{H}$ and $\mathcal{D}$?

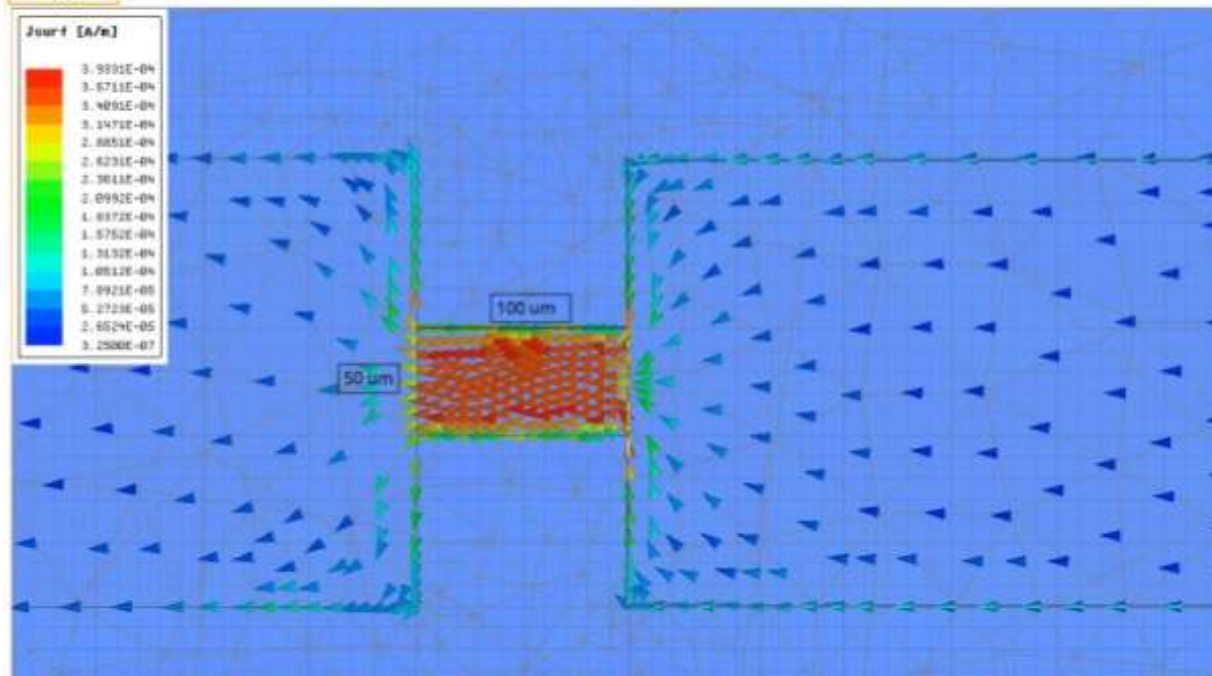


$$\hat{a}^\dagger \hat{b}^2 \xi(t)$$

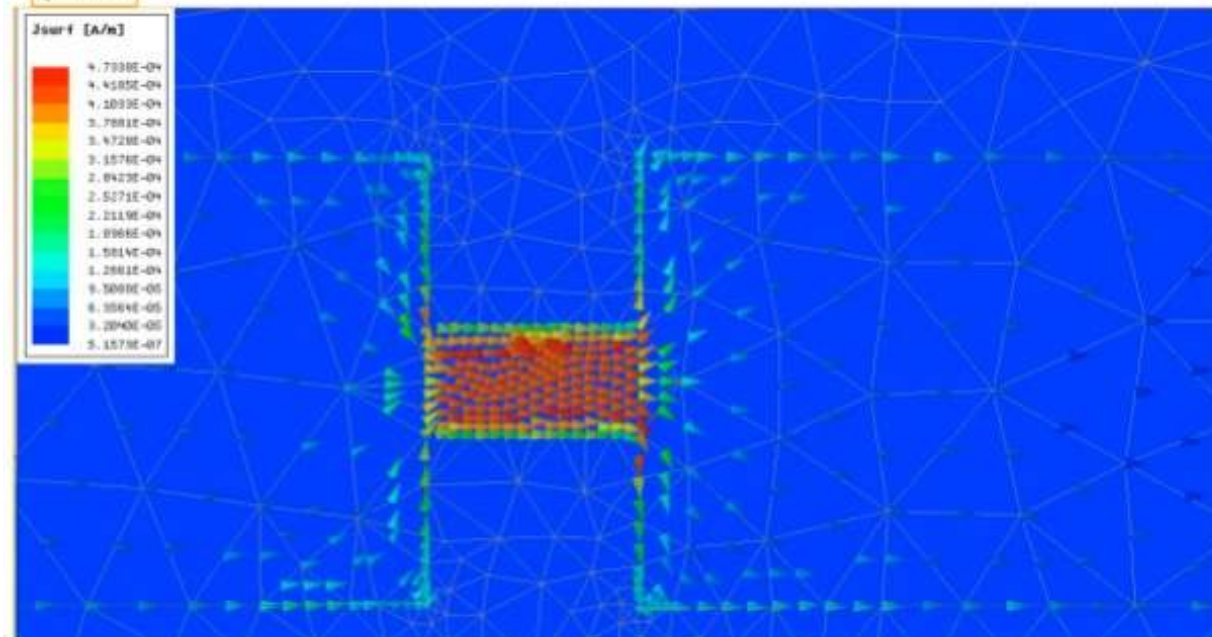
...

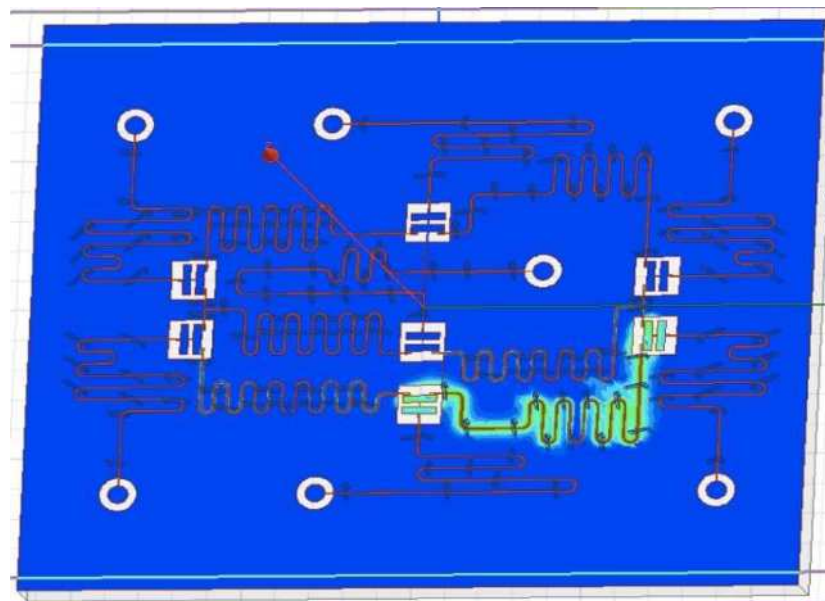
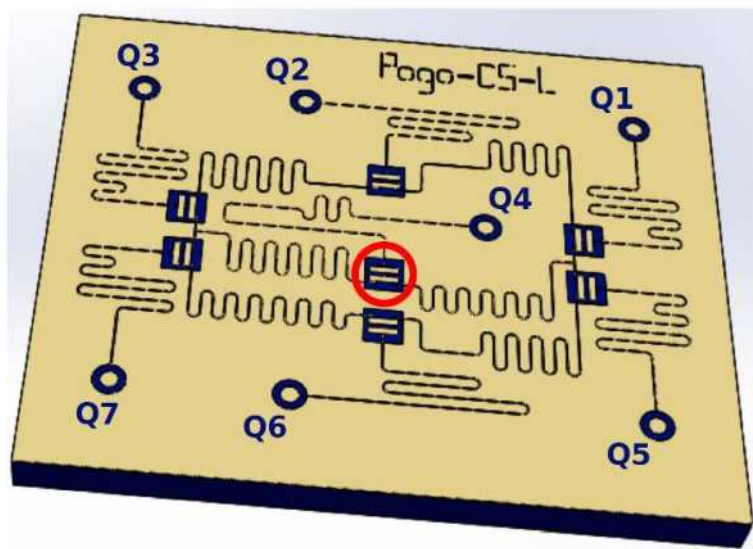


Cavity phase

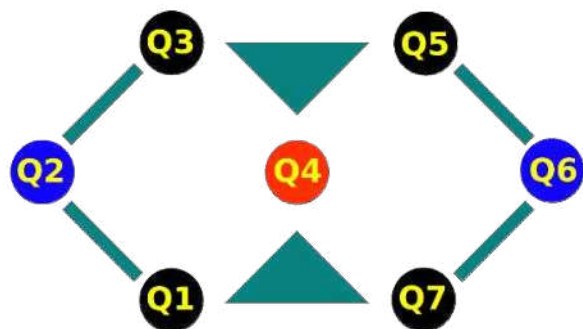


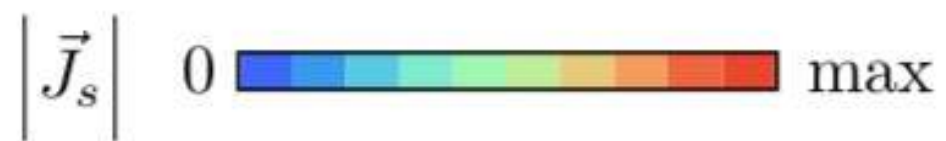
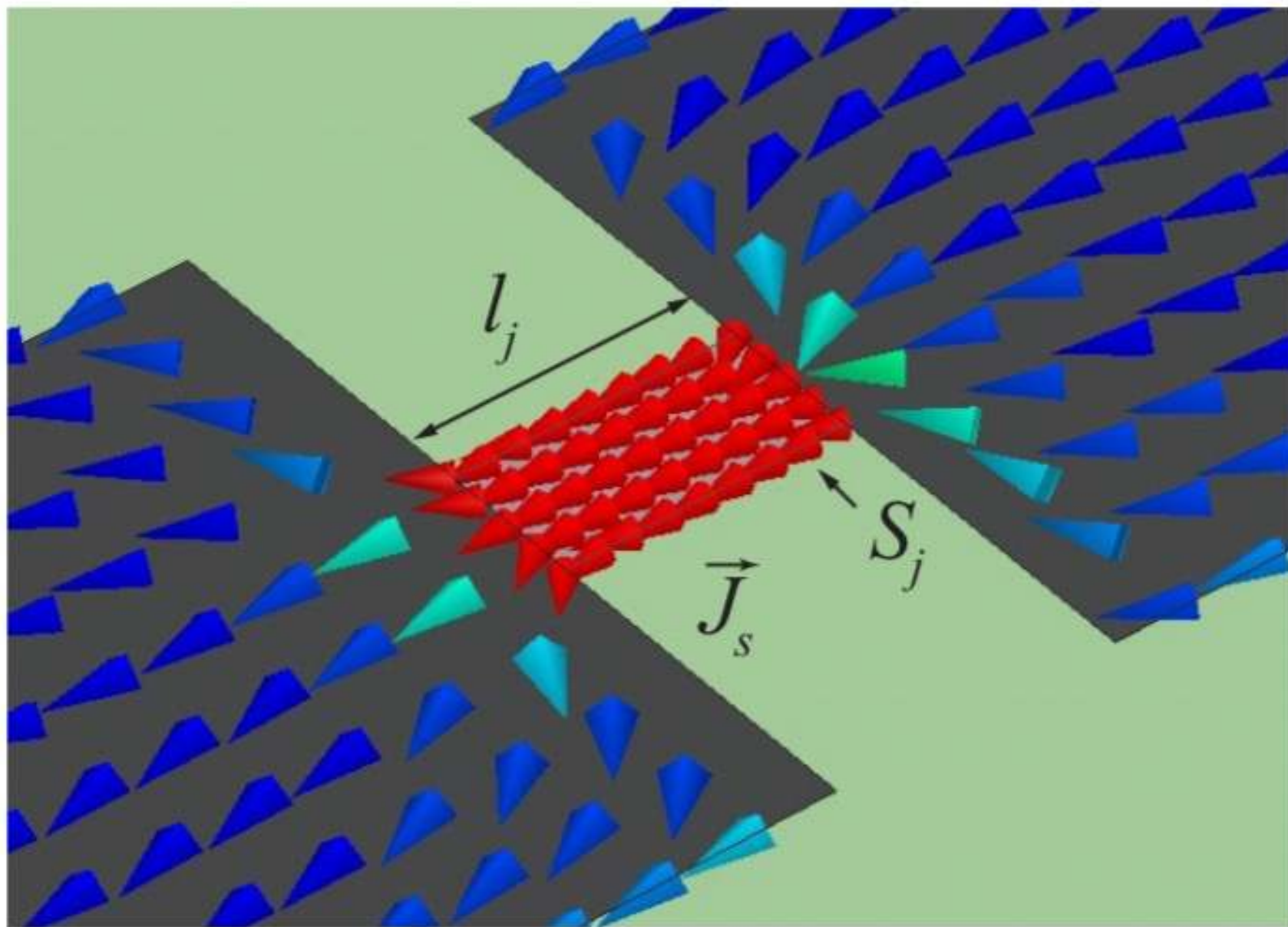
Qubit mode



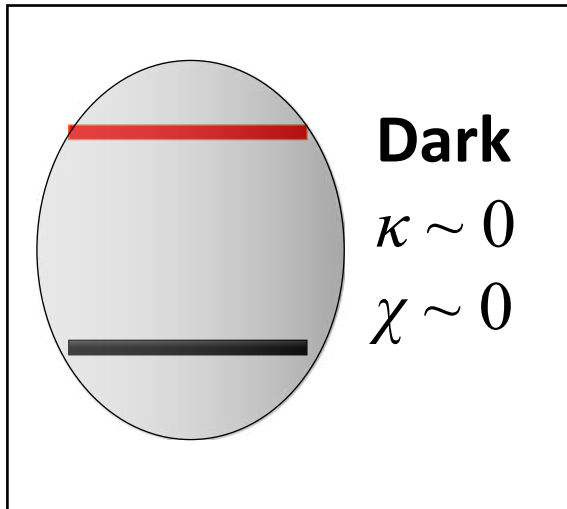


* Bronn, N.T. *et al.*, QST (2018)

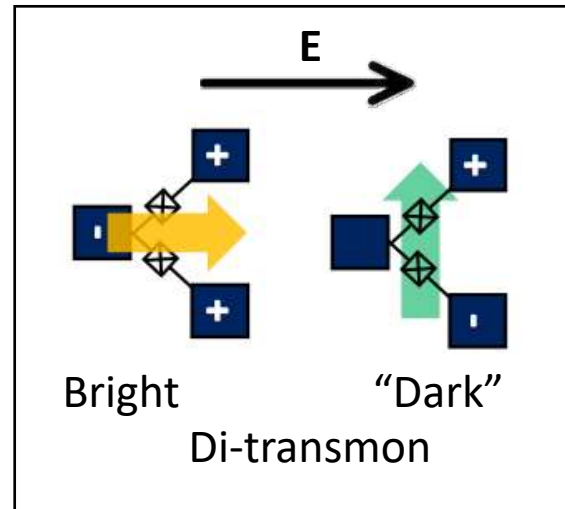




Step 2: Zero- χ qubit, but large- χ bright level

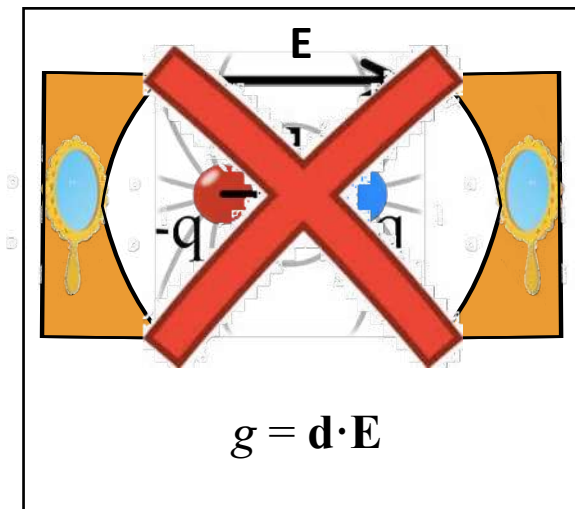


Goal: Dark mode



$$\kappa_r \sim 0$$

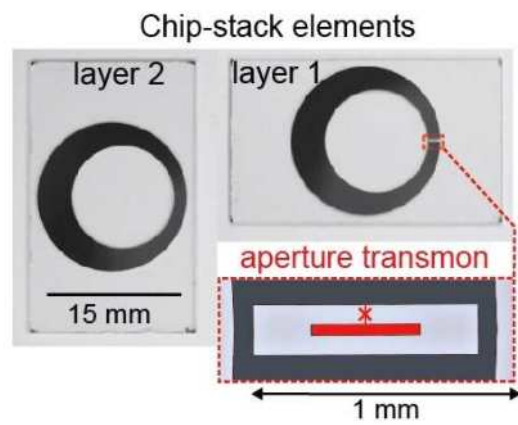
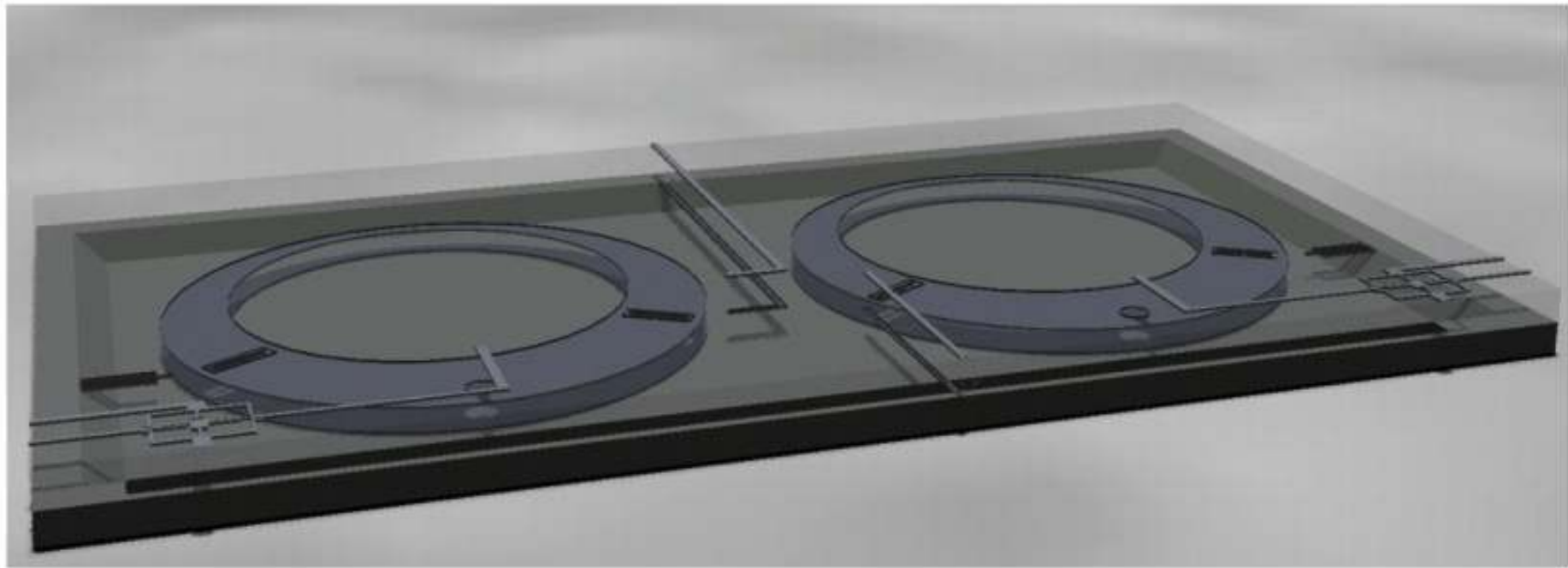
$$\chi_B \sim \chi_D \text{ (not 0!)}$$



κ is sort of given by dipole coupling, but χ is absolutely not!

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar^2 \omega_m \omega_n}{4E_j} p_{mj} p_{nj}$$

Mode ↗
 ↗ Junction
 Fixed
 Only thing to change
 Energy participation overlap

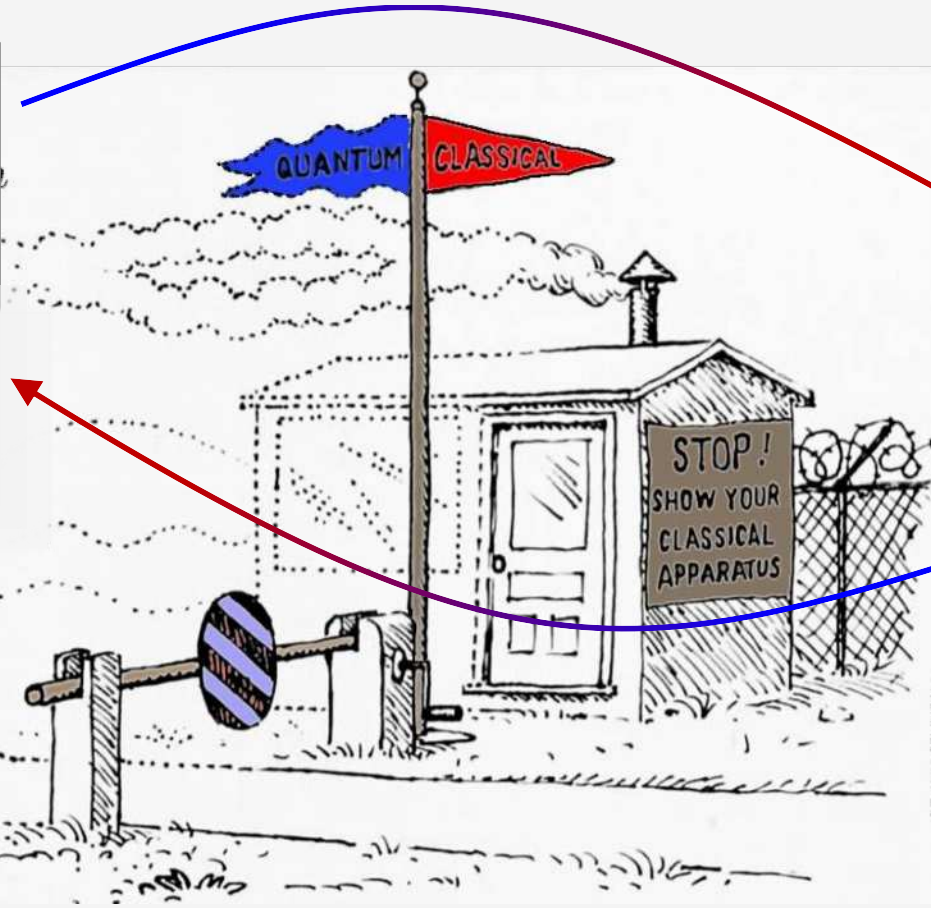
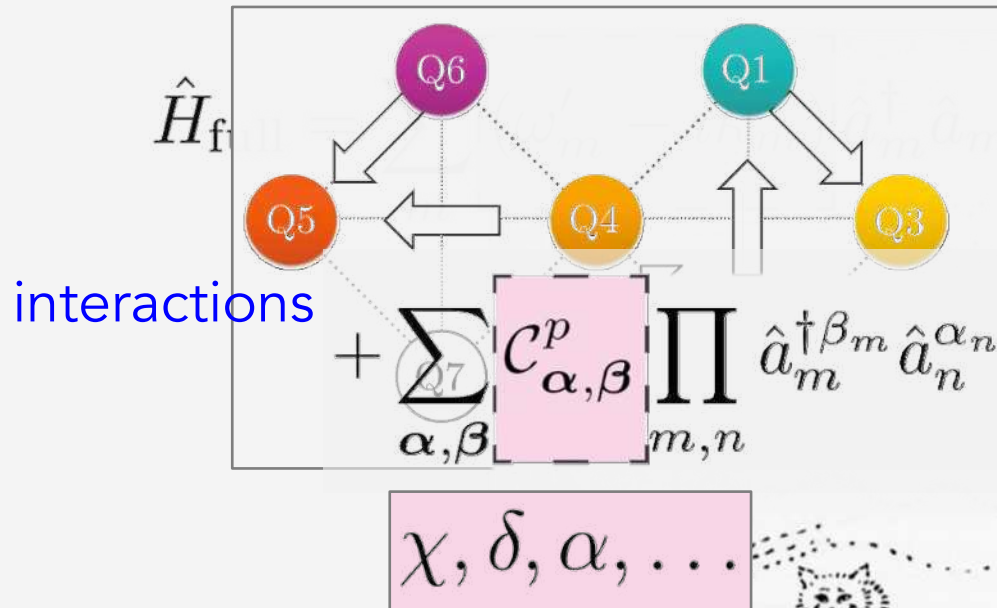


reduced to

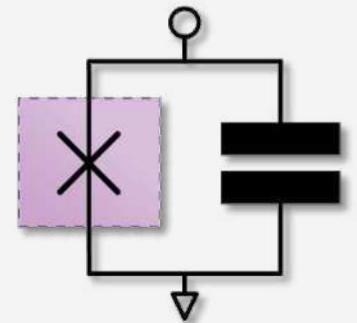
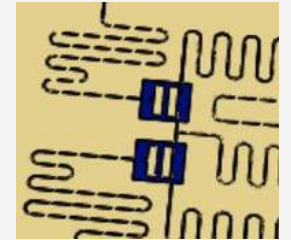
What fraction of the energy of a mode
is stored in the junction?

Quantum-classical bridge: Quantization

bosonic modes coupled
quantum system
to environment

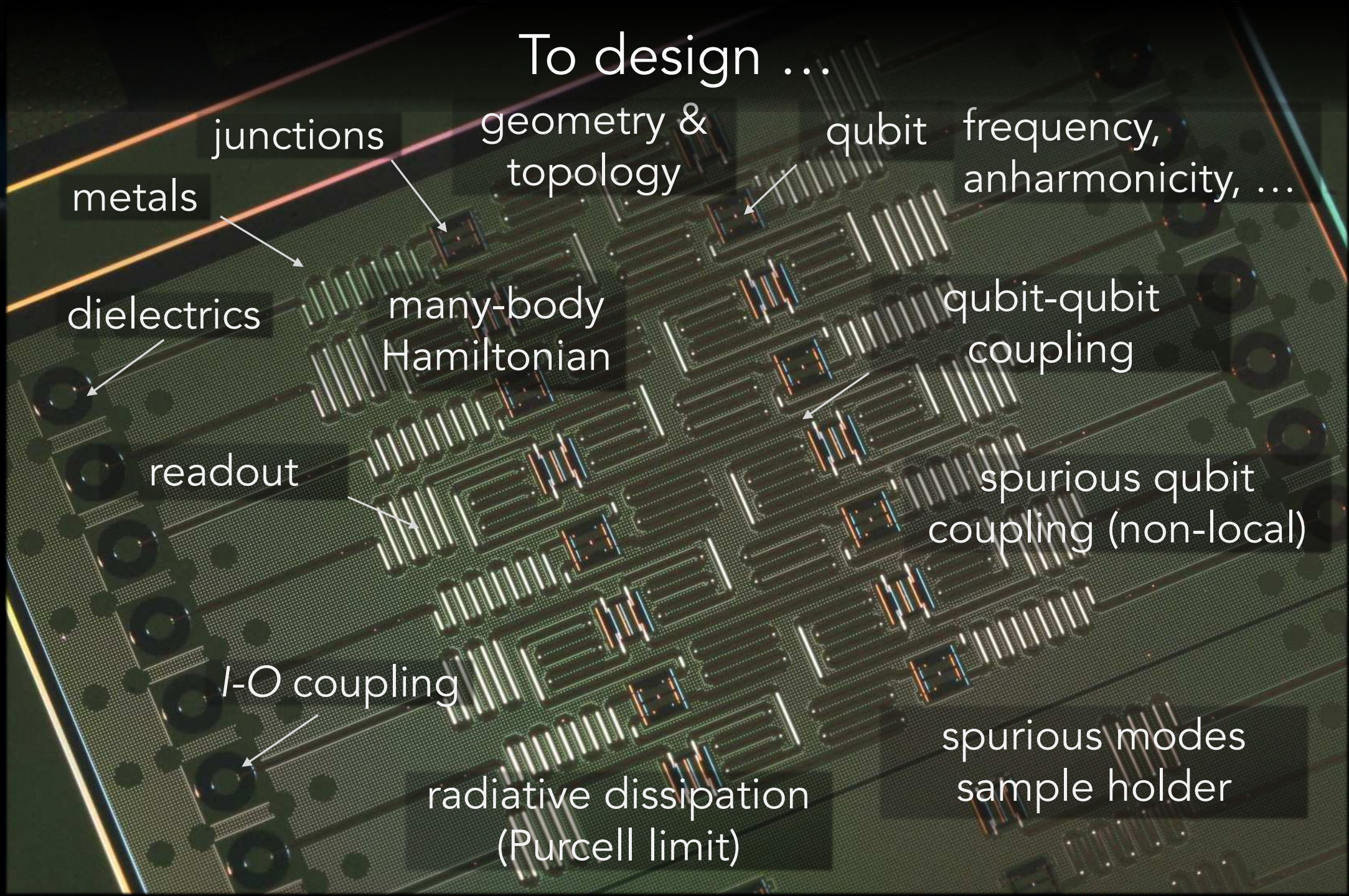
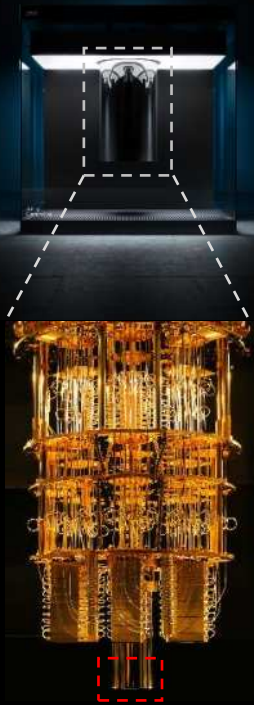


electromagnetic
fields,
circuits



geometry,
metals,
dielectrics, ...

To design ...



metals

junctions

geometry &
topology

qubit

frequency,
anharmonicity, ...

dielectrics

many-body
Hamiltonian

qubit-qubit
coupling

readout

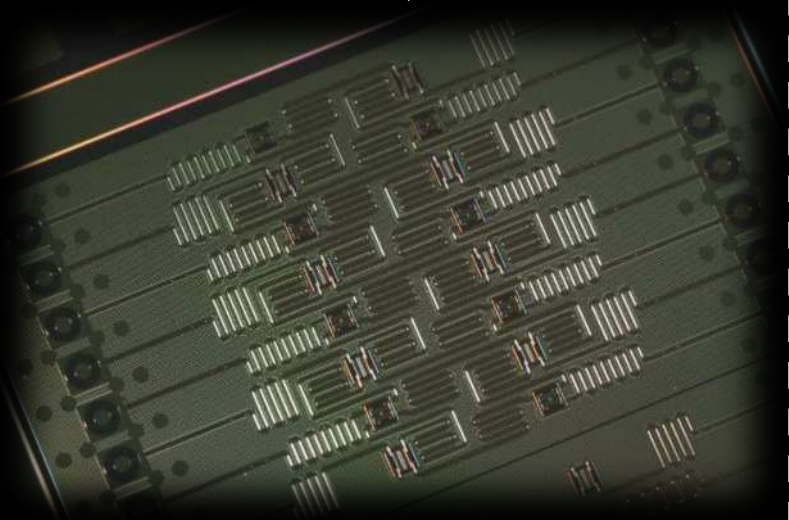
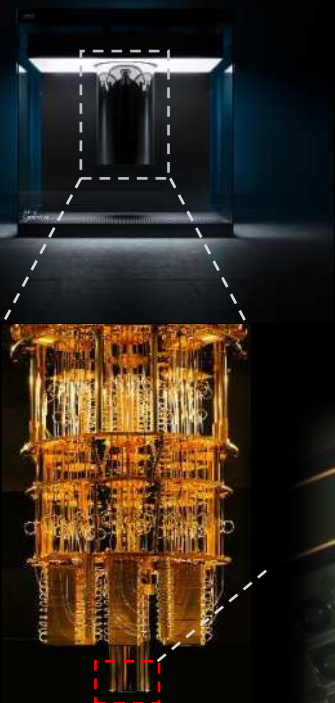
spurious qubit
coupling (non-local)

I-O coupling

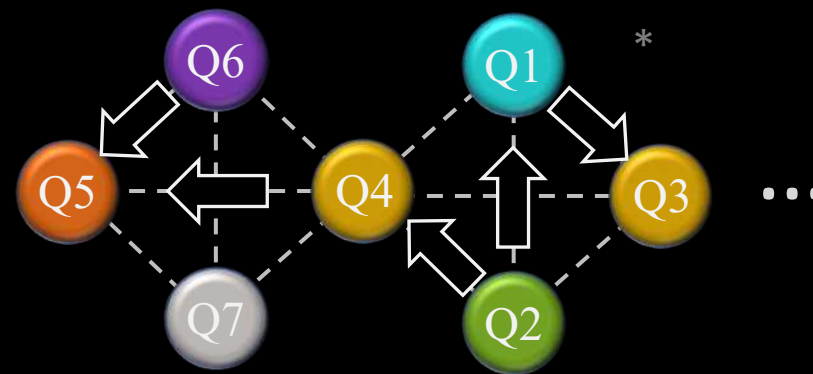
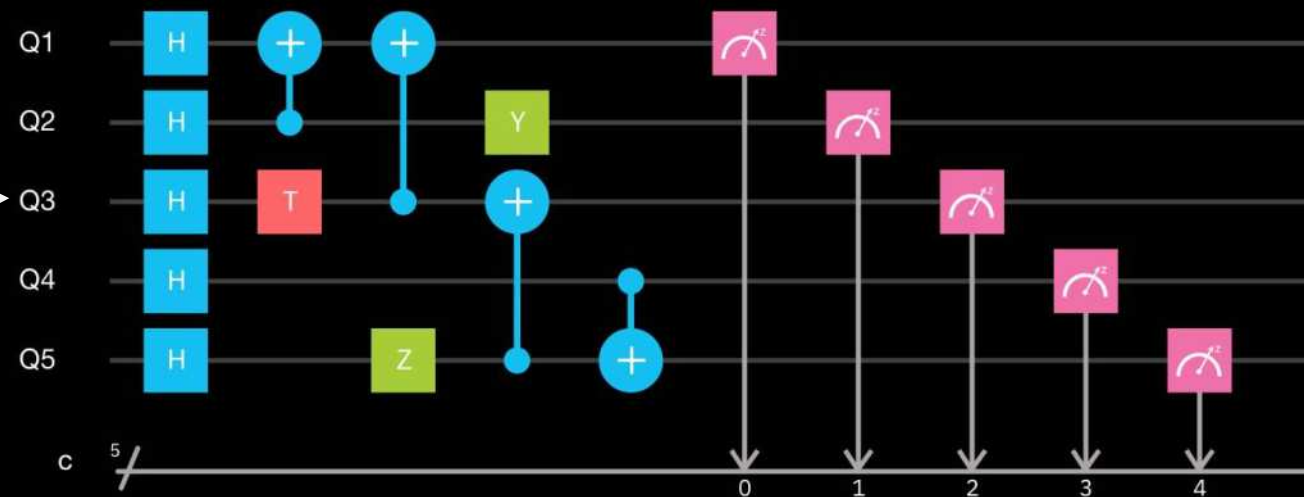
radiative dissipation
(Purcell limit)

spurious modes
sample holder

classical



quantum



* IBM quantum experience

Review of ideal CR

Ideal

$$\hat{H}_{\text{CR}} = \frac{1}{2} \omega_{zx} \hat{\sigma}_z \hat{\sigma}_x ,$$

$$\tau_{\text{CNOT}} = \frac{\pi}{2\omega_{zx}} ,$$

Unitary

$$\hat{U}_{\text{CR}}(\tau) = \cos\left(\frac{\omega_{zx}\tau}{2}\right) \hat{\sigma}_0 \hat{\sigma}_0 - i \sin\left(\frac{\omega_{zx}\tau}{2}\right) \hat{\sigma}_z \hat{\sigma}_x$$

CNOT

$$\text{CNOT} = \begin{pmatrix} \text{g g} & \text{g g} & \text{g e} & \text{e g} & \text{e e} \\ \text{g g} & 1 & 0 & 0 & 0 \\ \text{g e} & 0 & 1 & 0 & 0 \\ \text{e g} & 0 & 0 & 0 & 1 \\ \text{e e} & 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\text{CNOT} = [ZI]^{-1/2} [ZX]^{1/2} [IX]^{-1/2}$$

On target

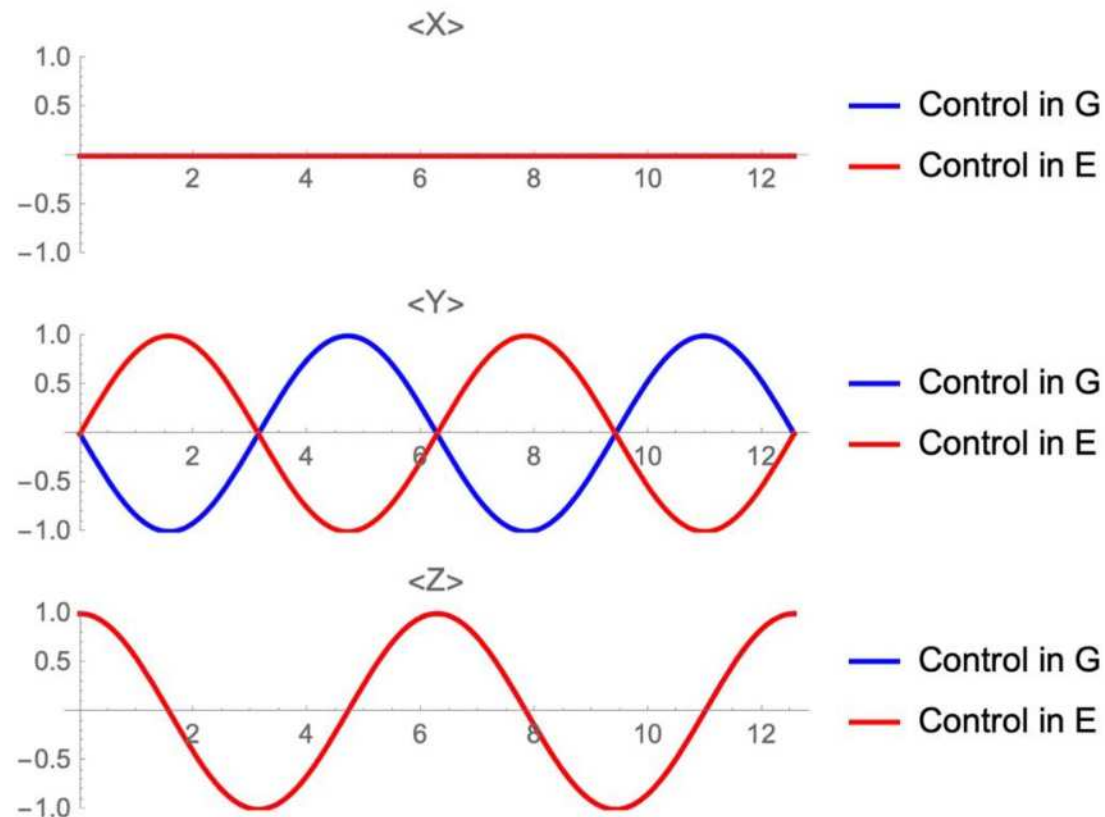


Figure 3. Ideal CR gate, ala Sheldon calibration.

Review of ideal CR

Ideal

$$\hat{H}_{\text{CR}} = \frac{1}{2}\omega_{zx}\hat{\sigma}_z\hat{\sigma}_x,$$

$$\tau_{\text{CNOT}} = \frac{\pi}{2\omega_{zx}},$$

Unitary

$$\hat{U}_{\text{CR}}(\tau) = \cos\left(\frac{\omega_{zx}\tau}{2}\right)\hat{\sigma}_0\hat{\sigma}_0 - i\sin\left(\frac{\omega_{zx}\tau}{2}\right)\hat{\sigma}_z\hat{\sigma}_x$$

Spurious IX term

$$\hat{H}_{\text{CR}} = \frac{1}{2}\omega_{zx}\hat{\sigma}_z\hat{\sigma}_x + \frac{1}{2}\omega_{ix}\hat{\sigma}_0\hat{\sigma}_x$$

$$\begin{aligned}\hat{U}_{\text{CR}}(\tau) = & \cos\left(\frac{\tau\omega_{ix}}{2}\right)\cos\left(\frac{\tau\omega_{zx}}{2}\right)\hat{I}\hat{I} - i\cos\left(\frac{\tau\omega_{ix}}{2}\right)\sin\left(\frac{\tau\omega_{zx}}{2}\right)\hat{Z}\hat{X} \\ & - i\sin\left(\frac{\tau\omega_{ix}}{2}\right)\cos\left(\frac{\tau\omega_{zx}}{2}\right)\hat{I}\hat{X} - \sin\left(\frac{\tau\omega_{ix}}{2}\right)\sin\left(\frac{\tau\omega_{zx}}{2}\right)\hat{Z}\hat{I},\end{aligned}$$

$$\frac{\omega_{zx} + \omega_{ix}}{2} \text{ and } \frac{\omega_{zx} - \omega_{ix}}{2}$$

On target

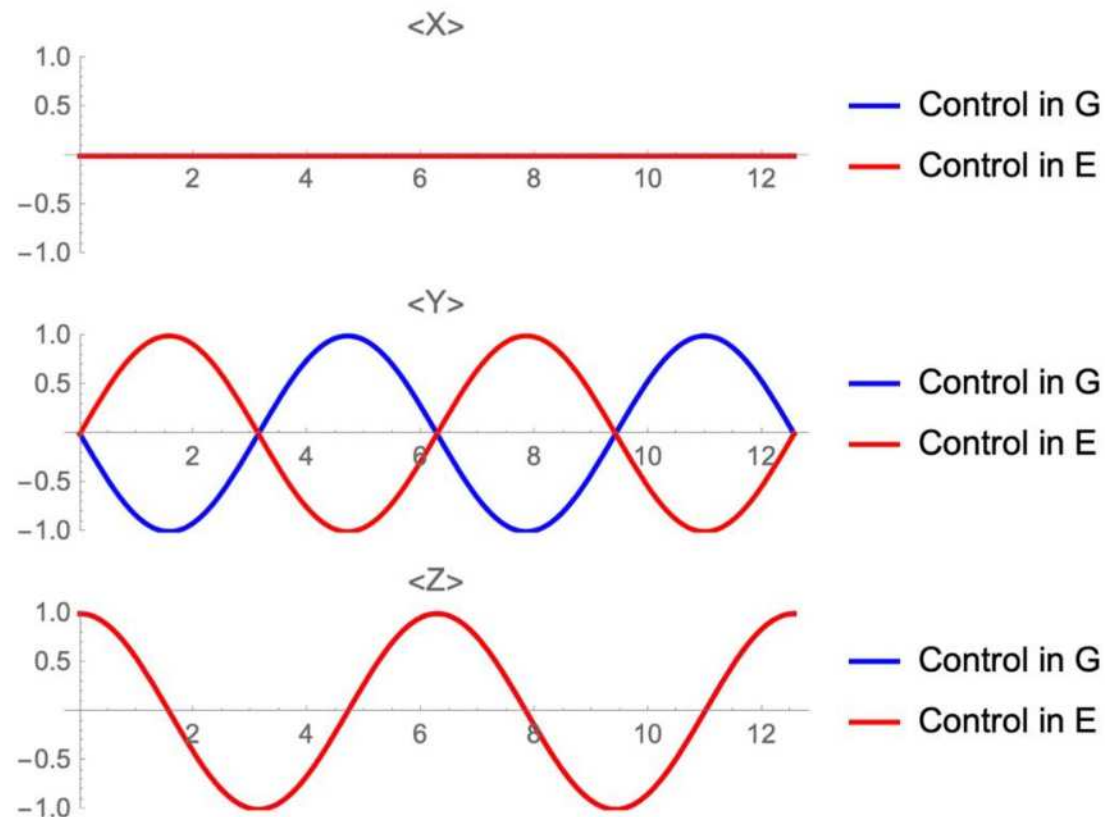
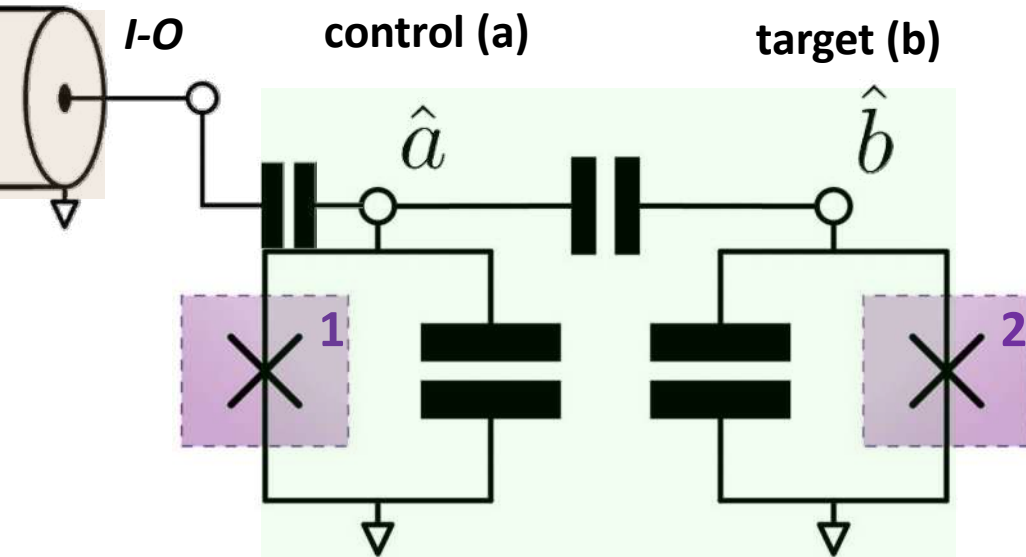


Figure 3. Ideal CR gate, ala Sheldon calibration.

Direct coupler: complete description



First approximations (removed later):

Ignore target junction? ($E_2 \rightarrow 0$)

YES, for ZX, no for IX.

Ignore target direct drive? ($\epsilon_b \rightarrow 0$)

YES, for ZX, no for IX.

$$\hat{H} = \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \quad \text{Eigen modes}$$

$$- \sum_{j=1}^2 \frac{E_j}{4!} \left(\phi_{aj} (\hat{a} + \hat{a}^\dagger) + \phi_{bj} (\hat{b} + \hat{b}^\dagger) \right)^4 + \dots \quad \text{Non-linear}$$

$$- i\epsilon_a(t) (\hat{a}^\dagger - \hat{a}) - i\epsilon_b(t) (\hat{b}^\dagger - \hat{b}) \quad \text{Drives}$$

Regime of interest

$$\begin{pmatrix} p_{a1} & p_{a2} \\ p_{b1} & p_{b2} \end{pmatrix} \approx \begin{pmatrix} 0.92 & 2 \times 10^{-4} \\ 2 \times 10^{-4} & 0.92 \end{pmatrix}$$

$$\phi_{mj}^2 = p_{mj} \frac{\hbar \omega_m}{2E_j}$$

χ s that are small, ≈ 200 KHz,

second quantization in eigen basis of linearized circuit

Direct coupler: control junction, no CX talk

Coupled System Hamiltonian
(drive-free)

$$\begin{aligned}\hat{H}_{4,\text{RWA1}} = H_a + H_b + H_{ab} = & (\omega_a - \Delta_a) \hat{a}^\dagger \hat{a} - \frac{\alpha_a}{2} \hat{a}^{\dagger 2} \hat{a}^2 \\ & + (\omega_b - \Delta_b) \hat{b}^\dagger \hat{b} - \frac{\alpha_b}{2} \hat{b}^{\dagger 2} \hat{b}^2 \\ & - \chi_{ab} \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b},\end{aligned}$$

Control single-mode

Target single-mode

Control-target

amplitudes

$$\Delta_{a/b} = \alpha_{a/b} + \frac{1}{2} \chi_{ab}, \quad \alpha_m = \frac{1}{2} \sum_{j=1}^{J=2} E_j \phi_{mj}^4, \quad \chi_{ab} = \frac{1}{2} \sum_{j=1}^{J=2} E_j \phi_{aj}^2 \phi_{bj}^2 \quad \{\text{p}=4, \text{RWA1}\}$$

Important corrections to {p=4,RWA2} and {p=6, RWA1}

Driven

$$\xi(t) = \frac{\epsilon}{\Delta_{ad}} e^{i\omega_b t}$$

$$\hat{\phi}_{j,\text{disp}} = \phi_a \left(\hat{a} e^{-i\omega_a t} + \hat{a}^\dagger e^{i\omega_a t} + \xi(t) + \xi^*(t) \right) + \phi_b \left(\hat{b} e^{-i\omega_b t} + \hat{b}^\dagger e^{i\omega_b t} \right)$$

CR Gate: weakly anharmonic system

{p=4, RWA1}

$$\hat{H}_{\text{CR},p=4,\text{RWA1}} = \underbrace{-\frac{E_j}{4!} 24 \phi_a^3 \phi_b \left[\left(\hat{a}^\dagger \hat{a} - \frac{1}{2} \right) (\xi \hat{b} + \xi^* \hat{b}^\dagger) \right]}_{\text{CR Gate Term (ZX)!}} \quad (\text{II.4})$$

$$- \frac{E_j}{4!} 24 |\xi|^2 \left[\phi_a^4 \hat{a}^\dagger \hat{a} + \phi_a^2 \phi_b^2 \hat{b}^\dagger \hat{b} \right] \quad \text{Stark shifts (Z)}$$

$$- \frac{E_j}{4!} 12 \xi \left(|\xi|^2 \phi_a^3 \phi_b + \phi_a \phi_b^3 \right) \hat{b} + \text{H.c.} \quad \text{Spurious target}_b \text{ X Y drive term (X,Y)!}$$

$$- \frac{E_j}{4!} (6 \xi^2 \phi_a^2 \phi_b^2) \hat{b}^{\dagger 2} + \text{H.c.} \quad \text{Spurious target}_b \text{ two-photon term}$$

$$- \frac{E_j}{4!} (12 \xi \phi_a \phi_b^2) \hat{b}^\dagger (\hat{b}^\dagger \hat{b}) + \text{H.c.} \quad \text{Spurious target}_b \text{ three-photon term}$$

Two-level manifold

$$\left(\hat{a}^\dagger \hat{a} - \frac{1}{2} \hat{I} \right) \rightarrow -\frac{1}{2} \hat{\sigma}_z \hat{\sigma}_0, \quad (\hat{b}^\dagger + \hat{b}) \rightarrow \hat{\sigma}_0 \hat{\sigma}_x, \quad (\hat{a}^\dagger \hat{a}) \rightarrow \frac{1}{2} (\hat{\sigma}_0 - \hat{\sigma}_z) \hat{\sigma}_0,$$

CR Gate: weakly anharmonic system

{p=4, RWA1, TLS}

$$\hat{H} = \frac{1}{2}\omega_{zx}\hat{\sigma}_z\hat{\sigma}_x$$

CR Gate Term (ZX)

$$+ \frac{1}{2}\omega_{ix}\hat{\sigma}_0\hat{\sigma}_x$$

Spurious target (IX)

$$+ \frac{1}{2}\omega_{zi}(\hat{1} - \hat{\sigma}_z\hat{\sigma}_0) + \frac{1}{2}\omega_{iz}(\hat{1} - \hat{\sigma}_0\hat{\sigma}_z)$$

Stark shifts (Z)

For control junction:

$$\omega_{zx} = -s_a s_b \frac{\sqrt{\hbar^4 \omega_a^3 \omega_b}}{4E_j} \sqrt{p_a^3 p_b} \xi \approx \sqrt{p_b} \times \frac{1}{2} \text{ GHz}.$$

for multiple junctions just sum this expression over j .

$$\phi_b \ll \phi_a,$$

$$\omega_{zx} = E_j \phi_a^3 \phi_b \xi$$

$$\omega_{zi} = -E_j \phi_a^4 |\xi|^2 = -2\alpha_a |\xi|^2$$

$$\omega_{ix} \approx -E_j \phi_a^3 \phi_b \xi |\xi|^2$$

$$\omega_{iz} = -E_j \phi_a^2 \phi_b^2 |\xi|^2 = -\chi_{ab} |\xi|^2$$

, $\omega_{zx} = \frac{J}{2\Delta}$, see Rigetti (2010),

CR Gate: weakly anharmonic system

{p=4, RWA1, TLS}

```
-----
phi_a = 0.467873
p_a = 0.92
omega_a = 2pi x 6000. MHz
E_J = 2pi x 12608. MHz,
alpha_{p=4,RWA1} = 2pi x 302. MHz
alpha_{p=4,RWA2} = 2pi x 65. MHz      = 65. MHz
alpha_{p=6,RWA1} = 2pi x -33. MHz
alpha_total = 2pi x 334. MHz
Stark shift (xi+xi*) xi=1: 2pi x {604., 264., 56.} MHz
-----
```

```
E_J = ( 12 610. ) MHz
      ( 13 660. )

phi_mj = ( 0.468 -0.007 )
          ( 0.007  0.442 )

alpha_m^{(p=4,RWA1)} = ( 302. ) MHz
                      ( 261. )

chi_{mm'}^{(p=4,RWA1)} = ( 604.20  0.24 ) MHz      Diagonal is 2x anhar
                        ( 0.24  521.10 )
```

(MHz)	$\xi_b = 0$	$\xi_b = 0.1$	Significant
$\omega_{xz}/2\pi$	8.76	8.78	N
$\omega_{iz}/2\pi$	-0.25	-3.9	so-so
$\omega_{ix}/2\pi$	-0.47	-27	Y
$\omega_{zi}/2\pi$	-604	-606	N

For $\xi = 1$. For standard params I used above too.

Conclusions / future directions

Direct coupling in EPR basis ($M=2, J=2$)

All qubit parameters & driven shifts

$\{p=4, \text{RWA2}\}$ and $\{p=6, \text{RWA1}\}$

CR rate rate and spurious terms in $\{p=4, \text{RWA1}\}$

Numerically verified (understood which terms matter)

Direct coupling

$\{p=4, \text{RWA2}\}$. And $\{p=6, \text{RWA1}\}$

anharmonic basis

idea: back out J

Bus coupling ($M=3, J=1$)

$\{p=4, \text{RWA1}\}$

eliminate bus

$\{p=4, \text{RWA2}\}$. And $\{p=6, \text{RWA1}\}$

anharmonic basis